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(54) VIDEO CODING METHOD AND DEVICE THEREFOR

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*H04N 19/176* (2014.01)
*H04N 19/70* (2014.01)

(52) **U.S. Cl.**
CPC ......... ***H04N 19/137*** (2014.11); ***H04N 19/172*** (2014.11); ***H04N 19/176*** (2014.11); ***H04N 19/70*** (2014.11)

(57) **ABSTRACT**

An image decoding method includes generating reference motion information for at least one block included in a current picture, based on at least one of: pixel data of at least one reconstructed picture, or motion information of the at least one reconstructed picture; obtaining motion information for a current block from the at least one block included in the current picture, based on the reference motion information and at least one syntax information obtained from a bitstream; and reconstructing the current block, based on the motion information for the current block. The method may include generating the reference motion information using a neural network. An image encoding method includes similar steps of generating reference motion information, determining motion information for a current block, and encoding the motion information into a bitstream, based on the reference motion information.

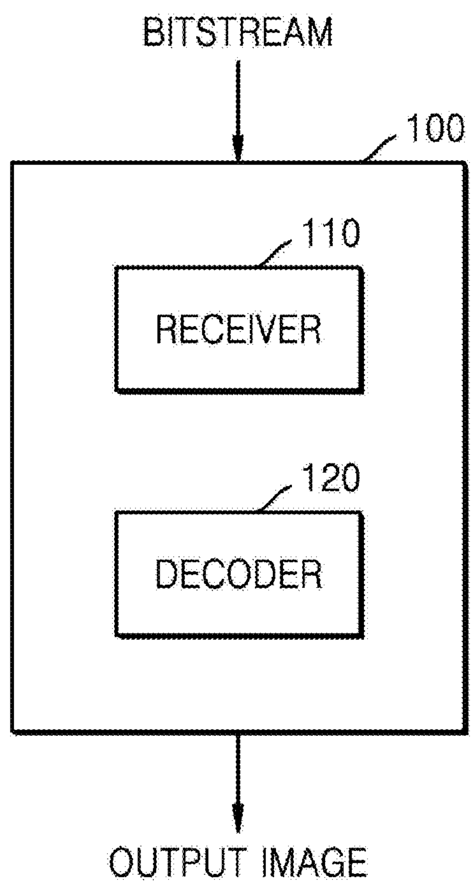


FIG. 1

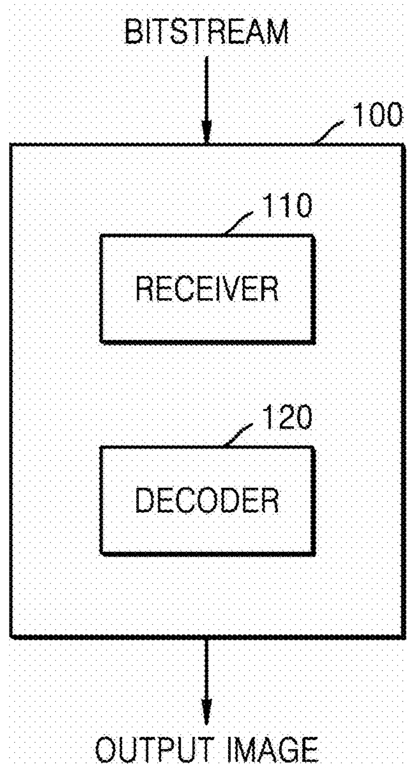


FIG. 2

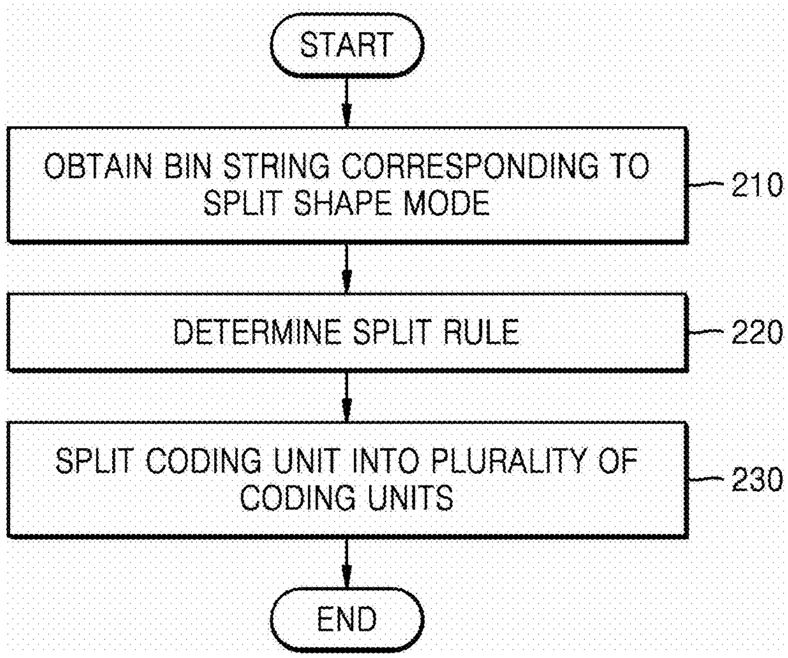

START
OBTAIN BIN STRING CORRESPONDING TO SPLIT SHAPE MODE
210
DETERMINE SPLIT RULE
220
SPLIT CODING UNIT INTO PLURALITY OF CODING UNITS
230
END

FIG. 3

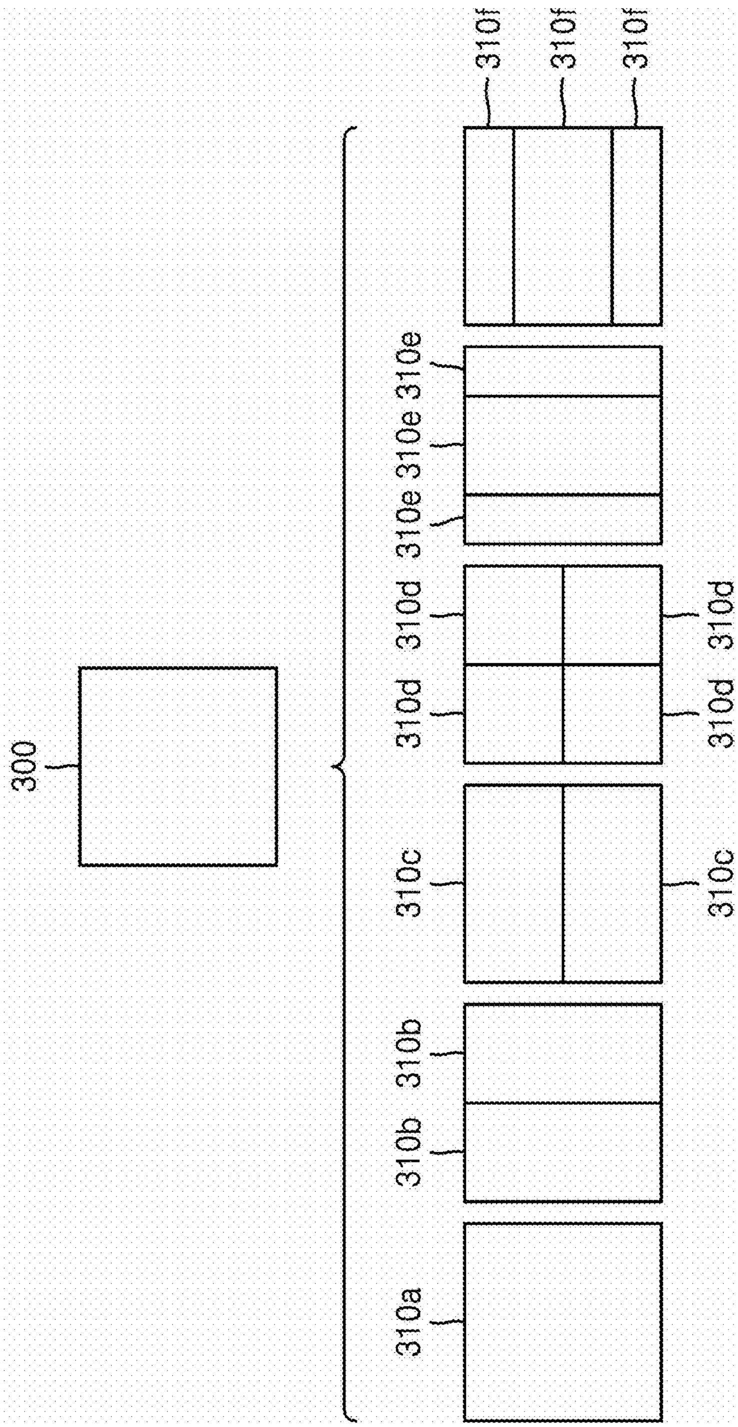

300
310a
310b
310b
310c
310c
310d
310d
310d
310d
310e
310e
310e
310f
310f
310f

FIG. 4

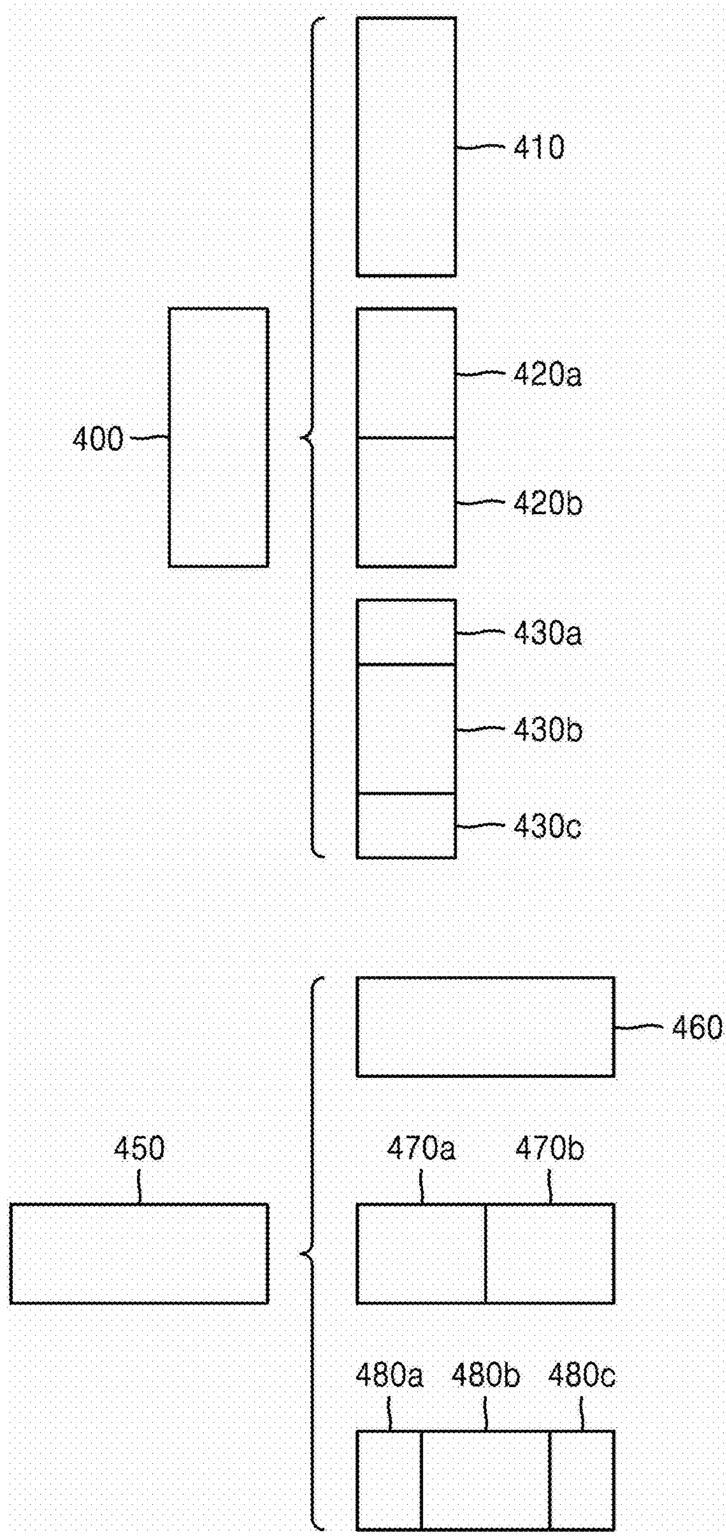

410
400
420a
420b
430a
430b
430c
460
450
470a
470b
480a
480b
480c

FIG. 5

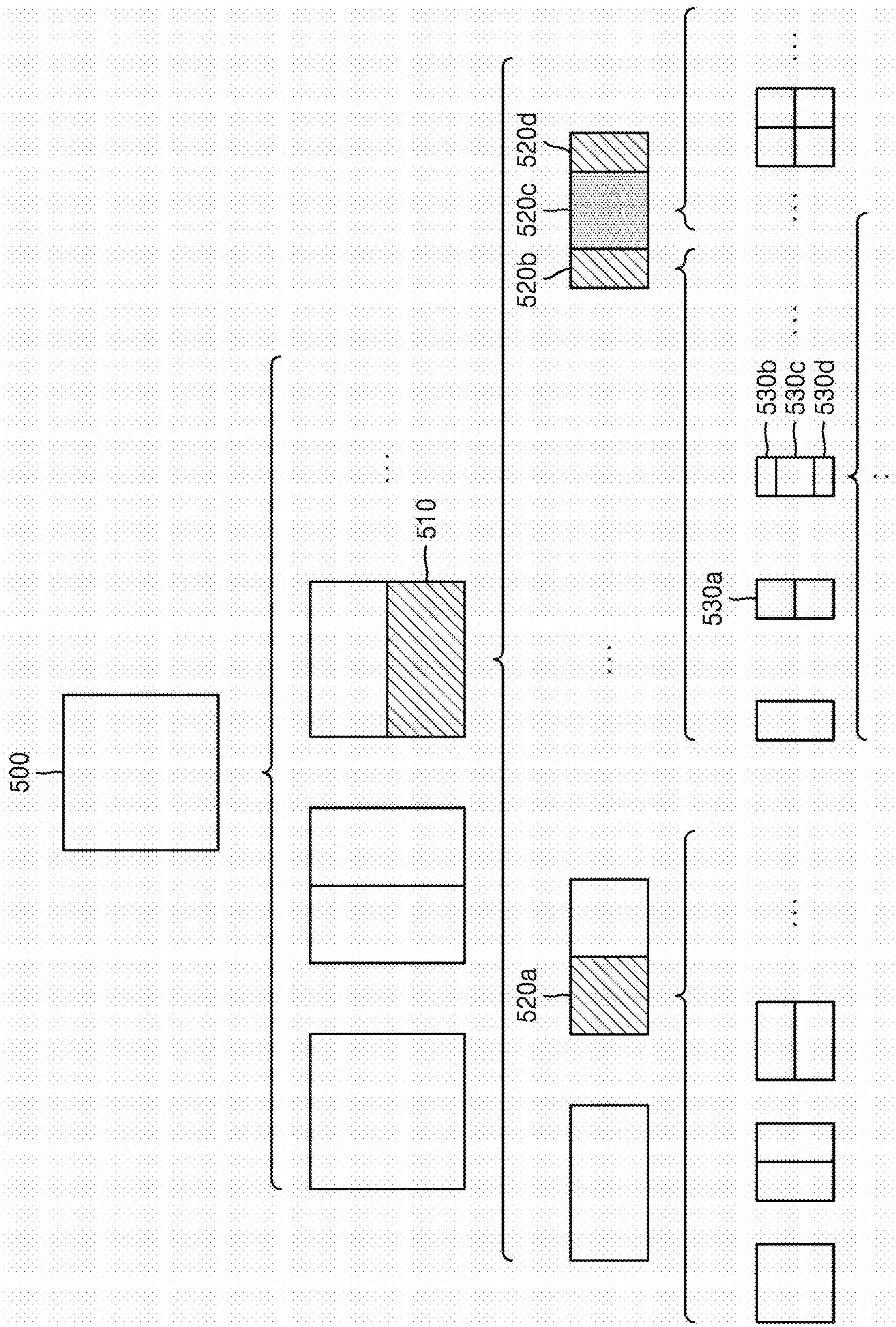


FIG. 6

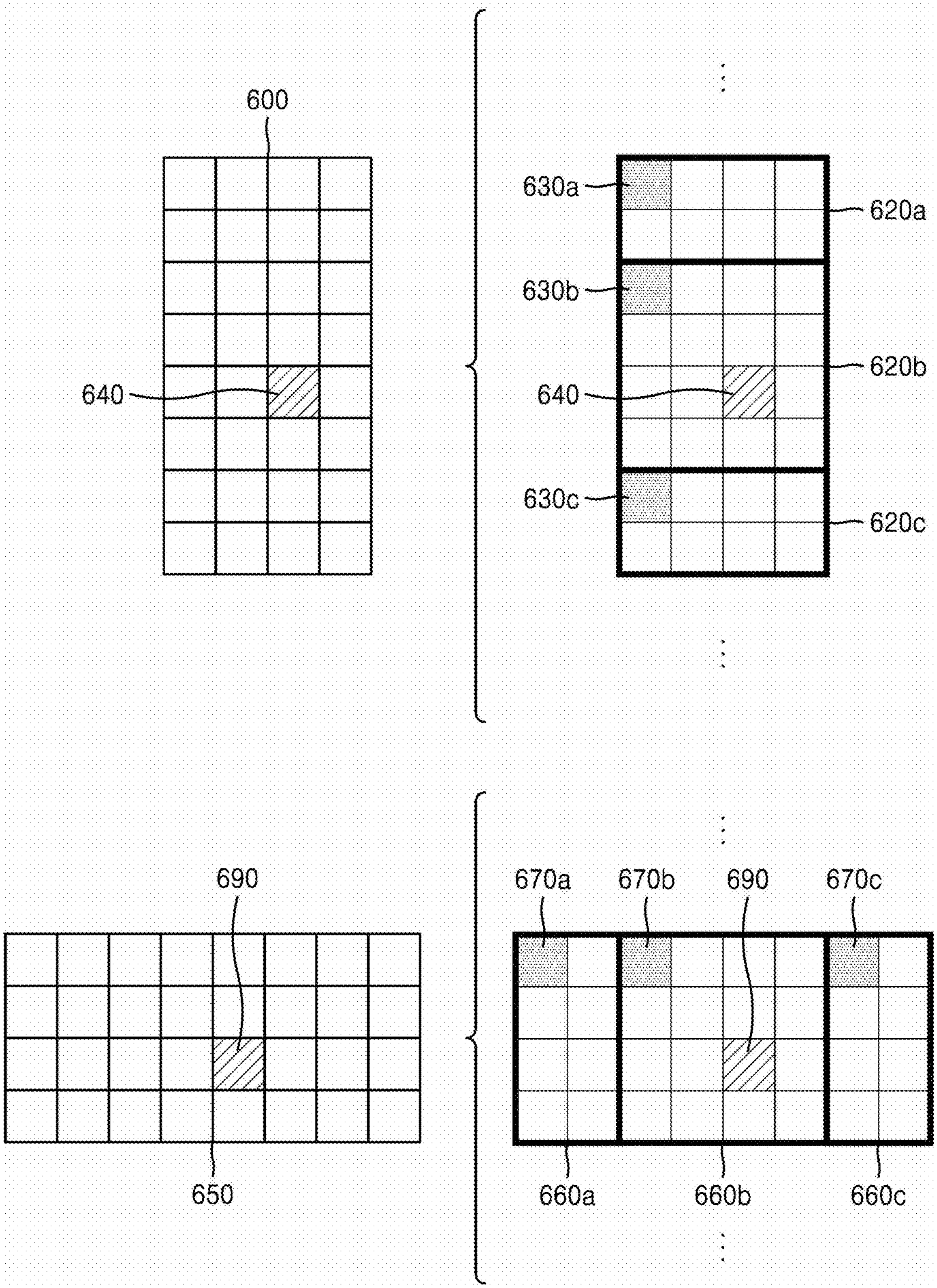
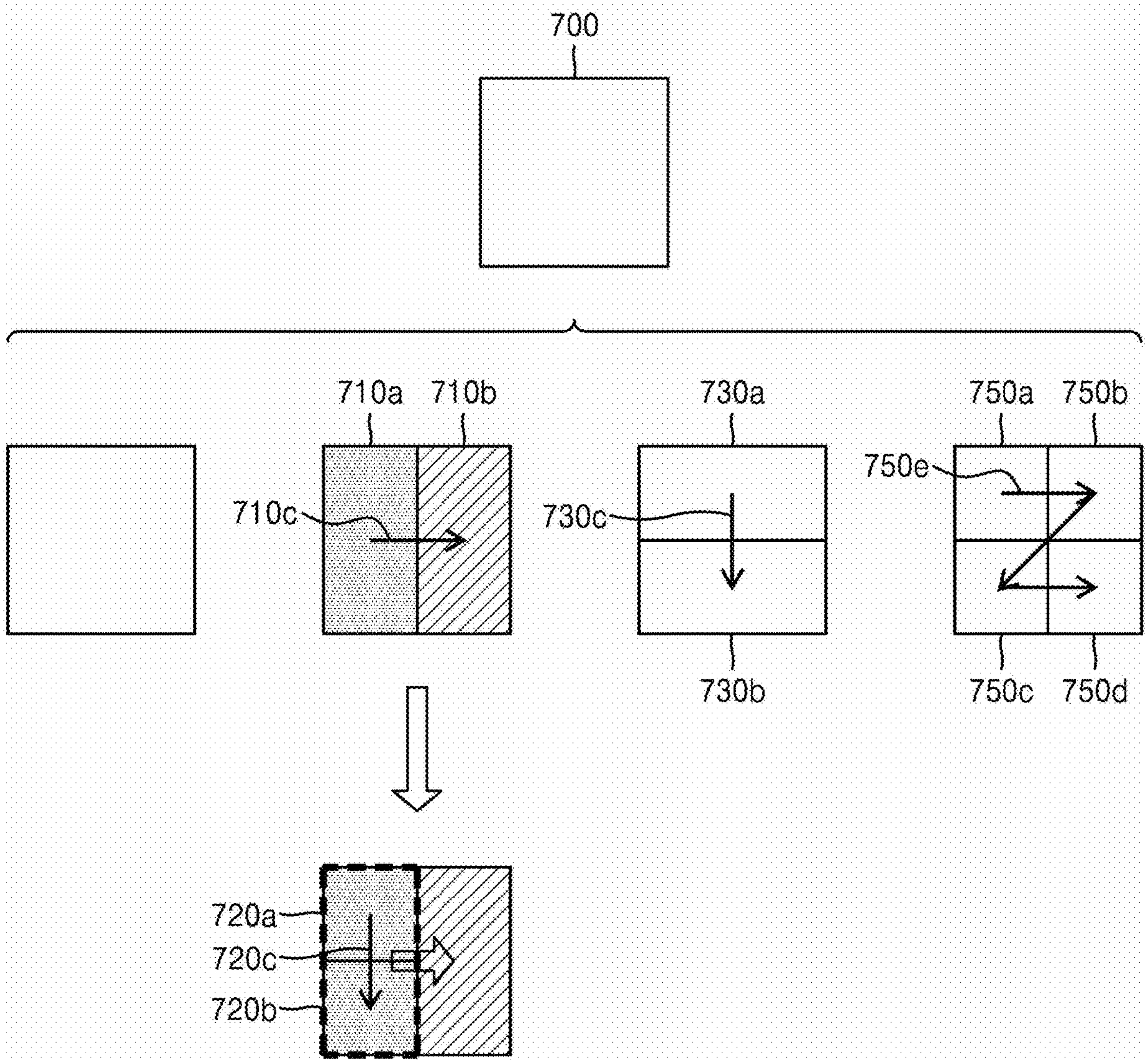

600
640
630a
620a
630b
620b
640
630c
620c
690
650
670a
670b
690
670c
660a
660b
660c

FIG. 7



700
710a
710b
710c
730a
730c
730b
750a
750b
750e
750c
750d
720a
720c
720b

FIG. 8

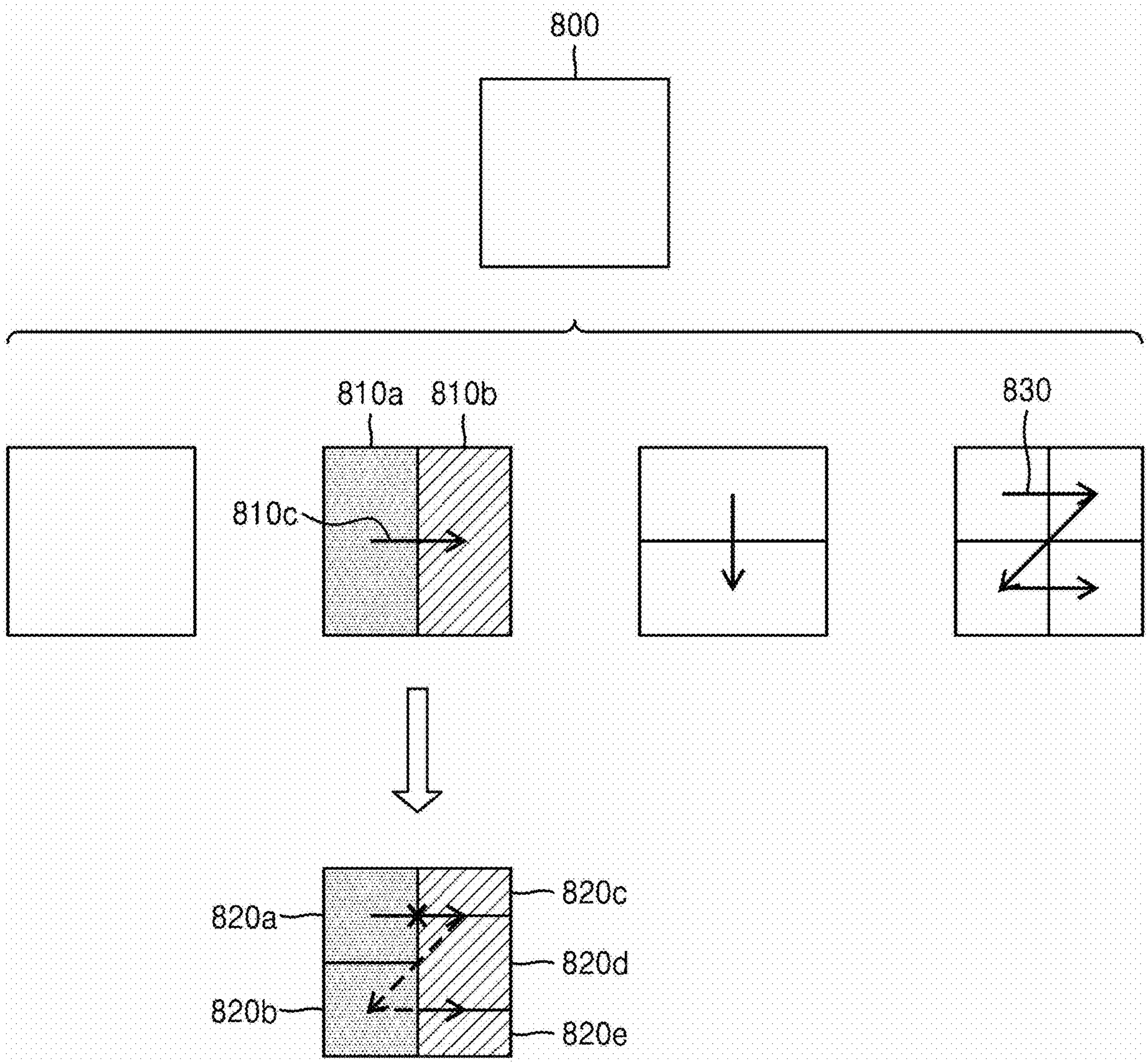


FIG. 9

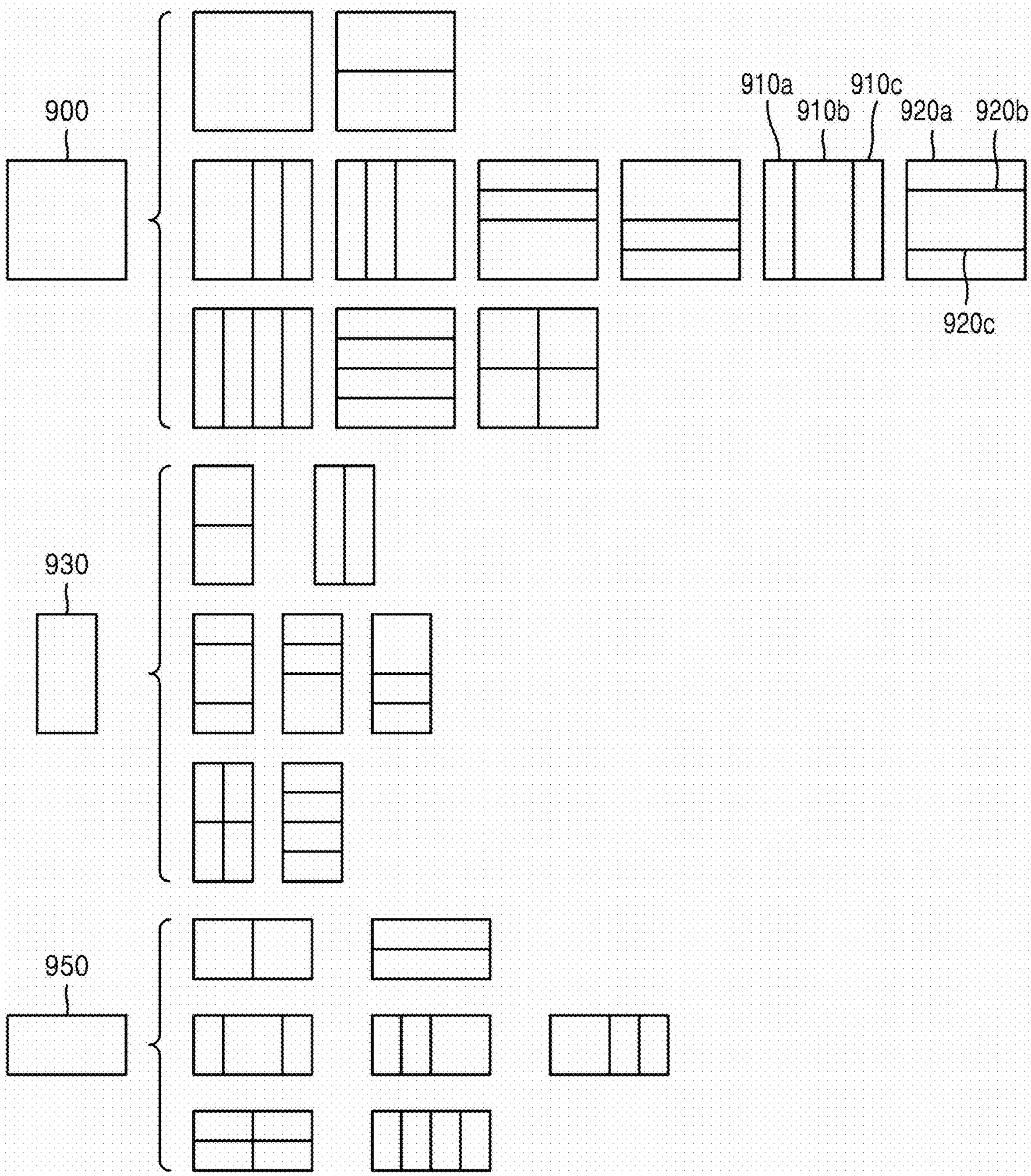


FIG. 10

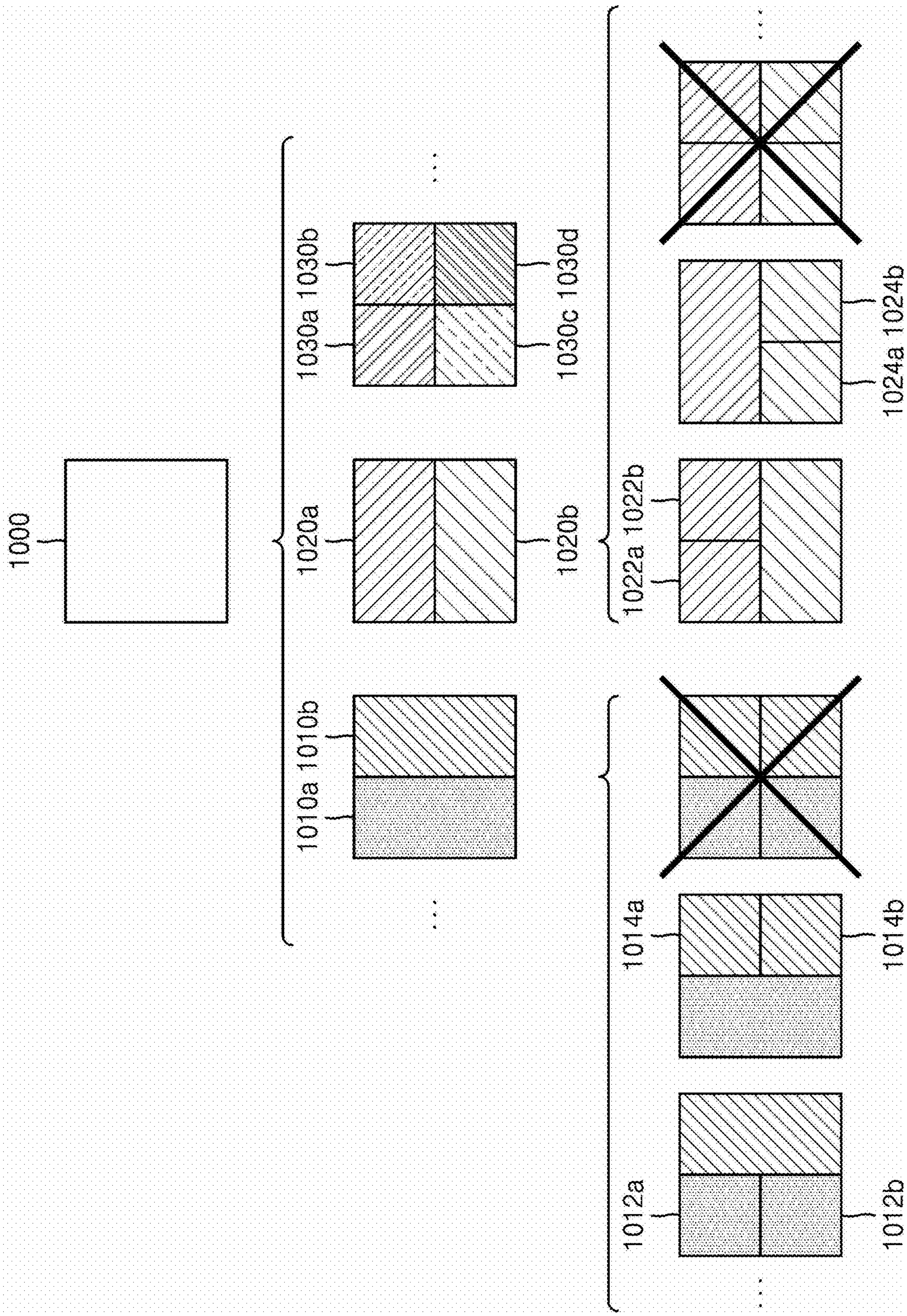

1000
1010a
1010b
1020a
1020b
1030a
1030b
1030c
1030d
1012a
1012b
1014a
1014b
1022a
1022b
1024a
1024b

FIG. 11

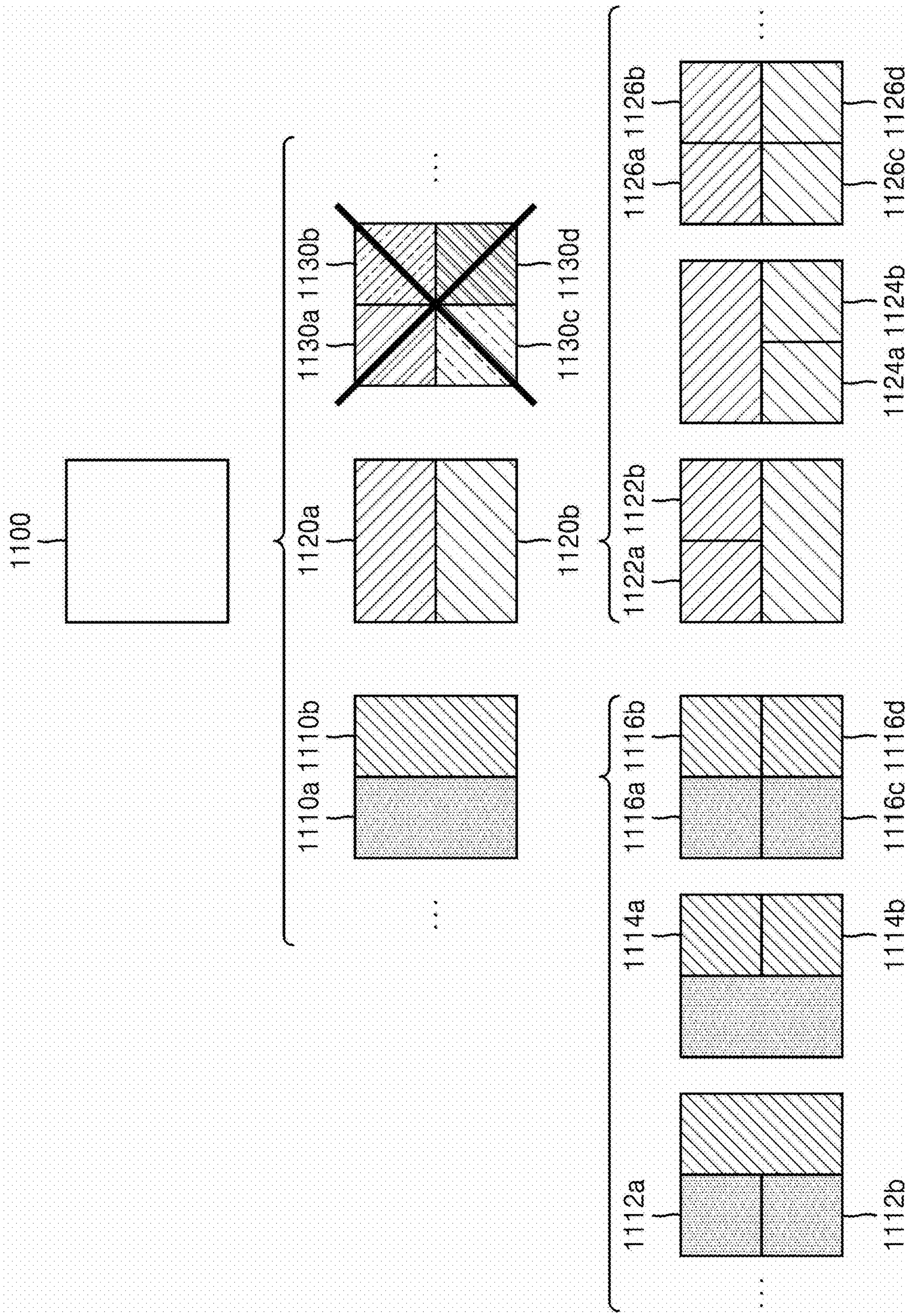

1100
1110a 1110b
1120a 1120b
1130a 1130b
1130c 1130d
1112a 1112b
1114a 1114b
1116a 1116b
1116c 1116d
1122a 1122b
1124a 1124b
1126a 1126b
1126c 1126d

FIG. 12

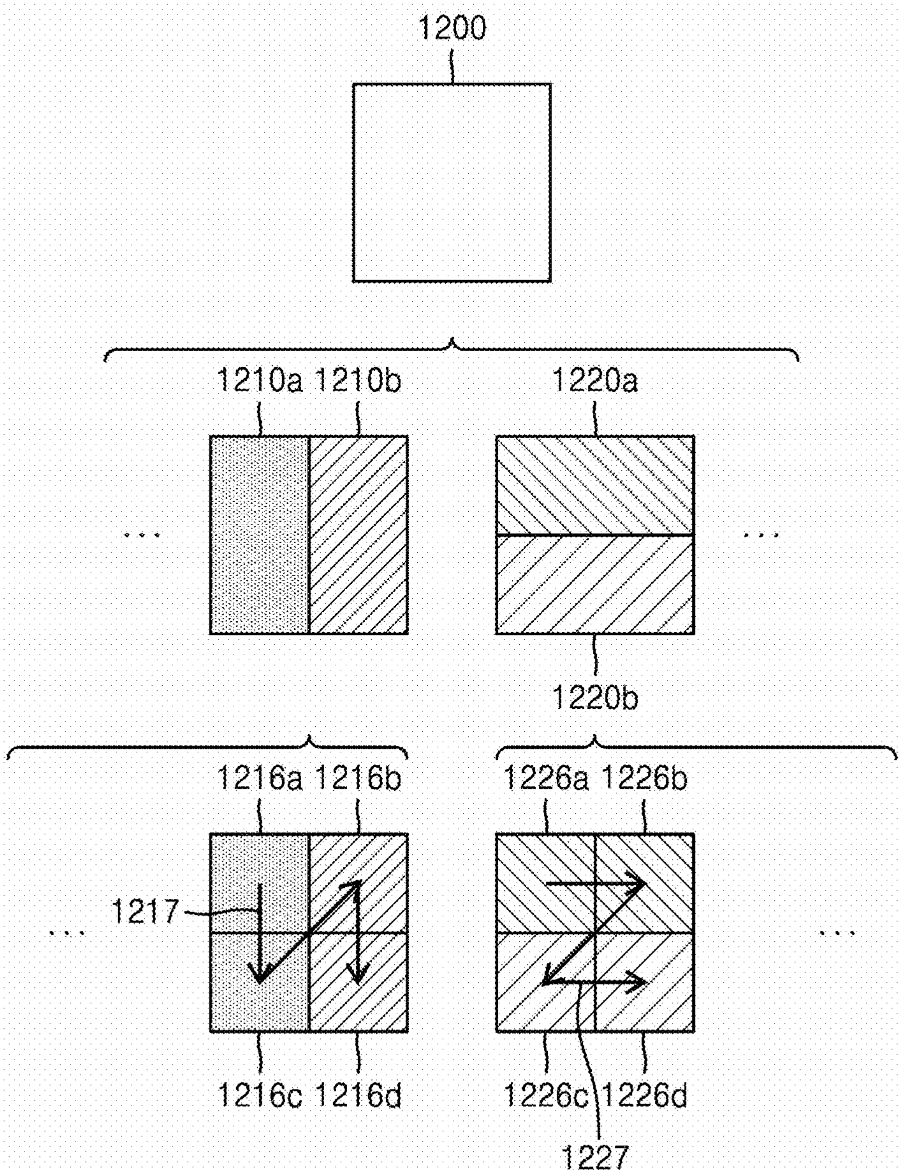

1200
1210a 1210b
1220a
1220b
1216a 1216b
1217
1216c 1216d
1226a 1226b
1226c 1226d
1227

FIG. 13

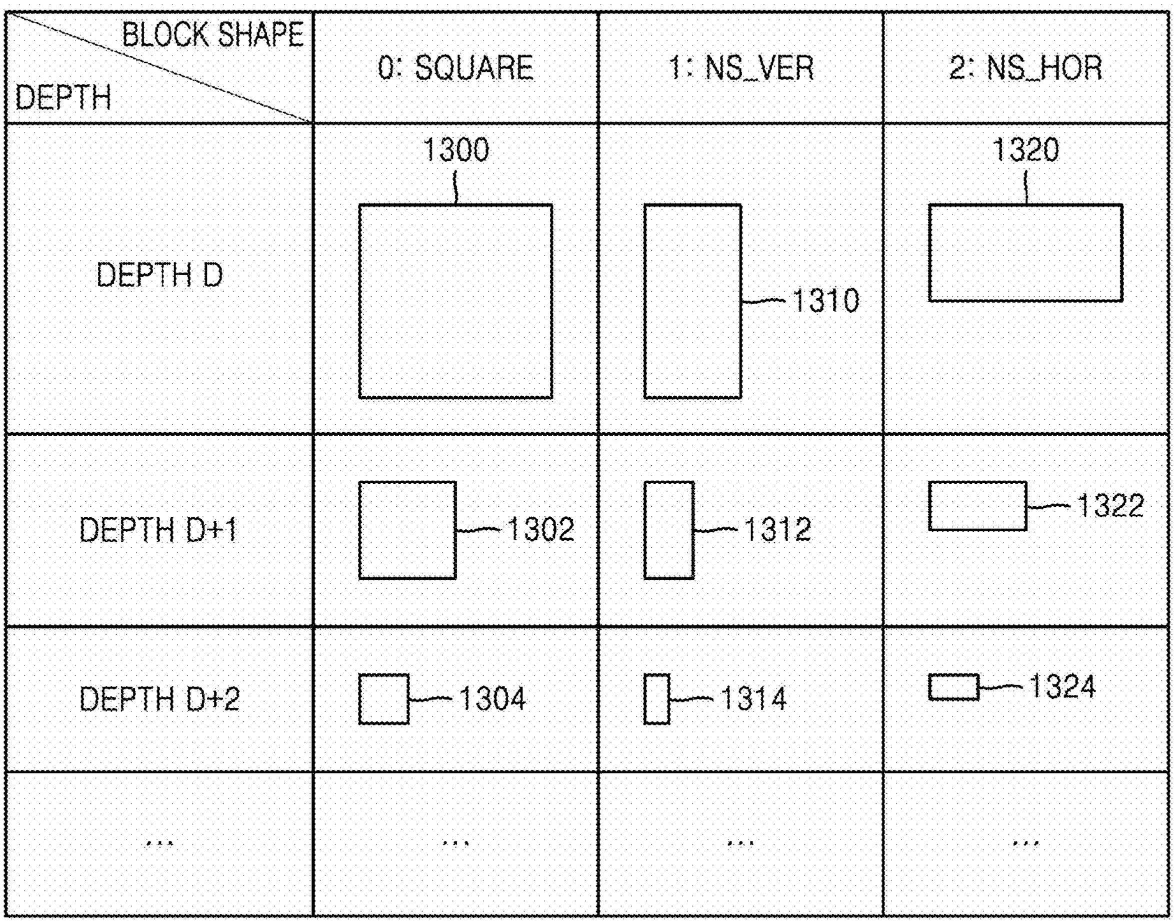

BLOCK SHAPE
DEPTH
0: SQUARE
1: NS_VER
2: NS_HOR
DEPTH D
1300
1310
1320
DEPTH D+1
1302
1312
1322
DEPTH D+2
1304
1314
1324
...
...
...
...

FIG. 14

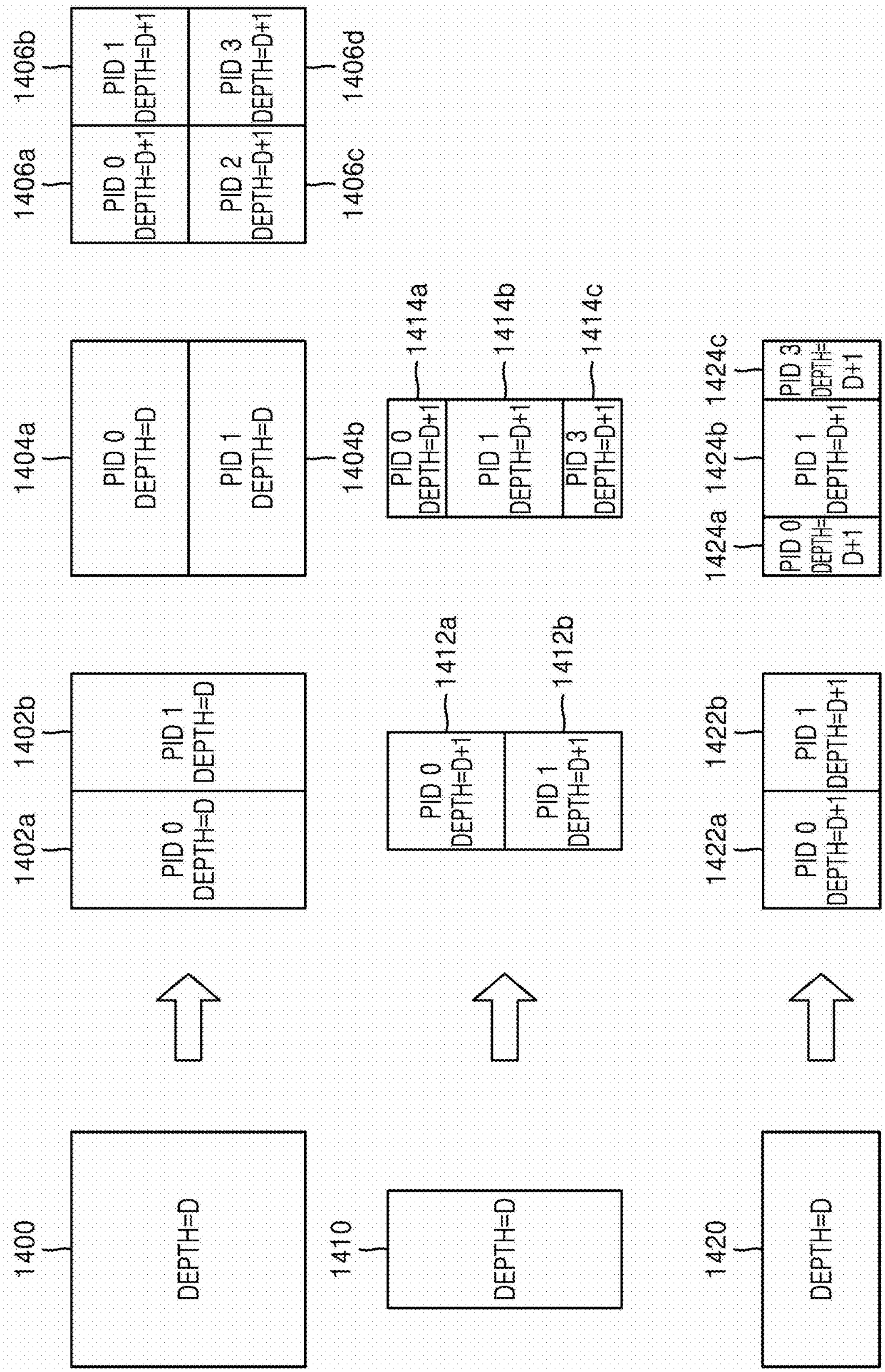

1400
DEPTH=D
1402a
PID 0
DEPTH=D
1402b
PID 1
DEPTH=D
1404a
PID 0
DEPTH=D
1404b
PID 1
DEPTH=D
1406a
PID 0
DEPTH=D+1
1406b
PID 1
DEPTH=D+1
1406c
PID 2
DEPTH=D+1
1406d
PID 3
DEPTH=D+1
1410
DEPTH=D
1412a
PID 0
DEPTH=D+1
1412b
PID 1
DEPTH=D+1
1414a
PID 0
DEPTH=D+1
1414b
PID 1
DEPTH=D+1
1414c
PID 3
DEPTH=D+1
1420
DEPTH=D
1422a
PID 0
DEPTH=D+1
1422b
PID 1
DEPTH=D+1
1424a
PID 0
DEPTH=
D+1
1424b
PID 1
DEPTH=D+1
1424c
PID 3
DEPTH=
D+1

FIG. 15

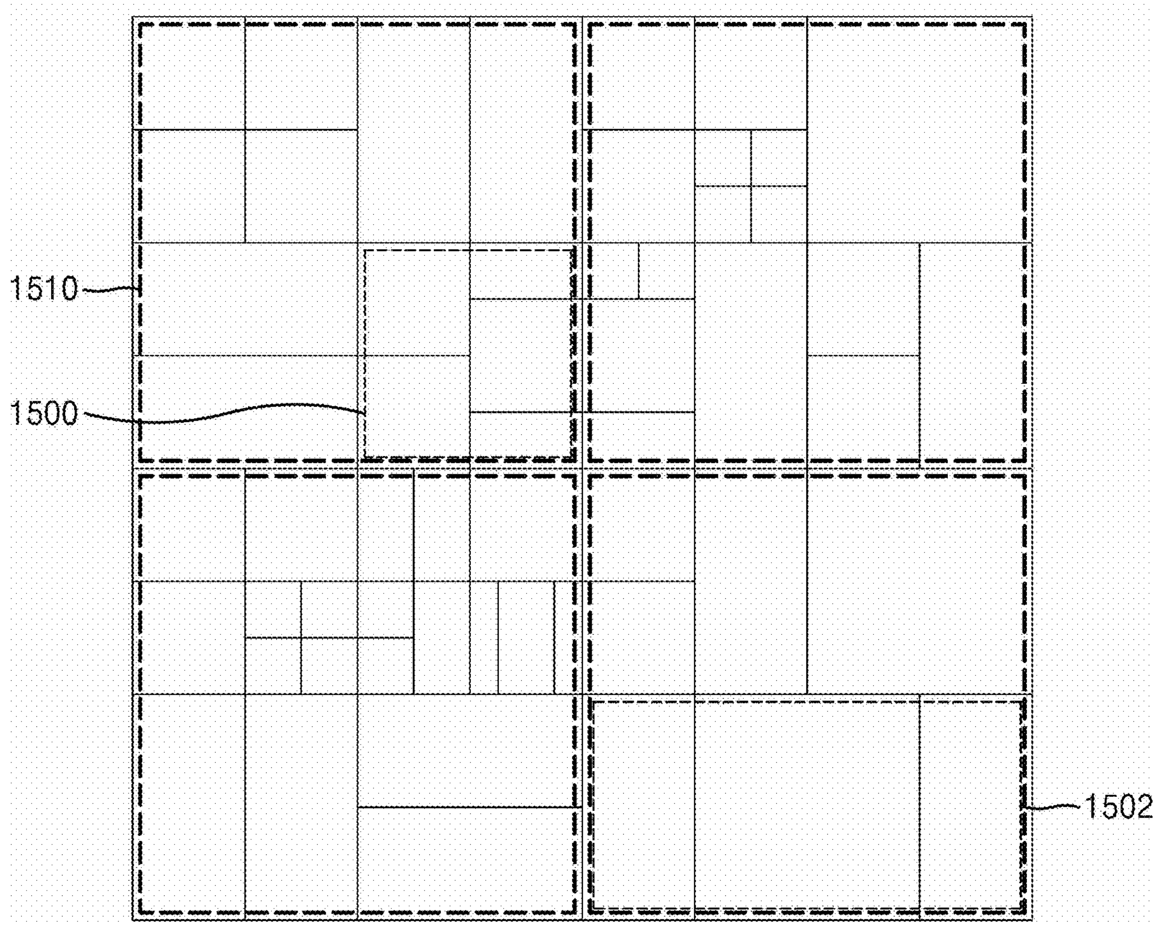


FIG. 16

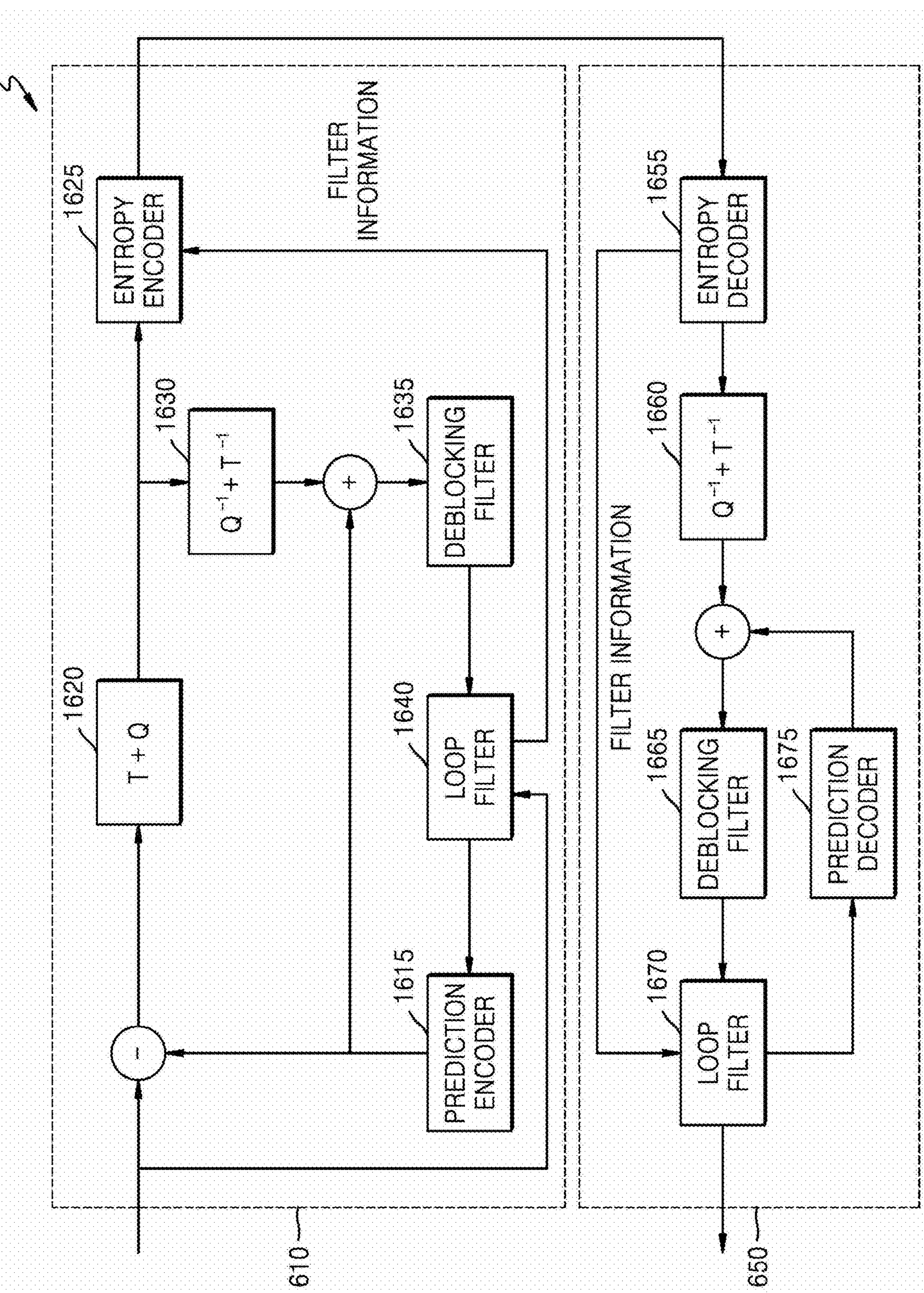

1600
1610
1615
PREDICTION ENCODER
1620
T + Q
1625
ENTROPY ENCODER
1630
$Q^{-1} + T^{-1}$
1635
DEBLOCKING FILTER
1640
LOOP FILTER
FILTER INFORMATION
1650
1655
ENTROPY DECODER
1660
$Q^{-1} + T^{-1}$
1665
DEBLOCKING FILTER
1670
LOOP FILTER
1675
PREDICTION DECODER
FILTER INFORMATION

FIG. 17

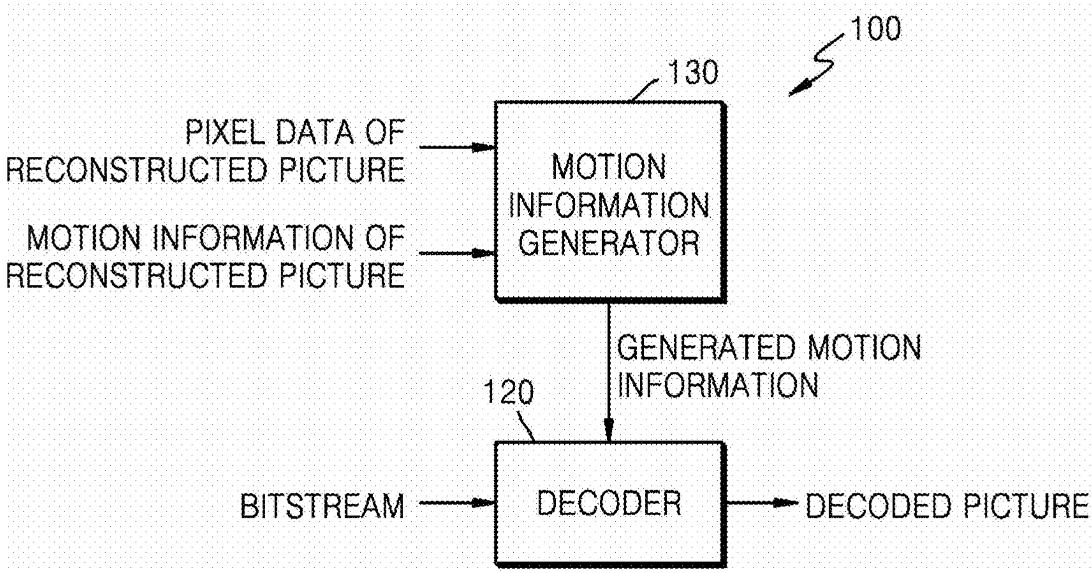

100
130
PIXEL DATA OF RECONSTRUCTED PICTURE
MOTION INFORMATION OF RECONSTRUCTED PICTURE
MOTION INFORMATION GENERATOR
GENERATED MOTION INFORMATION
120
BITSTREAM
DECODER
DECODED PICTURE

FIG. 18

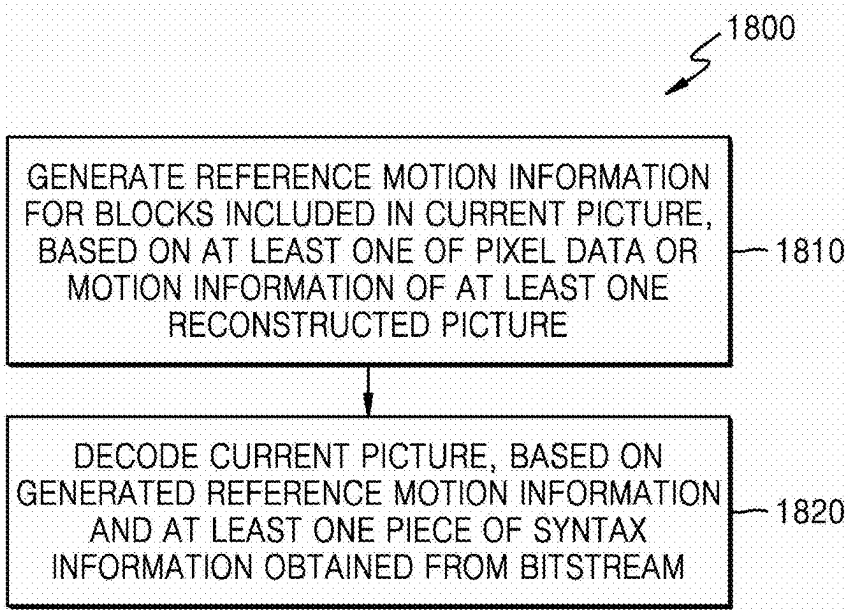

1800
GENERATE REFERENCE MOTION INFORMATION FOR BLOCKS INCLUDED IN CURRENT PICTURE, BASED ON AT LEAST ONE OF PIXEL DATA OR MOTION INFORMATION OF AT LEAST ONE RECONSTRUCTED PICTURE
1810
DECODE CURRENT PICTURE, BASED ON GENERATED REFERENCE MOTION INFORMATION AND AT LEAST ONE PIECE OF SYNTAX INFORMATION OBTAINED FROM BITSTREAM
1820

FIG. 19

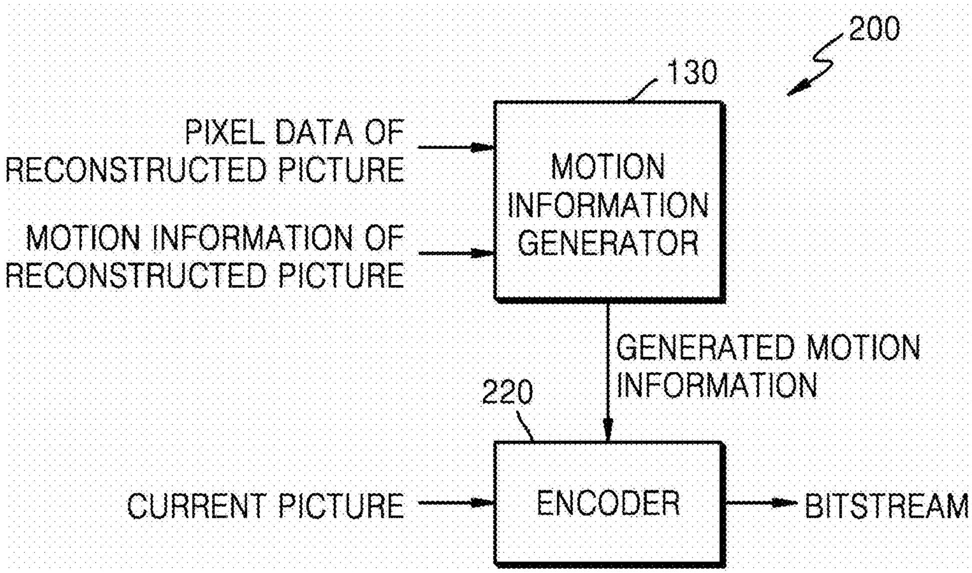

200
130
PIXEL DATA OF RECONSTRUCTED PICTURE
MOTION INFORMATION OF RECONSTRUCTED PICTURE
MOTION INFORMATION GENERATOR
GENERATED MOTION INFORMATION
220
CURRENT PICTURE
ENCODER
BITSTREAM

FIG. 20

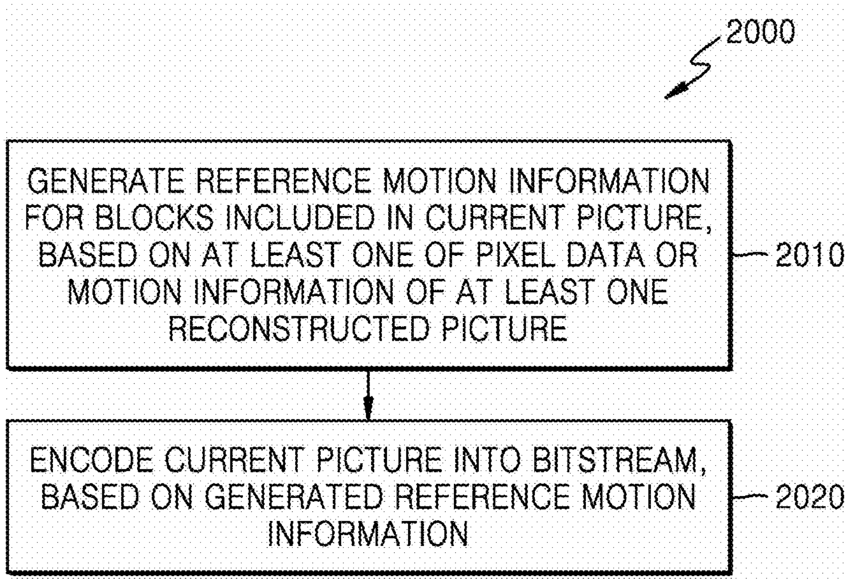

2000
GENERATE REFERENCE MOTION INFORMATION FOR BLOCKS INCLUDED IN CURRENT PICTURE, BASED ON AT LEAST ONE OF PIXEL DATA OR MOTION INFORMATION OF AT LEAST ONE RECONSTRUCTED PICTURE
2010
ENCODE CURRENT PICTURE INTO BITSTREAM, BASED ON GENERATED REFERENCE MOTION INFORMATION
2020

FIG. 21A

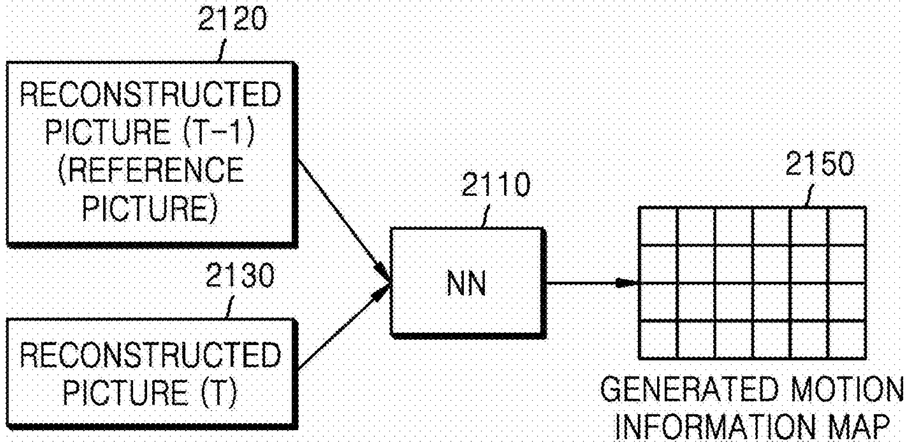

2120
RECONSTRUCTED PICTURE (T-1) (REFERENCE PICTURE)
2130
RECONSTRUCTED PICTURE (T)
2110
NN
2150
GENERATED MOTION INFORMATION MAP

FIG. 21B

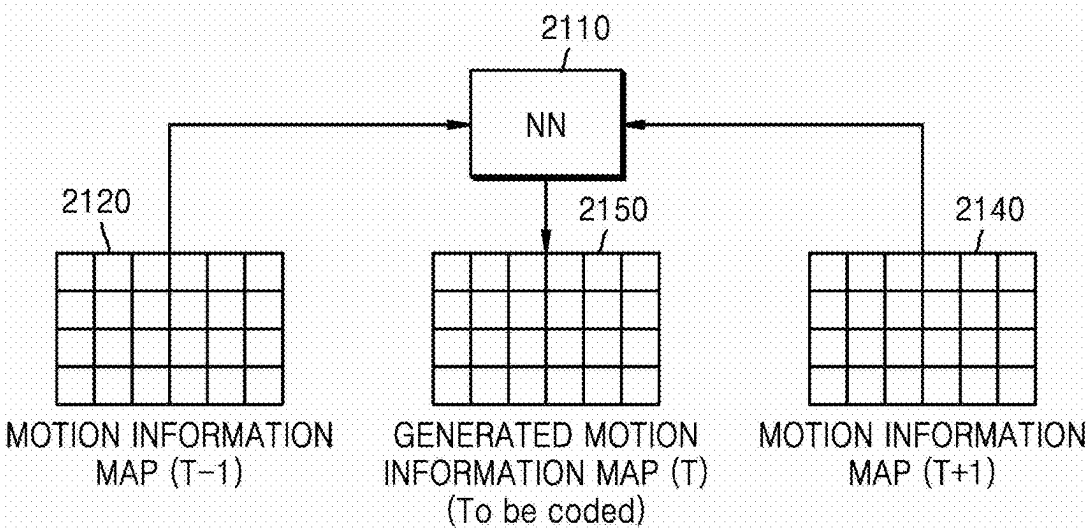

2110
NN
2120
2150
2140
MOTION INFORMATION MAP (T-1)
GENERATED MOTION INFORMATION MAP (T) (To be coded)
MOTION INFORMATION MAP (T+1)

FIG. 21C

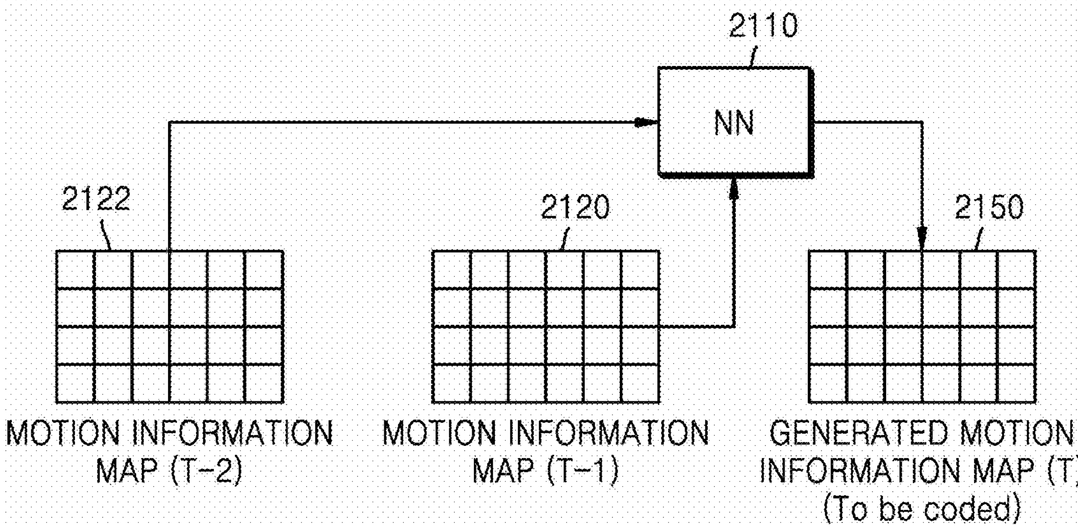

2110
NN
2122
2120
2150
MOTION INFORMATION MAP (T-2)
MOTION INFORMATION MAP (T-1)
GENERATED MOTION INFORMATION MAP (T)
(To be coded)

FIG. 22

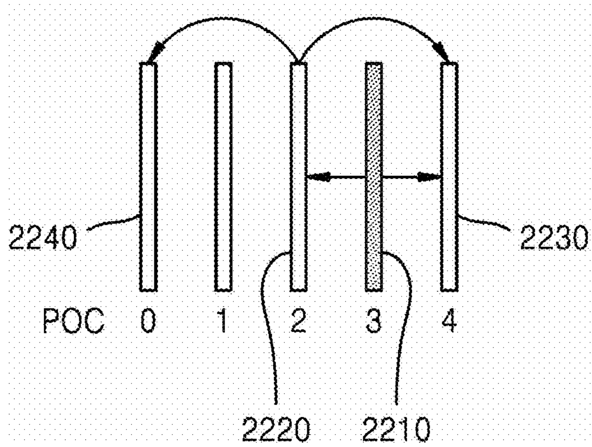


FIG. 23

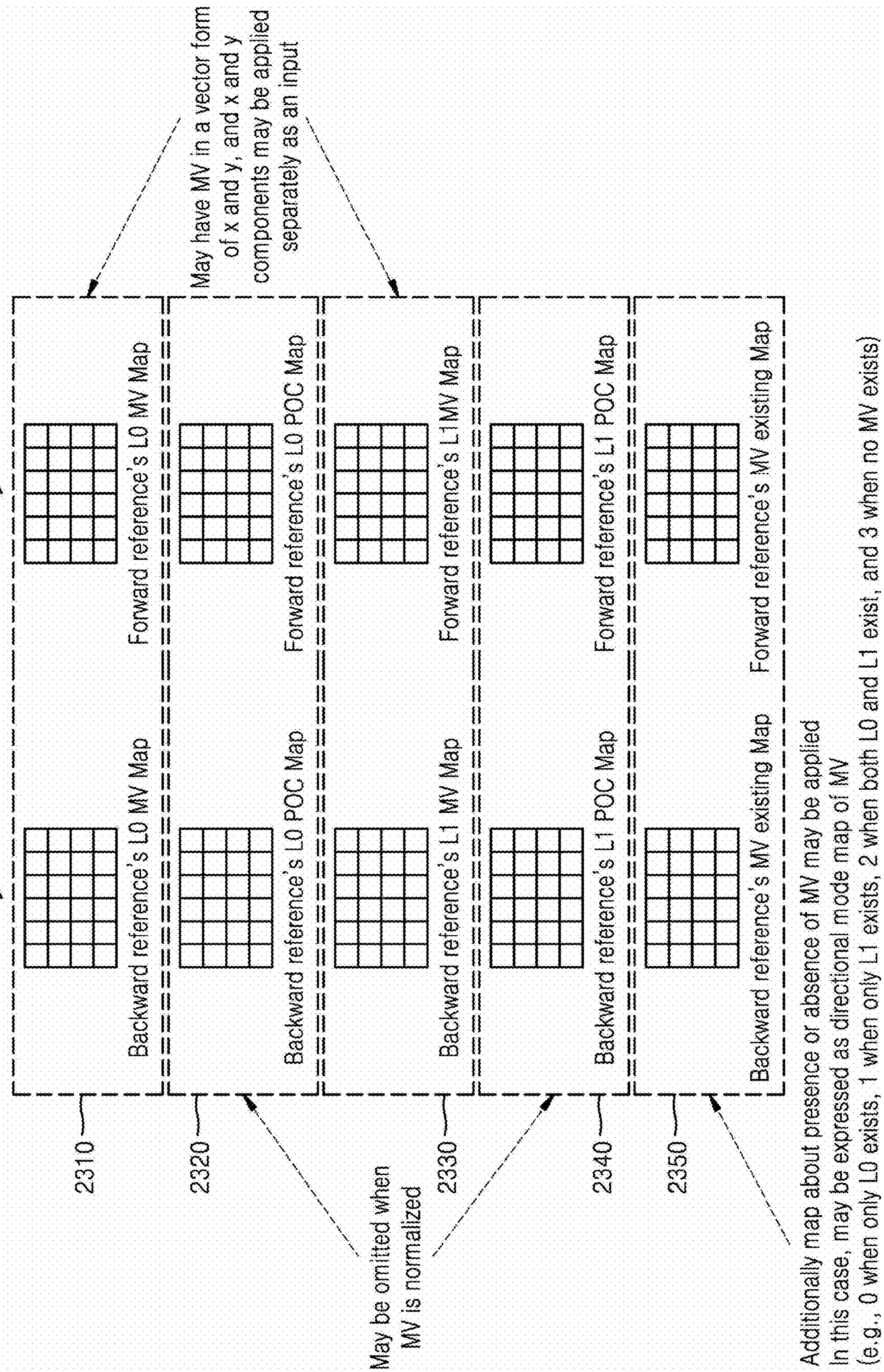

2120
2140
2310
2320
2330
2340
2350
Backward reference's L0 MV Map
Forward reference's L0 MV Map
Backward reference's L0 POC Map
Forward reference's L0 POC Map
Backward reference's L1 MV Map
Forward reference's L1MV Map
Backward reference's L1 POC Map
Forward reference's L1 POC Map
Backward reference's MV existing Map
Forward reference's MV existing Map
May have MV in a vector form of x and y, and x and y components may be applied separately as an input
May be omitted when MV is normalized
Additionally map about presence or absence of MV may be applied
In this case, may be expressed as directional mode map of MV
(e.g., 0 when only L0 exists, 1 when only L1 exists, 2 when both L0 and L1 exist, and 3 when no MV exists)

FIG. 24

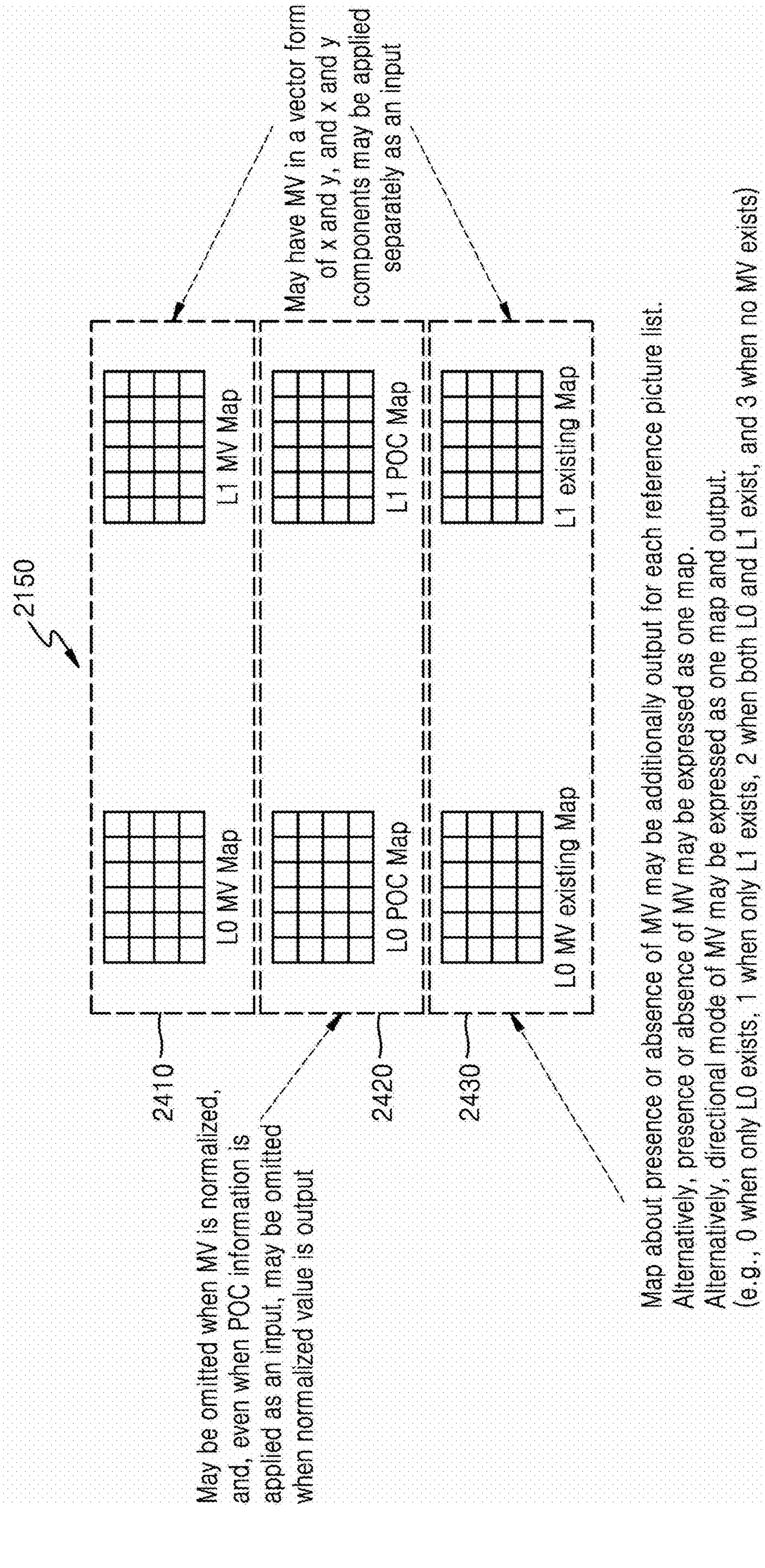

2150
2410
2420
2430
L0 MV Map
L1 MV Map
L0 POC Map
L1 POC Map
L0 MV existing Map
L1 existing Map
May have MV in a vector form of x and y, and x and y components may be applied separately as an input
May be omitted when MV is normalized, and, even when POC information is applied as an input, may be omitted when normalized value is output
Map about presence or absence of MV may be additionally output for each reference picture list.
Alternatively, presence or absence of MV may be expressed as one map.
Alternatively, directional mode of MV may be expressed as one map and output.
(e.g., 0 when only L0 exists, 1 when only L1 exists, 2 when both L0 and L1 exist, and 3 when no MV exists)

FIG. 25

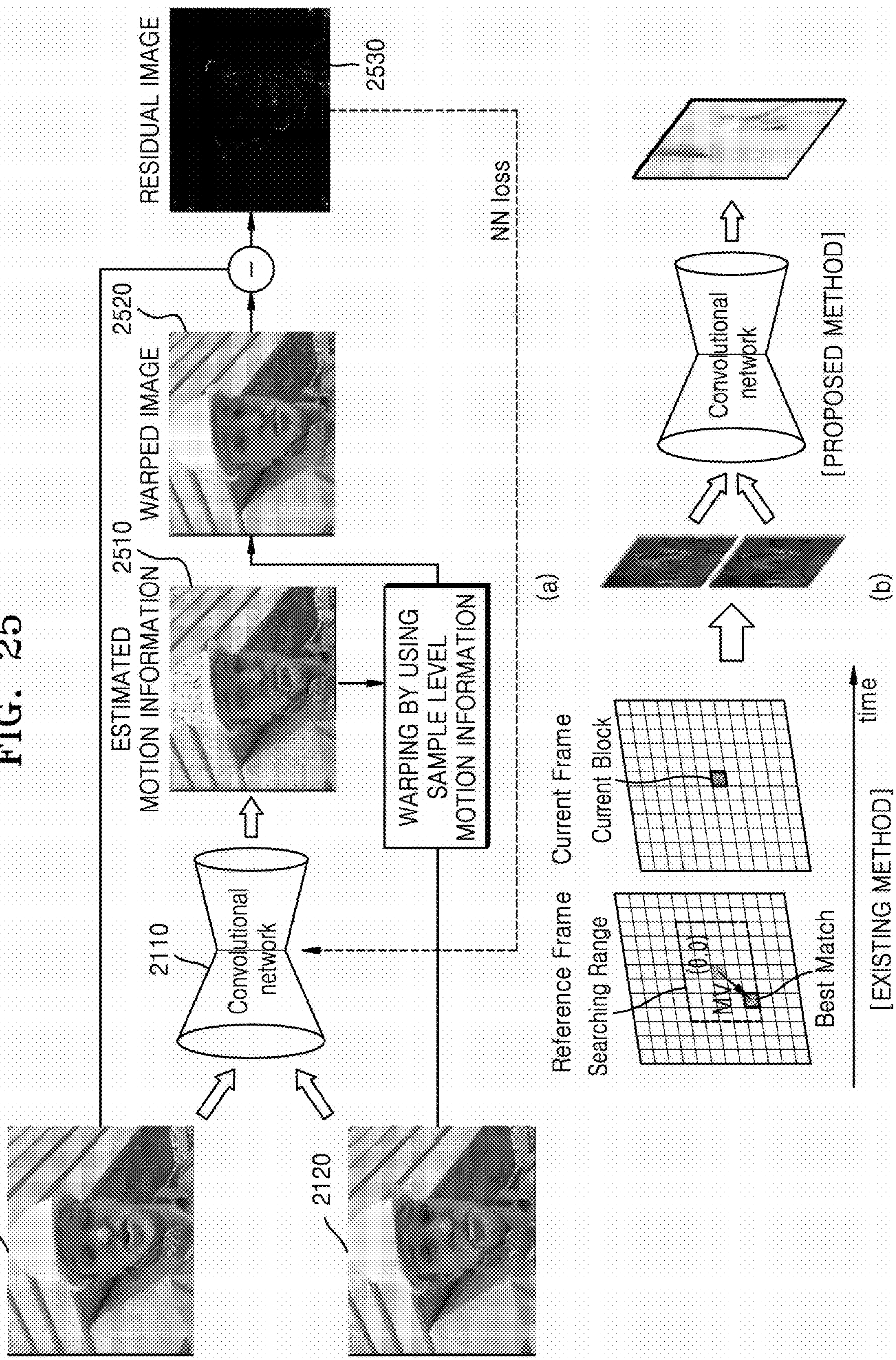

2130
2120
2110
Convolutional network
ESTIMATED MOTION INFORMATION
2510
WARPED IMAGE
2520
RESIDUAL IMAGE
2530
WARPING BY USING SAMPLE LEVEL MOTION INFORMATION
NN loss
(a)
Reference Frame
Current Frame
Searching Range
Current Block
(0,0)
MV
Best Match
time
[EXISTING METHOD]
Convolutional network
[PROPOSED METHOD]
(b)

FIG. 26

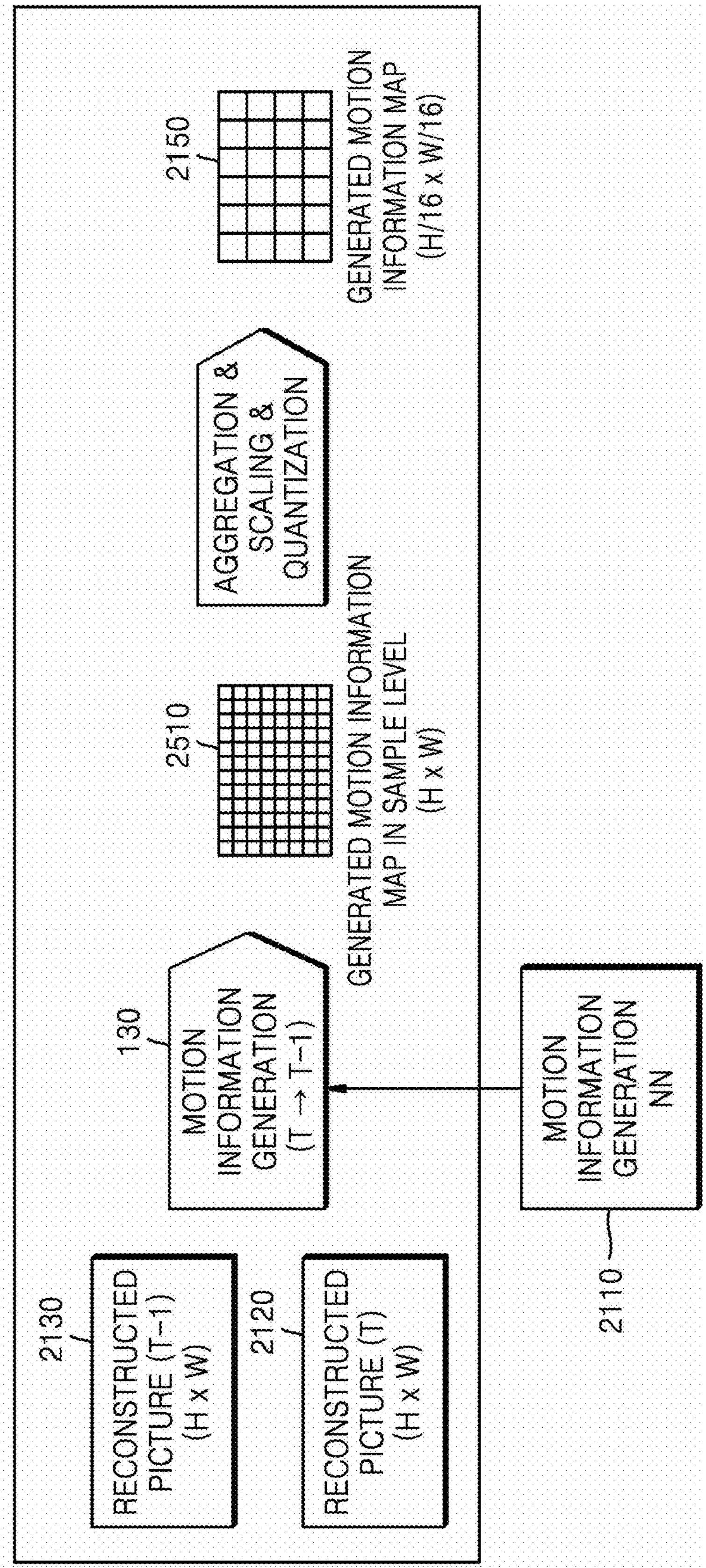

2130
RECONSTRUCTED PICTURE (T−1) (H x W)
2120
RECONSTRUCTED PICTURE (T) (H x W)
130
MOTION INFORMATION GENERATION (T → T−1)
MOTION INFORMATION GENERATION NN
2110
2510
GENERATED MOTION INFORMATION MAP IN SAMPLE LEVEL (H x W)
AGGREGATION & SCALING & QUANTIZATION
2150
GENERATED MOTION INFORMATION MAP (H/16 x W/16)

FIG. 27

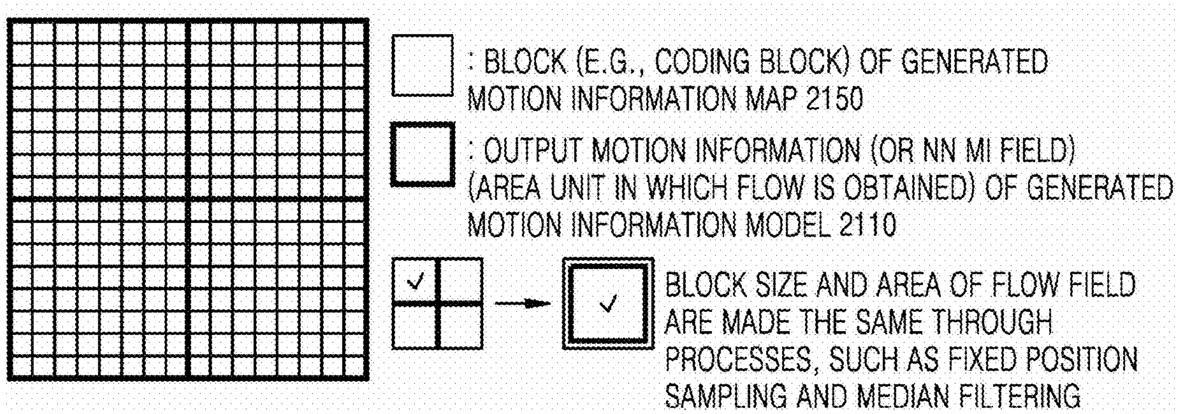

: BLOCK (E.G., CODING BLOCK) OF GENERATED MOTION INFORMATION MAP 2150
: OUTPUT MOTION INFORMATION (OR NN MI FIELD) (AREA UNIT IN WHICH FLOW IS OBTAINED) OF GENERATED MOTION INFORMATION MODEL 2110
BLOCK SIZE AND AREA OF FLOW FIELD ARE MADE THE SAME THROUGH PROCESSES, SUCH AS FIXED POSITION SAMPLING AND MEDIAN FILTERING

FIG. 28

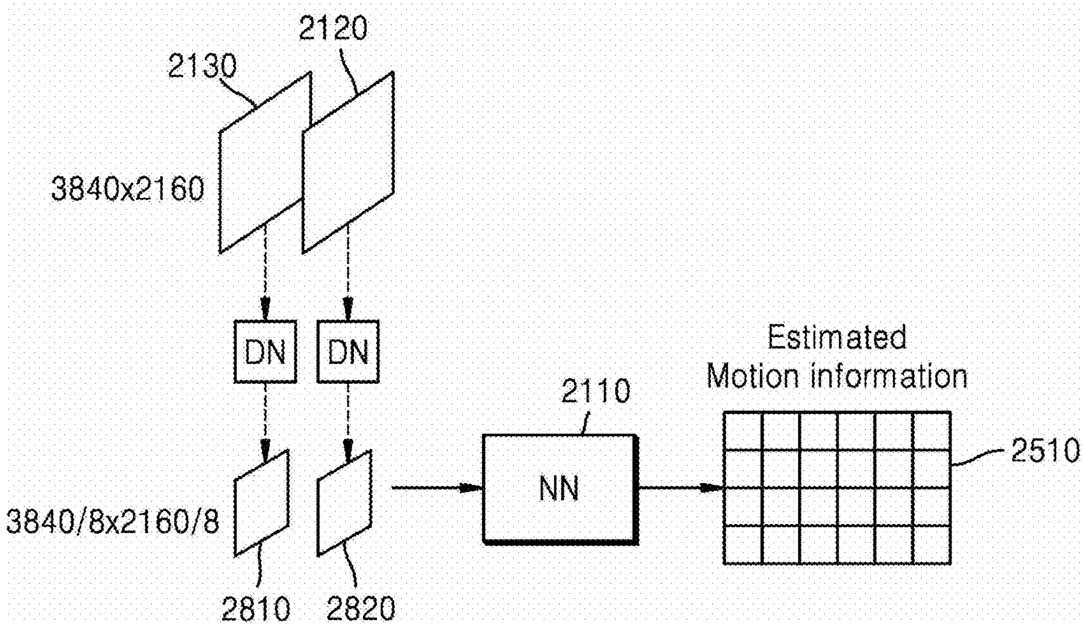

2120
2130
3840x2160
DN
DN
Estimated
Motion information
2110
NN
2510
3840/8x2160/8
2810
2820

FIG. 29

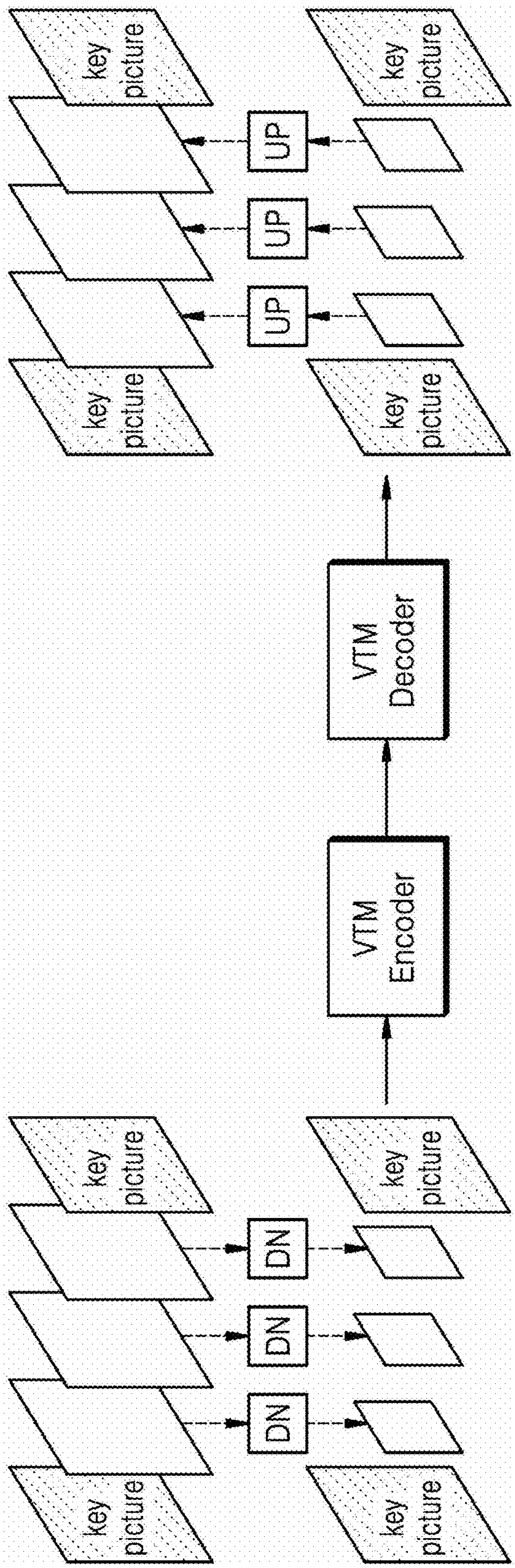

key picture
DN
DN
DN
key picture
key picture
key picture
VTM Encoder
VTM Decoder
key picture
UP
UP
UP
key picture
key picture
key picture

FIG. 30

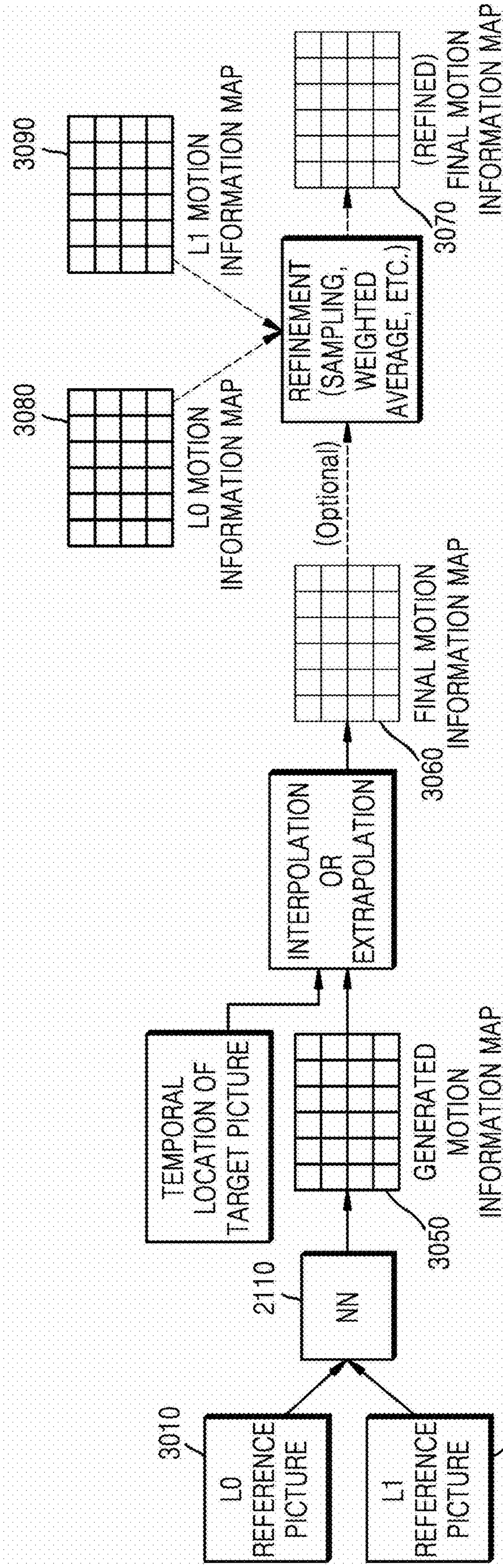

L0 REFERENCE PICTURE
3010
L1 REFERENCE PICTURE
3020
NN
2110
TEMPORAL LOCATION OF TARGET PICTURE
GENERATED MOTION INFORMATION MAP
3050
INTERPOLATION OR EXTRAPOLATION
FINAL MOTION INFORMATION MAP
3060
(Optional)
L0 MOTION INFORMATION MAP
3080
L1 MOTION INFORMATION MAP
3090
REFINEMENT (SAMPLING, WEIGHTED AVERAGE, ETC.)
(REFINED) FINAL MOTION INFORMATION MAP
3070

FIG. 31A

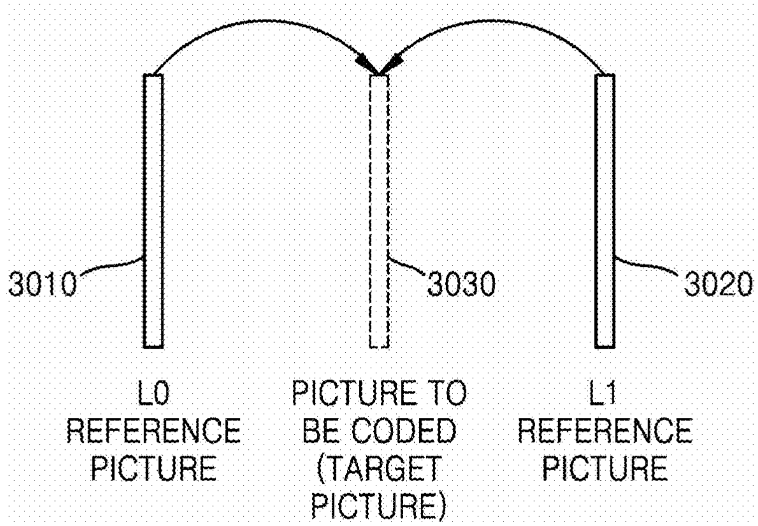

3010
3030
3020
L0 REFERENCE PICTURE
PICTURE TO BE CODED (TARGET PICTURE)
L1 REFERENCE PICTURE

FIG. 31B

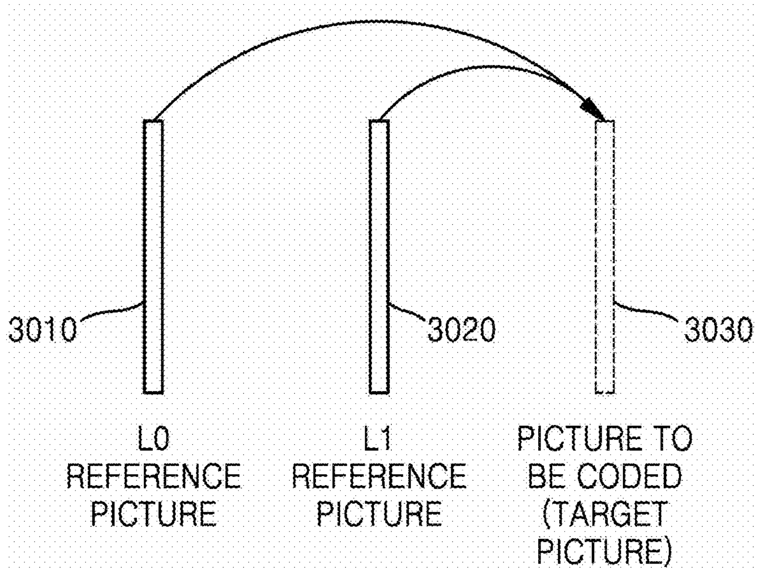

3010
3020
3030
L0 REFERENCE PICTURE
L1 REFERENCE PICTURE
PICTURE TO BE CODED (TARGET PICTURE)

FIG. 32A

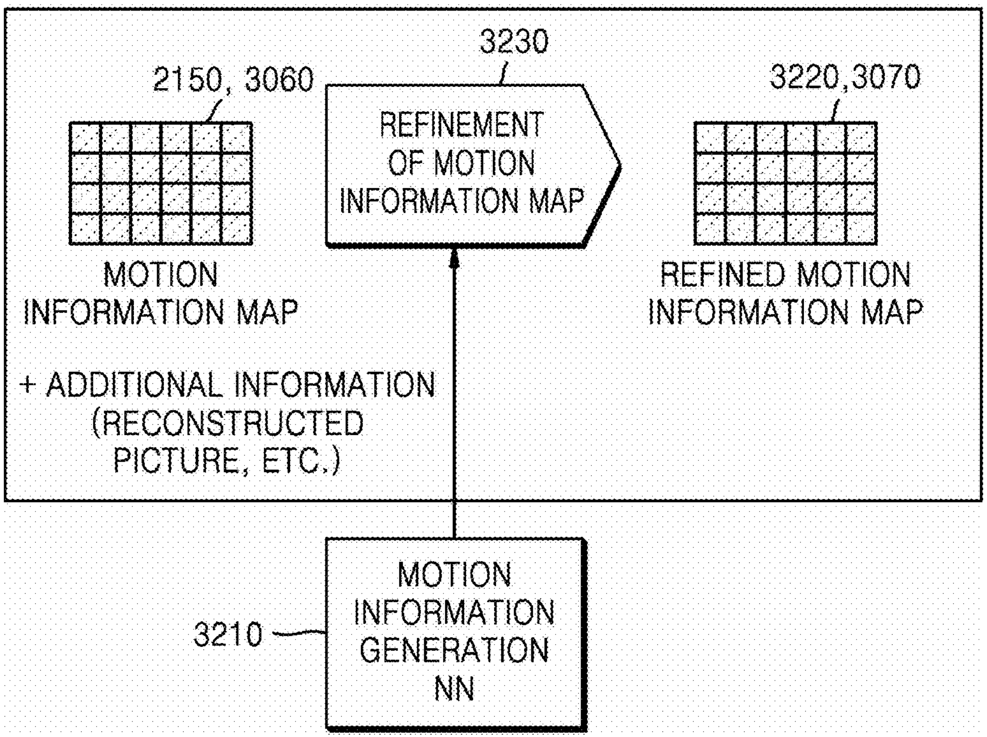

3230
2150, 3060
REFINEMENT OF MOTION INFORMATION MAP
3220,3070
MOTION INFORMATION MAP
REFINED MOTION INFORMATION MAP
+ ADDITIONAL INFORMATION (RECONSTRUCTED PICTURE, ETC.)
3210
MOTION INFORMATION GENERATION NN

FIG. 32B

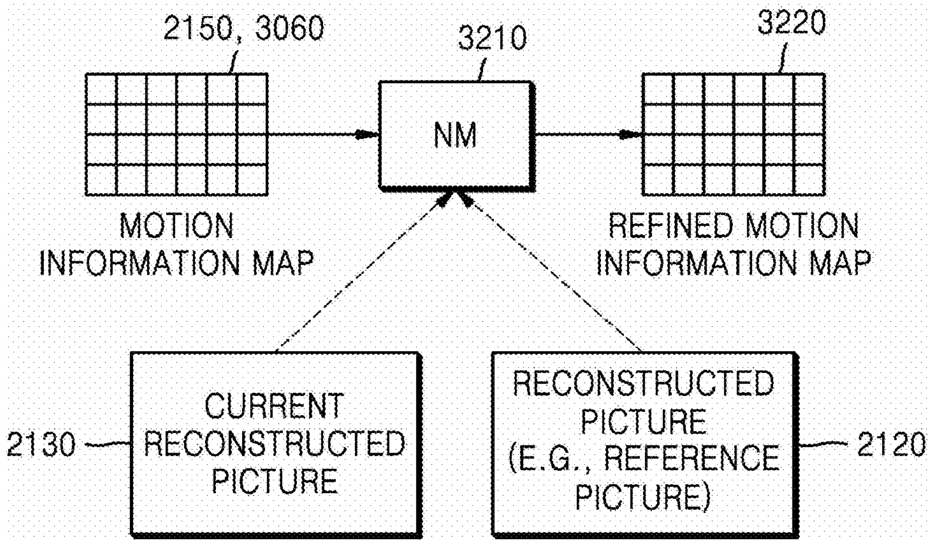

2150, 3060
3210
3220
NM
MOTION
INFORMATION MAP
REFINED MOTION
INFORMATION MAP
2130
CURRENT
RECONSTRUCTED
PICTURE
RECONSTRUCTED
PICTURE
(E.G., REFERENCE
PICTURE)
2120

FIG. 33

```
       CU (coding unit) syntax structure () {
3310     use_generated_mv_flag
         if(use_generated_mv_flag == 0) {
3320       // use of mode-related syntax of existing video coding standard
         }
         else { // use_generated_flag == 1
           if (it is determined in derived map that MV of current block does not exist) {
3330         // use intra syntax of existing video coding standard
           }
           else { // when MV is derived in advance
3340         MV information setting of current block // there is no additional data transmission
             // use of MV information corresponding to current block within MV information map
           }
         }
3350     // use of syntax related to existing residual coding
       }
```

FIG. 34A

```
CU (coding unit) syntax structure () {
   use_generated_mv_flag
   if(use_generated_mv_flag == 0) {
      // use of mode-related syntax of existing video coding standard
   }
   else { // use_generated_flag == 1
      if (it is determined in derived map that MV of current block does not exist) {
         // use intra syntax of existing video coding standard
      }
      else { // when MV is derived in advance
3410     mvd_flag // transmitted when transmission of MV difference is additionally needed, MV derived from NN is used as MV predictor (MVP)
3420     if(mvd_flag) information of MV difference is additionally transmitted, and transmission or non-transmission of L0 and L1 MVDs may be determined according to directional information
3430     else MV information setting of current block // there is no additional data transmission
      }
   }
   // use of syntax related to existing residual coding
}
```

FIG. 34B

```
CU (coding unit) syntax structure () {
   use_generated_mv_flag
   if(use_generated_mv_flag == 0) {
      // use of mode-related syntax of existing video coding standard
   }
   else { // use_generated_flag == 1
      if (it is determined in derived map that MV of current block does not exist) {
         // use intra syntax of existing video coding standard
      }
      else { // when MV is derived in advance
3440     mv_index // candicate may be generated so that one of MVs of current block location and neighboring block may be selected from generated MV map
3450     MV information setting of current block selected by mv_index // there is no additional data transmission
      }
   }
   // use of syntax related to existing residual coding
}
```

FIG. 35

```
CU (coding unit) syntax structure () {
    use_generated_mv_flag
    if(use_generated_mv_flag == 0) {
        // use of mode-related syntax of existing video coding standard
    }
    else { // use_generated_flag == 1
        if (it is determined in derived map that MV of current block does not exist) {
            // use intra syntax of existing video coding standard
        }
        else { // when MV is derived in advance
3510        no_mvd_flag
            if(no_mvd_flag) {
3440            mv_index // candicate may be generated so that one of MVs of current block location and neighboring block may be selected from generated MV map
3450            MV information setting of current block selected by mv_index // there is no additional data transmission
            }
            else {
3420            mvd transmission // information of MV difference is transmitted, and transmission or non-transmission of L0 and L1 MVDs may be determined according to directional information
                mvp_index // candicate may be generated so that one of MVs of current block location and neighboring block may be selected as MVP from generated MV map
            }
        }
    }
    // use of syntax related to existing residual coding
}
```

FIG. 36

```
CU (coding unit) syntax structure () {
    use_generated_mv_flag
    if(use_generated_mv_flag == 0) {
        // use of mode-related syntax of existing video coding standard
    }
    else { // use_generated_flag == 1
        // in this case, it is determined that intra is not selected (intra is processed when use_generated_mv_flag == 0)
        // coding may be performed using pre-derived MV. MV information setting of current block
        MV information setting of current block // there is no additional data transmission
        // in this case, when there is no MV corresponding to current block, another MV may be searched for and selected
    }
    // use of syntax related to existing residual coding
}
```

3610

# FIG. 37A

```
     CU (coding unit) syntax structure () {
3710   if(in case of inter picture) // transmission is allowed only in case of inter picture
3720     use_generated_mv_flag // may operate instead of existing skip mode
       if(use_generated_mv_flag == 0) {
3730     // use mode-related syntax of video coding standard except for existing skip mode
         // use of syntax related to existing residual coding
       }
       else { // use_generated_flag == 1
         // in this case, it is determined that intra is not selected (intra is processed when use_generated_mv_flag = 0)
3740     // coding may be performed using pre-derived MV
         MV information setting of current block // there is no additional data transmission
         // in this case, when there is no MV corresponding to current block, another MV may be searched for and selected
       }
     }
```

# FIG. 37B

```
CU (coding unit) syntax structure () {
   if(in case of inter picture) // transmission is allowed only in case of inter picture
     use_generated_mv_flag // may operate instead of existing skip mode
   if(use_generated_mv_flag == 0) {
     // use mode-related syntax of video coding standard except for existing skip mode
     // use of syntax related to existing residual coding
   }
   else { // use_generated_flag == 1
     // in this case, it is determined that intra is not selected (intra is processed when use_generated_mv_flag = 0)
     mv_index // candicate may be generated so that one of MVs of current block location and neighboring block may be selected from generated MV map
     MV information setting of current block selected by mv_index // there is no additional data transmission
   }
}
```

3750

3760

# FIG. 38

```
     CU (coding unit) syntax structure () {
3810   if(in case of inter picture) // transmission is allowed only in case of inter picture
         skip_flag // flag indicating whether in skip mode of existing codec
       if(skip_flag == 0) {
3820     cu_mode // determine whether inter or intra CU
         merge_flag // flag indicating whether in merge mode of existing codec
         if(merge_flag == 1) {
3830       use_generated_mv_flag
           if(use_generated_my_flag == 0)
3840         merge_data() // merge data (merge_index, etc.) of existing codec
           else
             // generated MV information corresponding to current block is used as MV information of current block
3850         // one of several candidates may be selected through the mv_index information
         }
         else {
3860       // use of existing advanced MVP (AMVP) syntax
         }
3870     // use of syntax related to existing residual coding
       }
       else { // skip_flag == 1
3830     use_generated_mv_flag
         if(use_generated_my_flag == 0)
3840       merge_data()
         else
           // generated MV information corresponding to current block is used as MV information of current block
3850       // one of several candidates may be selected through the mv_index information
       }
     }
```

FIG. 39A

5. The merging candidate list, mergeCandList, is constructed as follows:

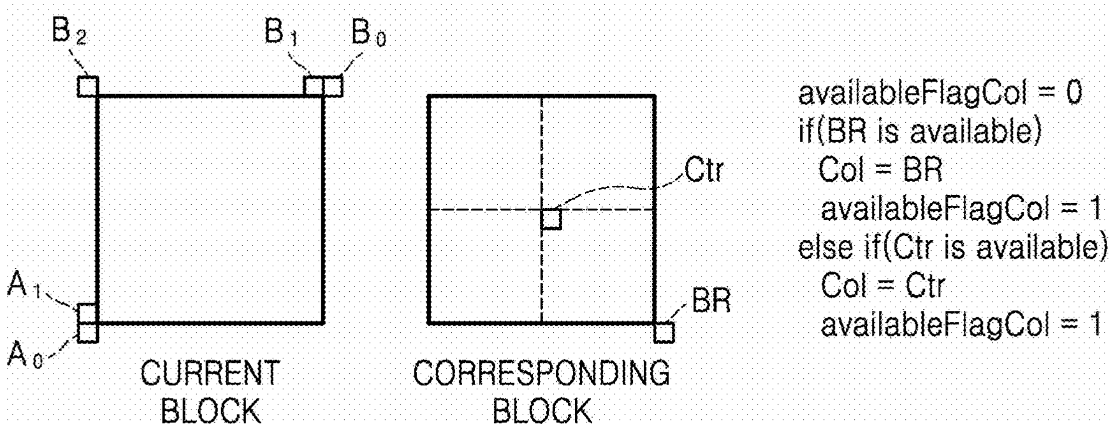
```
i = 0
if (availableFlagA1)
  mergeCandList [ i++ ] = A1
if (availableFlagB1)
  mergeCandList [ i++ ] = B1
if (availableFlagB0)
  mergeCandList [ i++ ] = B0
if (availableFlagA0)
  mergeCandList [ i++ ] = A0
if (availableFlagB2)
  mergeCandList [ i++ ] = B2
if (availableFlagCol)
  mergeCandList [ i++ ] = Col
```

FIG. 39B

5. The merging candidate list, mergeCandList, is constructed as follows:

```
i = 0
if (Genrated MV is available?)                      3910
   mergeCandList [ i++ ] = Generated MV
if (availableFlagA1 )
   mergeCandList [ i++ ] = A1
if (availableFlagB1 )
   mergeCandList [ i++ ] = B1
if (availableFlagB0 )
   mergeCandList [ i++ ] = B0
if (availableFlagA0 )
   mergeCandList [ i++ ] = A0
if (availableFlagB2 )
   mergeCandList [ i++ ] = B2
if (availableFlagCol)
   mergeCandList [ i++ ] = Col
```

FIG. 40



B2
B1
B0
A1
A0
CURRENT BLOCK
Ctr
BR
CORRESPONDING BLOCK
availableFlagCol = 0
if(BR is available)
Col = BR
availableFlagCol = 1
else if(Ctr is available)
Col = Ctr
availableFlagCol = 1

FIG. 41A

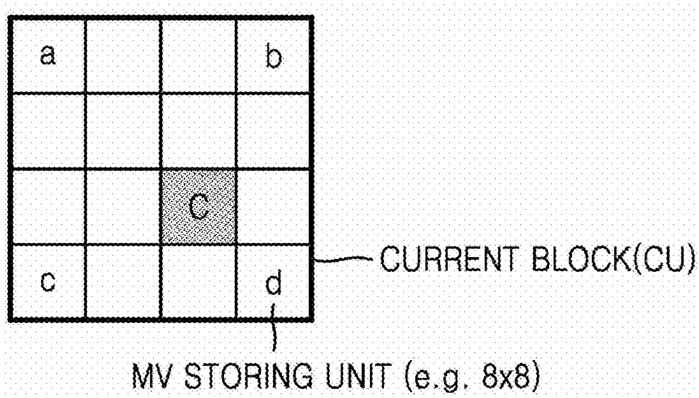


FIG. 41B

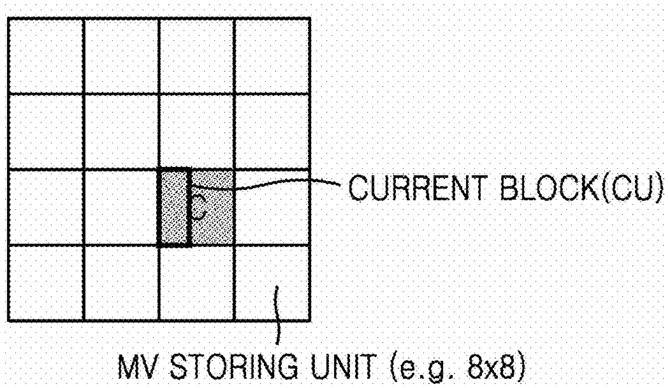


FIG. 41C

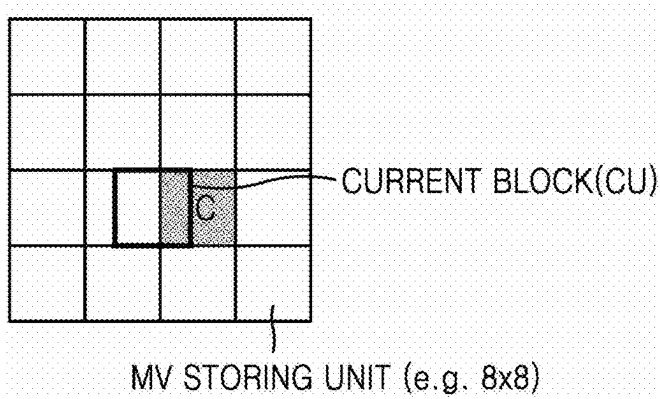


FIG. 41D

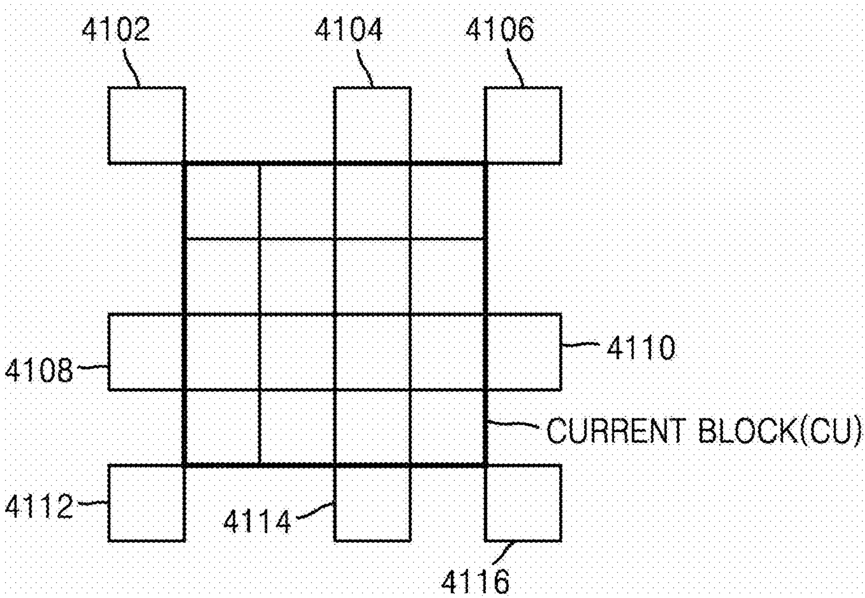


FIG. 42A

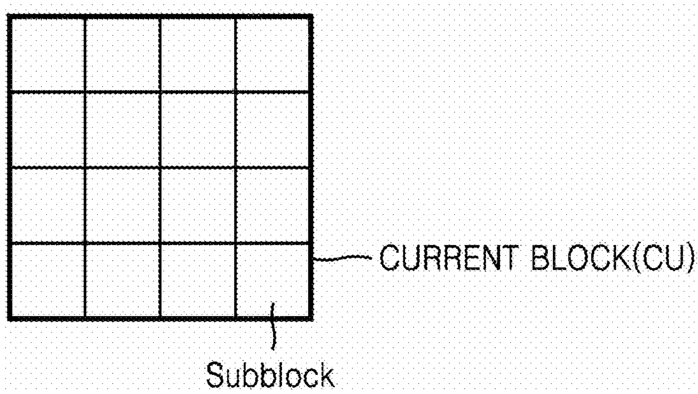


FIG. 42B

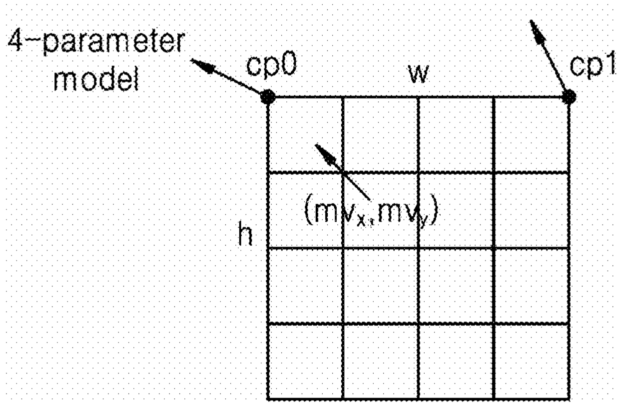
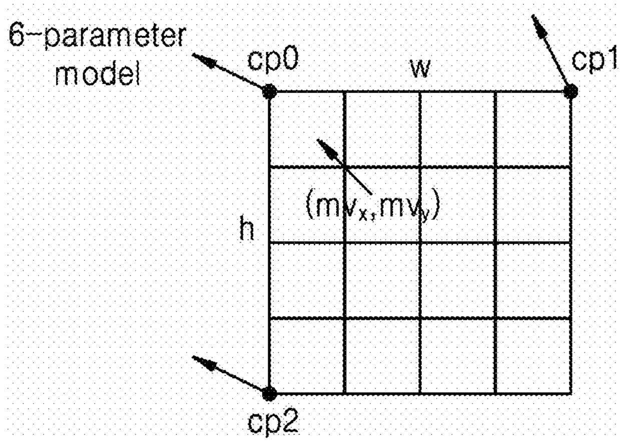


FIG. 42C



# VIDEO CODING METHOD AND DEVICE THEREFOR

## CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a by-pass continuation application of International Application No. PCT/KR2023/015509, filed on Oct. 10, 2023, which is based on and claims priority to Korean Patent Application No. 10-2022-0155825, filed on Nov. 18, 2022, in the Korean Patent Office, and Korean Patent Application No. 10-2023-0001329, filed on Jan. 4, 2023, in the Korean Patent Office, the disclosures of which are incorporated by reference herein in their entireties.

## 1. FIELD

[0002] The present disclosure relates to video coding, an image decoding method, an image decoding apparatus, an image encoding method, an image encoding apparatus, and a storage medium storing a bitstream generated by the image encoding method.

## 2. DESCRIPTION OF RELATED ART

[0003] The size of a coding unit included in a picture is usually determined by deciding whether to split or not, and then square coding units are determined through a recursive splitting process of uniformly splitting into four coding units of the same size. With the increasing use of high-resolution images, it has become problematic that using uniformly shaped square coding units may lead to degradation in the quality of reconstructed images. Accordingly, various methods and apparatuses for splitting a high-resolution image into various types of coding units have been proposed.

## SUMMARY

[0004] According to an aspect of the disclosure, an image decoding method includes generating reference motion information for at least one block included in a current picture, based on at least one of: pixel data of at least one reconstructed picture, or motion information of the at least one reconstructed picture; obtaining motion information for a current block among the at least one block included in the current picture, based on the reference motion information and at least one syntax information obtained from a bitstream; and reconstructing the current block, based on the motion information for the current block.

[0005] The generating of the reference motion information for the at least one block included in the current picture may include generating the reference motion information for the at least one block included in the current picture by inputting at least one of: the pixel data of the at least one reconstructed picture, or the motion information of the at least one reconstructed picture to an artificial neural network.

[0006] The motion information for the current block may include at least one of first motion vector information, second motion vector information, first prediction list utilization information, second prediction list utilization information, or prediction mode information.

[0007] The method may further include performing at least one of: aggregating the reference motion information, scaling the reference motion information, or transforming a format of the reference motion information.

[0008] The generating of the reference motion information for the at least one block included in the current picture may include down-sampling the at least one reconstructed picture; and generating the reference motion information for the at least one block included in the current picture, based on at least one of: pixel data of the downsampled reconstructed picture or motion information of the downsampled reconstructed picture.

[0009] The obtaining of the motion information for the current block may include, based on a time interval between the reconstructed picture and the current picture being equal to or greater than a certain value, excluding the reference motion information from a motion information candidate for the current block.

[0010] The generating of the reference motion information for the at least one block included in the current picture may include based on reference picture resampling being applied, downsampling the at least one reconstructed picture, based on a reference picture of low resolution.

[0011] The method may further include generating motion information for an additional picture by performing interpolation or extrapolation, based on the motion information of the at least one reconstructed picture.

[0012] The at least one syntax information may include information indicating whether the reference motion information is used to obtain the motion information for the current block.

[0013] The obtaining of the motion information for the current block may include constructing a merge candidate list including the reference motion information; and obtaining the motion information for the current block, based on the merge candidate list.

[0014] The reference motion information may include motion information at a location corresponding to: an x-coordinate based on adding half a width of the current block to an upper left sample location of the current block in a map of the reference motion information, and a y-coordinate based on adding half a height of the current block to the upper left sample location of the current block in the map of the reference motion information.

[0015] The obtaining of the motion information for the current block may include dividing the current block into a plurality of subblocks; and assigning the reference motion information to the plurality of subblocks, and the reconstructing of the current block may include performing motion compensation on a subblock, based on reference motion information for the subblock.

[0016] The obtaining of the motion information for the current block may include obtaining a plurality of control point motion vectors for the current block, based on the reference motion information, and the reconstructing of the current block may include performing motion compensation on a subblock of the current block based on the plurality of the control point motion vectors for the current block.

[0017] According to another aspect of the disclosure, an image encoding method includes generating reference motion information for at least one block included in a current picture, based on at least one of: pixel data of at least one reconstructed picture, or motion information of the at least one reconstructed picture; determining motion information for a current block among the at least one block included in the current picture; and encoding the motion information for the current block into a bitstream, based on the reference motion information.

[0018] According to another aspect of the disclosure, a non-transitory computer-readable storage medium storing a bitstream encoded by an image encoding method, the image encoding method comprising: generating reference motion information for at least one block included in a current picture, based on at least one of: pixel data of at least one reconstructed picture, or motion information of the at least one reconstructed picture; determining motion information for a current block among the at least one block included in the current picture; and encoding the motion information for the current block into a bitstream, based on the reference motion information.

## BRIEF DESCRIPTION OF THE DRAWINGS

**[0019]** The above and other aspects, features, and advantages of certain embodiments of the present disclosure are more apparent from the following description taken in conjunction with the accompanying drawings, in which:

**[0020]** A brief description of each drawing is provided for better understanding of the drawings cited herein.

**[0021]** FIG. **1** is a schematic block diagram of an image decoding apparatus according to an embodiment.

**[0022]** FIG. **2** is a flowchart of an image decoding method according to an embodiment.

**[0023]** FIG. **3** illustrates a process, performed by an image decoding apparatus, of determining at least one coding unit by splitting a current coding unit, according to an embodiment.

**[0024]** FIG. **4** illustrates a process, performed by an image decoding apparatus, of determining at least one coding unit by splitting a non-square coding unit, according to an embodiment.

**[0025]** FIG. **5** illustrates a process, performed by an image decoding apparatus, of splitting a coding unit based on at least one of block shape information or split shape mode information, according to an embodiment.

**[0026]** FIG. **6** illustrates a method, performed by an image decoding apparatus, of determining a preset coding unit from among an odd number of coding units, according to an embodiment.

**[0027]** FIG. **7** illustrates an order of processing a plurality of coding units when an image decoding apparatus determines the plurality of coding units by splitting a current coding unit, according to an embodiment.

**[0028]** FIG. **8** illustrates a process, performed by an image decoding apparatus, of determining that a current coding unit is to be split into an odd number of coding units, when the coding units are not processable in a preset order, according to an embodiment.

**[0029]** FIG. **9** illustrates a process, performed by an image decoding apparatus, of determining at least one coding unit by splitting a current coding unit, according to an embodiment.

**[0030]** FIG. **10** illustrates that a shape into which a second coding unit is splittable is restricted when the second coding unit having a non-square shape, which is determined when an image decoding apparatus splits a first coding unit, satisfies a preset condition, according to an embodiment.

**[0031]** FIG. **11** illustrates a process, performed by an image decoding apparatus, of splitting a square coding unit when split shape mode information indicates that the square coding unit is to not be split into four square coding units, according to an embodiment.

**[0032]** FIG. **12** illustrates that a processing order between a plurality of coding units may be changed depending on a process of splitting a coding unit, according to an embodiment.

**[0033]** FIG. **13** illustrates a process of determining a depth of a coding unit as a shape and size of the coding unit change, when the coding unit is recursively split such that a plurality of coding units are determined, according to an embodiment.

**[0034]** FIG. **14** illustrates depths that are determinable based on shapes and sizes of coding units, and part indexes (PIDs) for distinguishing the coding units, according to an embodiment.

**[0035]** FIG. **15** illustrates that a plurality of coding units are determined based on a plurality of preset data units included in a picture, according to an embodiment.

**[0036]** FIG. **16** is a block diagram of an image encoding and decoding system.

**[0037]** FIG. **17** is a block diagram of an image decoding apparatus according to the present disclosure.

**[0038]** FIG. **18** is a flowchart of an image decoding method according to the present disclosure.

**[0039]** FIG. **19** illustrates an image encoding apparatus according to the present disclosure.

**[0040]** FIG. **20** is a flowchart of an image encoding method according to the present disclosure.

**[0041]** FIGS. **21**A, **21**B, and **21**C illustrate a method of generating motion information, according to the present disclosure.

**[0042]** FIG. **22** illustrates a method of normalizing an input of a motion information generator, according to the present disclosure.

**[0043]** FIG. **23** illustrates an input motion information map according to the present disclosure.

**[0044]** FIG. **24** illustrates a generated motion information map according to the present disclosure.

**[0045]** FIG. **25** illustrates a motion information generating model and a learning method, according to the present disclosure.

**[0046]** FIG. **26** illustrates a method of generating motion information, according to the present disclosure.

**[0047]** FIG. **27** illustrates a process of aggregating motion information, according to the present disclosure.

**[0048]** FIG. **28** illustrates a method of downscaling an input reconstructed picture, according to the present disclosure.

**[0049]** FIG. **29** illustrates reference picture resampling (RPR).

**[0050]** FIGS. **30**, **31**A, and **31**B illustrate a method of utilizing a motion information generating method, according to the present disclosure, in frame rate up conversion (FRUC).

**[0051]** FIGS. **32**A and **32**B illustrate a method of refining a generated motion information map, according to the present disclosure.

**[0052]** FIGS. **33**, **34**A, **34**B, **35**, **36**, **37**A, **37**B, and **38** illustrate a syntax structure and a coding method according to the present disclosure.

**[0053]** FIGS. **39**A and **39**B illustrate use of generated motion information in a merge mode, according to the present disclosure.

**[0054]** FIG. **40** illustrates a location of a neighboring block usable as a merge candidate.

**[0055]** FIGS. **41**A through **41**D illustrate a method of determining a location corresponding to a current block in a generated motion information map, according to the present disclosure.

**[0056]** FIGS. **42**A through **42**C illustrate a method of performing subblock prediction by using generated motion information, according to the present disclosure.

## DETAILED DESCRIPTION

**[0057]** The embodiments described in the disclosure, and the configurations shown in the drawings, are only examples of embodiments, and various modifications may be made without departing from the scope and spirit of the disclosure.

**[0058]** As the disclosure allows for various changes and numerous embodiments, exemplary embodiments will be illustrated in the drawings and described in detail in the written description. This is not intended to limit the disclosure to particular modes of practice, and it is to be appreciated that all changes, equivalents, and substitutes that do not depart from the spirit and technical scope of the disclosure are encompassed in the disclosure.

**[0059]** In the description of embodiments, certain detailed explanations of the related art are omitted when it is deemed that they may unnecessarily obscure the essence of the disclosure. Also, numbers (e.g., first and second) used in the description of the specification are identifier codes for distinguishing one component from another.

**[0060]** Also, in the specification, it will be understood that when elements are "connected" or "coupled" to each other, the elements may be directly connected or coupled to each other, but may alternatively be connected or coupled to each other with an intervening element therebetween, unless specified otherwise.

**[0061]** In the specification, regarding a component represented as a "portion (unit)" or a "module", two or more components may be combined into one component or one component may be divided into two or more components according to subdivided functions. Each component described may additionally perform some or all of functions performed by another component, in addition to main functions of itself, and some of the main functions of each component may be performed entirely by another component.

**[0062]** The expressions "at least one of A, B and C" and "at least one of A, B, or C", both indicate "A", only "B", only "C", both "A and B", both "A and C", both "B and C", and all of "A, B, and C".

**[0063]** Also, the term 'image' or picture' used in some embodiments may refer to a still image of a video, or a moving image, i.e., a video itself.

**[0064]** Also, the term 'sample' used in some embodiments refers to data that is assigned to a sampling location of an image and is to be processed. For example, pixel values of an image in a spatial domain or transform coefficients in a transform domain may be samples. A unit including one or more samples may be defined as a block.

**[0065]** A 'current block' used in some embodiments may denote a block of a largest coding unit, coding unit, prediction unit, or transform unit of a current image that is to be encoded or decoded.

**[0066]** In this specification, when a motion vector is in the direction of list **0**, this may mean that the motion vector is used to indicate a block in a reference picture included in list **0** (or reference list **0** or reference picture list **0**), and when a motion vector is in the direction of list **1**, this may mean that the motion vector is used to indicate a block in a reference picture included in list **1** (or reference list **1** or reference picture list **1**). When a motion vector is in a uni-direction, this may mean that the motion vector is used to indicate a block in a reference picture included in list **0** or list **1**, and when a motion vector is in a bi-direction, this may mean that the motion vector includes a motion vector in the direction of list **0** and a motion vector in the direction of list **1**. List **0** may be briefly represented as L**0**, and list **1** may be briefly represented as L**1**.

**[0067]** In this specification, a 'binary split' of a block refers to a split that enables two subblocks, each of which a width or height is half a width or height of the block, to be generated. In detail, when a 'binary vertical split' is performed on a current block, a split is performed in a vertical direction (height direction) at a point being a half the width of the current block, so that two subblocks each having a width that is half the width of the current block and a height that is the same as the height of the current block may be generated. When a 'binary horizontal split' is performed on a current block, a split is performed in a horizontal direction (width direction) at a point being a half the height of the current block, so that two subblocks each having a height that is half the height of the current block and a width that is the same as the width of the current block may be generated.

**[0068]** In this specification, a 'ternary split' of a block refers to a split that enables three subblocks to be generated by splitting a width or height of the block at a ratio of 1:2:1. In detail, when a 'ternary vertical split' is performed on a current block, a split is performed in a vertical direction (height direction) at a point being the 1:2:1 ratio of the width of the current block, so that two subblocks each having a width that is ¼ the width of the current block and a height that is the same as the height of the current block, and one subblock having a width that is 2/4 the width of the current block and a height that is the same as the height of the current block may be generated. When a 'ternary vertical split' is performed on a current block, a split is performed in a horizontal direction (width direction) at a point being the 1:2:1 ratio of the height of the current block, so that two subblocks each having a width that is ¼ the height of the current block and a width that is the same as the width of the current block, and one subblock having a height that is 2/4 the height of the current block and a width that is the same as the width of the current block may be generated.

**[0069]** In this specification, a 'quadsplit' of a block refers to a split that enables four subblocks to be generated by splitting a width and height of the block at a ratio of 1:1. In detail, when a 'quadsplit' is performed on a current block, a split is performed in a vertical direction (height direction) at a point being half the width of the current block and a split is performed in a horizontal direction (width direction) at a point being a half the height of the current block, so that four subblocks each having a width that is ½ the width of the current block and a height that is ½ the height of the current block may be generated.

**[0070]** An image encoding method and apparatus and an image decoding method and apparatus according to an embodiment will be provided with reference to FIGS. **1** to **16**.

**[0071]** An image encoding method and apparatus and an image decoding method and apparatus according to an embodiment will be provided with reference to FIGS. **1** to **16**.

**[0072]** FIG. **1** is a schematic block diagram of an image decoding apparatus, according to an embodiment. The image decoding apparatus **100** may include a receiver **110** and a decoder **120**. The receiver **110** and the decoder **120** may include at least one processor. The image decoding apparatus **100** may be an electronic device, such as a smartphone, a tablet, a smart television, a set-top box, a computer or a laptop, or an augmented reality (AR) or virtual reality (VR) headset. The image decoding apparatus **100** may further include a display configured to display a result of decoding a bitstream.

**[0073]** Also, the receiver **110** and the decoder **120** may include a memory storing instructions to be executed by the at least one processor. The receiver **110** may receive a bitstream. The receiver **110** may be a communication interface including at least one of a wireless communication module such as a Wi-Fi receiver, a radio frequency modem, an antenna circuit, an HDMI input interface, a mobile communication module such as a 4G/5G modem, a memory interface for receiving a stored bitstream from memory, or related software and/or firmware. The bitstream may include information resulting from image encoding by an image encoding apparatus **200** which will be described below. Also, the bitstream may be transmitted from the image encoding apparatus **200**. The image decoding apparatus **100** may be connected to the image encoding apparatus **200** in a wired or wireless manner, and the receiver **110** may receive a bitstream in a wired or wireless manner. The receiver **110** may receive a bitstream from a storage medium, such as an optical medium or a hard disk. The decoder **120** may reconstruct an image based on information obtained from the received bitstream. The decoder **120** may obtain a syntax element for reconstructing an image from the bitstream.

**[0074]** The decoder **120** may reconstruct the image based on the syntax element.

**[0075]** An operation of the image decoding apparatus **100** will be described in more detail with reference to FIG. **2**.

**[0076]** FIG. **2** is a flow chart of an image decoding method according to an embodiment.

**[0077]** According to an embodiment of the present disclosure, the receiver **110** receives a bitstream. The image decoding apparatus **100** may perform an operation **210** of obtaining a bin string corresponding to a split shape mode of a coding unit from the bitstream. Then, the image decoding apparatus **100** may perform an operation **220** of determining a split rule of a coding unit. Also, the image decoding apparatus **100** may perform an operation **230** of splitting a coding unit into a plurality of coding units, based on at least one of the bin string corresponding to the split shape mode and the split rule. The image decoding apparatus **100** may determine a first range which is an allowable size range of a coding unit, according to a ratio of a height to a width of the coding unit, in order to determine the split rule.

**[0078]** The image decoding apparatus **100** may determine a second range which is an allowable size range of a coding unit, according to a split shape mode of the coding unit, in order to determine the split rule.

**[0079]** Splitting of a coding unit will be described in detail according to an embodiment of the disclosure. First, one picture may be split into one or more slices or one or more tiles. One slice or one tile may be a sequence of one or more largest coding units (coding tree units, CTUs).

**[0080]** As a concept compared to a largest coding unit (CTU), there is a largest coding block (coding tree block, CTB). A largest coding block (CTB) denotes an N×N block including N×N samples (N is an integer).

**[0081]** Each color component may be split into one or more largest coding blocks. When a picture has three sample arrays (sample arrays for Y, Cr, and Cb components), a largest coding unit (CTU) includes a largest coding block of a luma sample, two corresponding largest coding blocks of chroma samples, and syntax structures used to encode the luma sample and the chroma samples. When a picture is a monochrome picture, a largest coding unit includes a largest coding block of a monochrome sample and syntax structures used to encode the monochrome samples.

**[0082]** When a picture is a picture encoded in color planes separated according to color components, a largest coding unit includes syntax structures used to encode the picture and samples of the picture.

**[0083]** One largest coding block (CTB) may be split into M×N coding blocks including M×N samples (M and N are integers). When a picture has sample arrays for Y, Cr, and Cb components, a coding unit (CU) includes a coding block of a luma sample, two corresponding coding blocks of chroma samples, and syntax structures used to encode the luma sample and the chroma samples. When a picture is a monochrome picture, a coding unit includes a coding block of a monochrome sample and syntax structures used to encode the monochrome samples.

**[0084]** When a picture is a picture encoded in color planes separated according to color components, a coding unit includes syntax structures used to encode the picture and samples of the picture. As described above, a largest coding block and a largest coding unit are conceptually distinguished from each other, and a coding block and a coding unit are conceptually distinguished from each other. That is, a (largest) coding unit refers to a data structure including a (largest) coding block including a corresponding sample and a syntax structure corresponding to the (largest) coding block.

**[0085]** Because it is understood by one of ordinary skill in the art that a (largest) coding unit or a (largest) coding block refers to a block of a certain size including a certain number of samples, a largest coding block and a largest coding unit, or a coding block and a coding unit are mentioned in the following specification without being distinguished unless otherwise described. An image may be split into largest coding units (CTUs). A size of each largest coding unit may be determined based on information obtained from a bitstream. A shape of each largest coding unit may be a square shape of the same size.

**[0086]** An embodiment is not limited thereto. For example, information about a maximum size of a luma coding block may be obtained from a bitstream.

**[0087]** For example, the maximum size of the luma coding block indicated by the information about the maximum size of the luma coding block may be one of 4×4, 8×8, 16×16, 32×32, 64×64, 128×128, and 256×256. For example, information about a luma block size difference and a maximum size of a luma coding block that may be split into two may be obtained from a bitstream. The information about the luma block size difference may refer to a size difference between a luma largest coding unit and a largest luma coding block that may be split into two. Accordingly, when the information about the maximum size of the luma coding block that may be split into two and the information about the luma block size difference obtained from the bitstream are combined with each other, a size of the luma largest coding unit may be determined. A size of a chroma largest coding unit may be determined by using the size of the luma largest coding unit.

**[0088]** For example, when a Y:Cb:Cr ratio is 4:2:0 according to a color format, a size of a chroma block may be half a size of a luma block, and a size of a chroma largest coding unit may be half a size of a luma largest coding unit. According to an embodiment, because information about a maximum size of a luma coding block that is binary splittable is obtained from a bitstream, the maximum size of the luma coding block that is binary splittable may be variably determined. In contrast, a maximum size of a luma coding block that is ternary splittable may be fixed.

**[0089]** For example, the maximum size of the luma coding block that is ternary splittable in an I-picture may be 32×32, and the maximum size of the luma coding block that is ternary splittable in a P-picture or a B-picture may be 64×64. Also, a largest coding unit may be hierarchically split into coding units based on split shape mode information obtained from a bitstream.

**[0090]** At least one of information indicating whether quad splitting is performed, information indicating whether multi-splitting is performed, split direction information, and split type information may be obtained as the split shape mode information from the bitstream.

**[0091]** For example, the information indicating whether quad splitting is performed may indicate whether a current coding unit is quad split (QUAD_SPLIT) or not.

**[0092]** When the current coding unit is not quad split, the information indicating whether multi-splitting is performed may indicate whether the current coding unit is no longer split (NO_SPLIT) or binary/ternary split.

**[0093]** When the current coding unit is binary split or ternary split, the split direction information indicates that the current coding unit is split in one of a horizontal direction and a vertical direction.

**[0094]** When the current coding unit is split in the horizontal direction or the vertical direction, the split type information indicates that the current coding unit is binary split or ternary split. A split mode of the current coding unit may be determined according to the split direction information and the split type information.

**[0095]** A split mode when the current coding unit is binary split in the horizontal direction may be determined to be a binary horizontal split mode (SPLIT_BT_HOR), a split mode when the current coding unit is ternary split in the horizontal direction may be determined to be a ternary horizontal split mode (SPLIT_TT_HOR), a split mode when the current coding unit is binary split in the vertical direction may be determined to be a binary vertical split mode (SPLIT_BT_VER), and a split mode when the current coding unit is ternary split in the vertical direction may be determined to be a ternary vertical split mode (SPLIT_TT_VER). The image decoding apparatus **100** may obtain, from the bitstream, the split shape mode information from one bin string. A form of the bitstream received by the image decoding apparatus **100** may include fixed length binary code, unary code, truncated unary code, pre-determined binary code, or the like. The bin string is information in a binary number. The bin string may include at least one bit. The image decoding apparatus **100** may obtain the split shape mode information corresponding to the bin string, based on the split rule.

**[0096]** The image decoding apparatus **100** may determine whether to quad split a coding unit, whether not to split a coding unit, a split direction, and a split type, based on one bin string. The coding unit may be smaller than or the same as the largest coding unit. For example, because a largest coding unit is a coding unit having a maximum size, the largest coding unit is one of coding units. When split shape mode information about a largest coding unit indicates that splitting is not performed, a coding unit determined in the largest coding unit has the same size as that of the largest coding unit. When split shape mode information about a largest coding unit indicates that splitting is performed, the largest coding unit may be split into coding units. Also, when split shape mode information about a coding unit indicates that splitting is performed, the coding unit may be split into smaller coding units. The splitting of the image is not limited thereto, and the largest coding unit and the coding unit may not be distinguished.

**[0097]** The splitting of the coding unit will be described in more detail with reference to FIGS. **3** to **16**. Also, one or more prediction blocks for prediction may be determined from a coding unit. The prediction block may be the same as or smaller than the coding unit. Also, one or more transform blocks for transformation may be determined from a coding unit.

**[0098]** The transform block may be equal to or smaller than the coding unit.

**[0099]** The shapes and sizes of the transform block and prediction block may not be related to each other. In another embodiment, prediction may be performed by using a coding unit as a prediction unit.

**[0100]** Also, transformation may be performed by using a coding unit as a transform block. A current block and a neighboring block of the disclosure may indicate one of the largest coding unit, the coding unit, the prediction block, and the transform block. Also, the current block or the current coding unit is a block that is currently being decoded or encoded or a block that is currently being split. The neighboring block may be a block reconstructed before the current block. The neighboring block may be adjacent to the current block spatially or temporally.

**[0101]** The neighboring block may be located at one of the lower left, left, upper left, top, upper right, right, lower right of the current block.

**[0102]** FIG. **3** illustrates a process in which the image decoding apparatus determines at least one coding unit by splitting a current coding unit, according to an embodiment. A block shape may include 4N×4N, 4N×2N, 2N×4N, 4N×N, N×4N, 32N×N, N×32N, 16N×N, N×16N, 8N×N, or N×8N. Here, N may be a positive integer.

**[0103]** Block shape information is information indicating at least one of a shape, a direction, a ratio of width and height, and size of a coding unit. The shape of the coding unit may include a square and a non-square. When the width and height of the coding unit are the same (i.e., when the block shape of the coding unit is 4N×4N), the image decoding apparatus **100** may determine the block shape information of the coding unit as a square.

**[0104]** The image decoding apparatus **100** may determine the shape of the coding unit to be a non-square. When the width and the height of the coding unit are different from each other (i.e., when the block shape of the coding unit is 4N×2N, 2N×4N, 4N×N, N×4N, 32N×N, N×32N, 16N×N, N×16N, 8N×N, or N×8N), the image decoding apparatus **100** may determine the block shape information of the coding unit as a non-square. When the shape of the coding unit is a non-square, the image decoding apparatus **100** may determine the ratio of the width and height among the block shape information of the coding unit as at least one of 1:2, 2:1, 1:4, 4:1, 1:8, 8:1, 1:16, 16:1, 1:32, and 32:1. Also, the image decoding apparatus **100** may determine whether the coding unit is in a horizontal direction or a vertical direction, based on the length of the width and the length of the height of the coding unit.

**[0105]** Also, the image decoding apparatus **100** may determine the size of the coding unit, based on at least one of the length of the width, the length of the height, and the area of the coding unit. According to an embodiment, the image decoding apparatus **100** may determine the shape of the coding unit by using the block shape information, and may determine a splitting method of the coding unit by using split shape mode information.

**[0106]** That is, a coding unit splitting method indicated by the split shape mode information may be determined based on a block shape indicated by the block shape information used by the image decoding apparatus **100**. The image decoding apparatus **100** may obtain the split shape mode information from a bitstream. An embodiment is not limited thereto, and the image decoding apparatus **100** and the image encoding apparatus **200** may determine pre-agreed split shape mode information, based on the block shape information. The image decoding apparatus **100** may determine the pre-agreed split shape mode information with respect to a largest coding unit or a smallest coding unit. For example, the image decoding apparatus **100** may determine the split shape mode information with respect to the largest coding unit to be a quad split. Also, the image decoding apparatus **100** may determine the split shape mode information with respect to the smallest coding unit to be "not to perform splitting". In particular, the image decoding apparatus **100** may determine the size of the largest coding unit to be 256×256. The image decoding apparatus **100** may determine the pre-agreed split shape mode information to be a quad split. The quad split is a split shape mode in which the width and the height of the coding unit are both bisected. The image decoding apparatus **100** may obtain a coding unit of a 128×128 size from the largest coding unit of a 256×256 size, based on the split shape mode information. Also, the image decoding apparatus **100** may determine the size of the smallest coding unit to be 4×4.

**[0107]** The image decoding apparatus **100** may obtain split shape mode information indicating "not to perform splitting" with respect to the smallest coding unit. According to an embodiment, the image decoding apparatus **100** may use the block shape information indicating that the current coding unit has a square shape. For example, the image decoding apparatus **100** may determine whether not to split a square coding unit, whether to vertically split the square coding unit, whether to horizontally split the square coding unit, or whether to split the square coding unit into four coding units, based on the split shape mode information.

**[0108]** Referring to FIG. **3**, when the block shape information of a current coding unit **300** indicates a square shape, the decoder **120** may not split a coding unit **310***a* having the same size as the current coding unit **300**, based on the split shape mode information indicating not to perform splitting, or may determine coding units **310***b*, **310***c*, **310***d*, **310***e*, or **310***f* split based on the split shape mode information indicating a certain splitting method. Referring to FIG. **3**, according to an embodiment, the image decoding apparatus **100** may determine two coding units **310***b* obtained by splitting the current coding unit **300** in a vertical direction, based on the split shape mode information indicating to perform splitting in a vertical direction. The image decoding apparatus **100** may determine two coding units **310***c* obtained by splitting the current coding unit **300** in a horizontal direction, based on the split shape mode information indicating to perform splitting in a horizontal direction. The image decoding apparatus **100** may determine four coding units **310***d* obtained by splitting the current coding unit **300** in vertical and horizontal directions, based on the split shape mode information indicating to perform splitting in vertical and horizontal directions. According to an embodiment, the image decoding apparatus **100** may determine three coding units **310***e* obtained by splitting the current coding unit **300** in a vertical direction, based on the split shape mode information indicating to perform ternary splitting in a vertical direction. The image decoding apparatus **100** may determine three coding units **310***f* obtained by splitting the current coding unit **300** in a horizontal direction, based on the split shape mode information indicating to perform ternary splitting in a horizontal direction. Splitting methods of the square coding unit are not limited to the above-described methods, and the split shape mode information may indicate various methods.

**[0109]** Certain splitting methods of splitting the square coding unit will be described in detail below through various embodiments.

**[0110]** FIG. **4** illustrates a process in which the image decoding apparatus determines at least one coding unit by splitting a non-square coding unit, according to an embodiment. According to an embodiment, the image decoding apparatus **100** may use block shape information indicating that a current coding unit has a non-square shape. The image decoding apparatus **100** may determine whether not to split the non-square current coding unit or whether to split the non-square current coding unit by using a certain splitting method, based on split shape mode information. Referring to FIG. **4**, when the block shape information of a current coding unit **400** or **450** indicates a non-square shape, the image decoding apparatus **100** may determine a coding unit **410** or **460** having the same size as the current coding unit **400** or **450**, based on the split shape mode information indicating not to perform splitting, or may determine coding units **420***a* and **420***b*, **430***a* to **430***c*, **470***a* and **470***b*, or **480***a* to **480***c* split based on the split shape mode information indicating a certain splitting method.

**[0111]** Certain splitting methods of splitting a non-square coding unit will be described in detail below through various embodiments. According to an embodiment, the image decoding apparatus **100** may determine a splitting method of a coding unit by using the split shape mode information and, in this case, the split shape mode information may indicate the number of one or more coding units generated by splitting a coding unit.

**[0112]** Referring to FIG. **4**, when the split shape mode information indicates to split the current coding unit **400** or **450** into two coding units, the image decoding apparatus **100**

may determine two coding units **420***a* and **420***b*, or **470***a* and **470***b* included in the current coding unit **400** or **450**, by splitting the current coding unit **400** or **450** based on the split shape mode information. According to an embodiment, when the image decoding apparatus **100** splits the non-square current coding unit **400** or **450** based on the split shape mode information, the image decoding apparatus **100** may consider the location of a long side of the non-square current coding unit **400** or **450** to split a current coding unit.

**[0113]** For example, the image decoding apparatus **100** may determine a plurality of coding units by splitting the current coding unit **400** or **450** in a direction of splitting a long side of the current coding unit **400** or **450**, in consideration of the shape of the current coding unit **400** or **450**. According to an embodiment, when the split shape mode information indicates to split (ternary split) a coding unit into an odd number of blocks, the image decoding apparatus **100** may determine an odd number of coding units included in the current coding unit **400** or **450**.

**[0114]** For example, when the split shape mode information indicates to split the current coding unit **400** or **450** into three coding units, the image decoding apparatus **100** may split the current coding unit **400** or **450** into three coding units **430***a*, **430***b*, and **430***c*, or **480***a*, **480***b*, and **480***c*. According to an embodiment, a ratio of the width and height of the current coding unit **400** or **450** may be 4:1 or 1:4. When the ratio of the width and height is 4:1, the block shape information may indicate a horizontal direction because the length of the width is longer than the length of the height. When the ratio of the width and height is 1:4, the block shape information may indicate a vertical direction because the length of the width is shorter than the length of the height. The image decoding apparatus **100** may determine to split a current coding unit into an odd number of blocks, based on the split shape mode information. Also, the image decoding apparatus **100** may determine a split direction of the current coding unit **400** or **450**, based on the block shape information of the current coding unit **400** or **450**. For example, when the current coding unit **400** is in the vertical direction, the image decoding apparatus **100** may determine the coding units **430***a*, **430***b*, and **430***c* by splitting the current coding unit **400** in the horizontal direction.

**[0115]** Also, when the current coding unit **450** is in the horizontal direction, the image decoding apparatus **100** may determine the coding units **480***a*, **480***b*, and **480***c* by splitting the current coding unit **450** in the vertical direction. According to an embodiment, the image decoding apparatus **100** may determine an odd number of coding units included in the current coding unit **400** or **450**, and not all the determined coding units may have the same size. For example, a certain coding unit **430***b* or **480***b* from among the determined odd number of coding units **430***a*, **430***b*, and **430***c*, or **480***a*, **480***b*, and **480***c* may have a size different from the size of the other coding units **430***a* and **430***c*, or **480***a* and **480***c*.

**[0116]** That is, coding units which may be determined by splitting the current coding unit **400** or **450** may have multiple sizes and, in some cases, all of the odd number of coding units **430***a*, **430***b*, and **430***c*, or **480***a*, **480***b*, and **480***c* may have different sizes. According to an embodiment, when the split shape mode information indicates to split a coding unit into the odd number of blocks, the image decoding apparatus **100** may determine the odd number of coding units included in the current coding unit **400** or **450**, and moreover, may put a certain restriction on at least one of the odd number of coding units generated by splitting the current coding unit **400** or **450**. Referring to FIG. **4**, the image decoding apparatus **100** may set a decoding process regarding the coding unit **430***b* or **480***b* located at the center among the three coding units **430***a*, **430***b*, and **430***c*, or **480***a*, **480***b*, and **480***c* generated as the current coding unit **400** or **450** is split to be different from that of the other coding units **430***a* and **430***c*, or **480***a* and **480***c*.

**[0117]** For example, the image decoding apparatus **100** may restrict the coding unit **430***b* or **480***b* at the center location to be no longer split or to be split a certain number of times, unlike the other coding units **430***a* and **430***c*, or **480***a* and **480***c*.

**[0118]** FIG. **5** illustrates a process in which the image decoding apparatus splits a coding unit based on at least one of block shape information and split shape mode information, according to an embodiment. According to an embodiment, the image decoding apparatus **100** may determine to split or not to split a square first coding unit **500** into coding units, based on at least one of the block shape information and the split shape mode information. According to an embodiment, when the split shape mode information indicates to split the first coding unit **500** in a horizontal direction, the image decoding apparatus **100** may determine a second coding unit **510** by splitting the first coding unit **500** in a horizontal direction. A first coding unit, a second coding unit, and a third coding unit used according to an embodiment are terms used to understand a relation before and after splitting a coding unit. For example, a second coding unit may be determined by splitting a first coding unit, and a third coding unit may be determined by splitting the second coding unit.

**[0119]** It will be understood that the relation of the first coding unit, the second coding unit, and the third coding unit follows the above descriptions. According to an embodiment, the image decoding apparatus **100** may determine to split or not to split the determined second coding unit **510** into coding units, based on the split shape mode information. Referring to FIG. **5**, the image decoding apparatus **100** may or may not split the non-square second coding unit **510**, which is determined by splitting the first coding unit **500**, into one or more third coding units **520***a*, **520***b*, **520***c*, and **520***d* based on the split shape mode information. The image decoding apparatus **100** may obtain the split shape mode information, and may obtain a plurality of various-shaped second coding units (e.g., **510**) by splitting the first coding unit **500**, based on the obtained split shape mode information, and the second coding unit **510** may be split by using a splitting method of the first coding unit **500** based on the split shape mode information. According to an embodiment, when the first coding unit **500** is split into the second coding units **510** based on the split shape mode information of the first coding unit **500**, the second coding unit **510** may also be split into the third coding units (e.g., **520***a*, or **520***b*, **520***c*, and **520***d*) based on the split shape mode information of the second coding unit **510**. That is, a coding unit may be recursively split based on the split shape mode information of each coding unit.

**[0120]** Accordingly, a square coding unit may be determined by splitting a non-square coding unit, and a non-square coding unit may be determined by recursively splitting the square coding unit. Referring to FIG. **5**, a certain coding unit (e.g., a coding unit located at a center location, or a square coding unit) from among an odd number of third coding units **520***b*, **520***c*, and **520***d* determined by splitting the non-square second coding unit **510** may be recursively split. According to an embodiment, the non-square third coding unit **520***b* from among the odd number of third coding units **520***b*, **520***c*, and **520***d* may be split in a horizontal direction into a plurality of fourth coding units. A non-square fourth coding unit **530***b* or **530***d* from among the plurality of fourth coding units **530***a*, **530***b*, **530***c*, and **530***d* may be re-split into a plurality of coding units. For example, the non-square fourth coding unit **530***b* or **530***d* may be re-split into an odd number of coding units.

**[0121]** A method that may be used to recursively split a coding unit will be described below through various embodiments. According to an embodiment, the image decoding apparatus **100** may split each of the third coding units **520***a*, **520***b*, **520***c*, and **520***d* into coding units, based on the split shape mode information. Also, the image decoding apparatus **100** may determine not to split the second coding unit **510** based on the split shape mode information. According to an embodiment, the image decoding apparatus **100** may split the non-square second coding unit **510** into the odd number of third coding units **520***b*, **520***c*, and **520***d*. The image decoding apparatus **100** may put a certain restriction on a certain third coding unit from among the odd number of third coding units **520***b*, **520***c*, and **520***d*.

**[0122]** For example, the image decoding apparatus **100** may restrict the third coding unit **520***c* at a center location from among the odd number of third coding units **520***b*, **520***c*, and **520***d* to be no longer split or to be split a settable number of times. Referring to FIG. **5**, the image decoding apparatus **100** may restrict the third coding unit **520***c*, which is at the center location from among the odd number of third coding units **520***b*, **520***c*, and **520***d* included in the non-square second coding unit **510**, to be no longer split, to be split by using a certain splitting method (e.g., split into four coding units or split by using a splitting method of the second coding unit **510**), or to be split a certain number of times (e.g., split n times (where n>0)).

**[0123]** The restrictions on the third coding unit **520***c* at the center location are not limited to the above-described examples, and may include various restrictions for decoding the third coding unit **520***c* at the center location differently from the other third coding units **520***b* and **520***d*.

**[0124]** According to an embodiment, the image decoding apparatus **100** may obtain the split shape mode information, which is used to split a current coding unit, from a certain location in the current coding unit.

**[0125]** FIG. **6** illustrates a method in which the image decoding apparatus determines a certain coding unit from among an odd number of coding units, according to an embodiment. Referring to FIG. **6**, split shape mode information of a current coding unit **600** or **650** may be obtained from a sample of a certain location (e.g., a sample **640** or **690** of a center location) from among a plurality of samples included in the current coding unit **600** or **650**. The certain location in the current coding unit **600**, from which at least one piece of the split shape mode information may be obtained, is not limited to the center location in FIG. **6**, and may include various locations included in the current coding unit **600** (e.g., top, bottom, left, right, upper left, lower left, upper right, lower right locations, or the like).

**[0126]** The image decoding apparatus **100** may obtain the split shape mode information from the certain location and may determine to split or not to split the current coding unit into various-shaped and various-sized coding units. According to an embodiment, when the current coding unit is split into a certain number of coding units, the image decoding apparatus **100** may select one of the coding units.

**[0127]** Various methods may be used to select one of a plurality of coding units, as will be described below through various embodiments.

**[0128]** According to an embodiment, the image decoding apparatus **100** may split the current coding unit into a plurality of coding units, and may determine a coding unit at a certain location. According to an embodiment, the image decoding apparatus **100** may use information indicating locations of the odd number of coding units, to determine a coding unit at a center location from among the odd number of coding units. Referring to FIG. **6**, the image decoding apparatus **100** may determine the odd number of coding units **620***a*, **620***b*, and **620***c* or the odd number of coding units **660***a*, **660***b*, and **660***c* by splitting the current coding unit **600** or the current coding unit **650**. The image decoding apparatus **100** may determine the middle coding unit **620***b* or the middle coding unit **660***b* by using information about locations of the odd number of coding units **620***a*, **620***b*, and **620***c* or the odd number of coding units **660***a*, **660***b*, and **660***c*. For example, the image decoding apparatus **100** may determine the coding unit **620***b* of the center location by determining the locations of the coding units **620***a*, **620***b*, and **620***c* based on information indicating locations of certain samples included in the coding units **620***a*, **620***b*, and **620***c*.

**[0129]** In detail, the image decoding apparatus **100** may determine the coding unit **620***b* at the center location by determining the locations of the coding units **620***a*, **620***b*, and **620***c* based on information indicating locations of upper left samples **630***a*, **630***b*, and **630***c* of the coding units **620***a*, **620***b*, and **620***c*. According to an embodiment, the information indicating the locations of the upper left samples **630***a*, **630***b*, and **630***c*, which are respectively included in the coding units **620***a*, **620***b*, and **620***c*, may include information about locations or coordinates of the coding units **620***a*, **620***b*, and **620***c* in a picture. According to an embodiment, the information indicating the locations of the upper left samples **630***a*, **630***b*, and **630***c*, which are included in the coding units **620***a*, **620***b*, and **620***c*, respectively, may include information indicating widths or heights of the coding units **620***a*, **620***b*, and **620***c* included in the current coding unit **600**, and the widths or heights may correspond to information indicating differences between the coordinates of the coding units **620***a*, **620***b*, and **620***c* in the picture.

**[0130]** That is, the image decoding apparatus **100** may determine the coding unit **620***b* at the center location by directly using the information about the locations or coordinates of the coding units **620***a*, **620***b*, and **620***c* in the picture, or by using the information about the widths or heights of the coding units, which correspond to the difference values between the coordinates. According to an embodiment, information indicating the location of the upper left sample **630***a* of the upper coding unit **620***a* may include coordinates (xa, ya), information indicating the location of the upper left sample **630***b* of the middle coding unit **620***b* may include coordinates (xb, yb), and information indicating the location of the upper left sample **630***c* of the lower coding unit **620***c* may include coordinates (xc, yc). The image decoding apparatus **100** may determine the middle coding unit **620***b* by using the coordinates of the upper left samples **630***a*, **630***b*, and **630***c* which are included in the coding units **620***a*, **620***b*, and **620***c*, respectively. For example, when the coordinates of the upper left samples **630***a*, **630***b*, and **630***c* are sorted in an ascending or descending order, the coding unit **620***b* including the coordinates (xb, yb) of the sample **630***b* at a center location may be determined as a coding unit at a center location from among the coding units **620***a*, **620***b*, and **620***c* determined by splitting the current coding unit **600**. The coordinates indicating the locations of the upper left samples **630***a*, **630***b*, and **630***c* may include coordinates indicating absolute locations in the picture, or may use coordinates (dxb, dyb) indicating a relative location of the upper left sample **630***b* of the middle coding unit **620***b* and coordinates (dxc, dyc) indicating a relative location of the upper left sample **630***c* of the lower coding unit **620***c* with respect to the location of the upper left sample **630***a* of the upper coding unit **620***a*.

**[0131]** A method of determining a coding unit at a certain location by using coordinates of a sample included in the coding unit as information indicating a location of the sample is not limited to the above-described method, and may include various arithmetic methods capable of using the coordinates of the sample. According to an embodiment, the image decoding apparatus **100** may split the current coding unit **600** into a plurality of coding units **620***a*, **620***b*, and **620***c*, and may select one of the coding units **620***a*, **620***b*, and **620***c* based on a certain criterion.

**[0132]** For example, the image decoding apparatus **100** may select the coding unit **620***b*, which has a size different from that of the others, from among the coding units **620***a*, **620***b*, and **620***c*. According to an embodiment, the image decoding apparatus **100** may determine the width or height of each of the coding units **620***a*, **620***b*, and **620***c* by using the coordinates (xa, ya) that is the information indicating the location of the upper left sample **630***a* of the upper coding unit **620***a*, the coordinates (xb, yb) that is the information indicating the location of the upper left sample **630***b* of the middle coding unit **620***b*, and the coordinates (xc, yc) that are the information indicating the location of the upper left sample **630***c* of the lower coding unit **620***c*. The image decoding apparatus **100** may determine the respective sizes of the coding units **620***a*, **620***b*, and **620***c* by using the coordinates (xa, ya), (xb, yb), and (xc, yc) indicating the locations of the coding units **620***a*, **620***b*, and **620***c*. According to an embodiment, the image decoding apparatus **100** may determine the width of the upper coding unit **620***a* to be the width of the current coding unit **600**. The image decoding apparatus **100** may determine the height of the upper coding unit **620***a* to be yb-ya. According to an embodiment, the image decoding apparatus **100** may determine the width of the middle coding unit **620***b* to be the width of the current coding unit **600**. The image decoding apparatus **100** may determine the height of the middle coding unit **620***b* to be yc-yb. According to an embodiment, the image decoding apparatus **100** may determine the width or height of the lower coding unit **620***c* by using the width or height of the current coding unit **600** and the widths or heights of the upper and middle coding units **620***a* and **620***b*. The image decoding apparatus **100** may determine a coding unit, which has a size different from that of the others, based on the determined widths and heights of the coding units **620***a*, **620***b*, and **620***c*. Referring to FIG. **6**, the image decoding apparatus **100** may determine the middle coding unit **620***b*, which has a size different from the size of the upper and lower coding units **620***a* and **620***c*, as the coding unit of the certain location.

**[0133]** The above-described process in which the image decoding apparatus **100** determines a coding unit having a size different from the size of the other coding units corresponds to an example of determining a coding unit at a certain location by using the sizes of coding units, which are determined based on coordinates of samples, and thus various processes of determining a coding unit at a certain location by comparing the sizes of coding units, which are determined based on coordinates of certain samples, may be used. The image decoding apparatus **100** may determine the width or height of each of the coding units **660***a*, **660***b*, and **660***c* by using coordinates (xd, yd) that are information indicating a location of an upper left sample **670***a* of the left coding unit **660***a*, coordinates (xe, ye) that are information indicating a location of an upper left sample **670***b* of the middle coding unit **660***b*, and coordinates (xf, yf) that are information indicating a location of an upper left sample **670***c* of the right coding unit **660***c*.

**[0134]** The image decoding apparatus **100** may determine the respective sizes of the coding units **660***a*, **660***b*, and **660***c* by using the coordinates (xd, yd), (xe, ye), and (xf, yf) indicating the locations of the coding units **660***a*, **660***b*, and **660***c*. According to an embodiment, the image decoding apparatus **100** may determine the width of the left coding unit **660***a* to be xe-xd. The image decoding apparatus **100** may determine the height of the left coding unit **660***a* to be the height of the current coding unit **650**. According to an embodiment, the image decoding apparatus **100** may determine the width of the middle coding unit **660***b* to be xf-xe. The image decoding apparatus **100** may determine the height of the middle coding unit **660***b* to be the height of the current coding unit **650**. According to an embodiment, the image decoding apparatus **100** may determine the width or height of the right coding unit **660***c* by using the width or height of the current coding unit **650** and the widths or heights of the left and middle coding units **660***a* and **660***b*. The image decoding apparatus **100** may determine a coding unit, which has a size different from that of the others, based on the determined widths and heights of the coding units **660***a*, **660***b*, and **660***c*. Referring to FIG. **6**, the image decoding apparatus **100** may determine the middle coding unit **660***b*, which has a size different from the size of the left and right coding units **660***a* and **660***c*, as the coding unit of the certain location.

**[0135]** The above-described process in which the image decoding apparatus **100** determines a coding unit having a size different from the size of the other coding units corresponds to an example of determining a coding unit at a certain location by using the sizes of coding units, which are determined based on coordinates of samples, and thus various processes of determining a coding unit at a certain location by comparing the sizes of coding units, which are determined based on coordinates of certain samples, may be used.

**[0136]** Locations of samples considered to determine locations of coding units are not limited to the above-described upper left locations, and information about arbitrary locations of samples included in the coding units may be used. According to an embodiment, the image decoding apparatus **100** may select a coding unit at a certain location from among an odd number of coding units determined by splitting the current coding unit, by considering the shape of the current coding unit. For example, when the current coding unit has a non-square shape, a width of which is longer than a height, the image decoding apparatus **100** may determine the coding unit at the certain location in a horizontal direction. That is, the image decoding apparatus **100** may determine one of coding units at different locations in a horizontal direction and may put a restriction on the coding unit. When the current coding unit has a non-square shape, a height of which is longer than a width, the image decoding apparatus **100** may determine the coding unit at the certain location in a vertical direction.

**[0137]** That is, the image decoding apparatus **100** may determine one of coding units at different locations in a vertical direction and may put a restriction on the coding unit. According to an embodiment, the image decoding apparatus **100** may use information indicating respective locations of an even number of coding units, to determine the coding unit at the certain location from among the even number of coding units. The image decoding apparatus **100** may determine an even number of coding units by splitting (binary splitting) the current coding unit, and may determine the coding unit at the certain location by using the information about the locations of the even number of coding units.

**[0138]** An operation related thereto may correspond to the operation of determining a coding unit at a certain location (e.g., a center location) from among an odd number of coding units, which has been described in detail above with reference to FIG. **6**, and thus detailed descriptions thereof will be omitted. According to an embodiment, when a non-square current coding unit is split into a plurality of coding units, certain information about a coding unit at a certain location may be used in a splitting operation to determine the coding unit at the certain location from among the plurality of coding units.

**[0139]** For example, the image decoding apparatus **100** may use at least one of block shape information and split shape mode information, which is stored in a sample included in a middle coding unit, in a splitting operation to determine a coding unit at a center location from among the plurality of coding units determined by splitting the current coding unit. Referring to FIG. **6**, the image decoding apparatus **100** may split the current coding unit **600** into the plurality of coding units **620***a*, **620***b*, and **620***c* based on the split shape mode information, and may determine the coding unit **620***b* at a center location from among the plurality of the coding units **620***a*, **620***b*, and **620***c*. Furthermore, the image decoding apparatus **100** may determine the coding unit **620***b* at the center location, in consideration of a location from which the split shape mode information is obtained. That is, the split shape mode information of the current coding unit **600** may be obtained from the sample **640** at a center location of the current coding unit **600** and, when the current coding unit **600** is split into the plurality of coding units **620***a*, **620***b*, and **620***c* based on the split shape mode information, the coding unit **620***b* including the sample **640** may be determined as the coding unit at the center location.

**[0140]** Information used to determine the coding unit at the center location is not limited to the split shape mode information, and various types of information may be used to determine the coding unit at the center location. According to an embodiment, certain information for identifying the coding unit at the certain location may be obtained from a certain sample included in a coding unit to be determined. Referring to FIG. **6**, the image decoding apparatus **100** may use the split shape mode information, which is obtained from a sample at a certain location in the current coding unit **600** (e.g., a sample at a center location of the current coding unit **600**) to determine a coding unit at a certain location from among the plurality of the coding units **620***a*, **620***b*, and **620***c* determined by splitting the current coding unit **600** (e.g., a coding unit at a center location from among a plurality of split coding units). That is, the image decoding apparatus **100** may determine the sample at the certain location by considering a block shape of the current coding unit **600**, may determine the coding unit **620***b* including a sample, from which certain information (e.g., the split shape mode information) may be obtained, from among the plurality of coding units **620***a*, **620***b*, and **620***c* determined by splitting the current coding unit **600**, and may put a certain restriction on the coding unit **620***b*. Referring to FIG. **6**, according to an embodiment, the image decoding apparatus **100** may determine the sample **640** at the center location of the current coding unit **600** as the sample from which the certain information may be obtained, and may put a certain restriction on the coding unit **620***b* including the sample **640**, in a decoding process.

**[0141]** The location of the sample from which the certain information may be obtained is not limited to the above-described location, and may include arbitrary locations of samples included in the coding unit **620***b* to be determined for a restriction. According to an embodiment, the location of the sample from which the certain information may be obtained may be determined based on the shape of the current coding unit **600**. According to an embodiment, the block shape information may indicate whether the current coding unit has a square or non-square shape, and the location of the sample from which the certain information may be obtained may be determined based on the shape. For example, the image decoding apparatus **100** may determine a sample located at a boundary for splitting at least one of a width and height of the current coding unit in half, as the sample from which the certain information may be obtained, by using at least one of information about the width of the current coding unit and information about the height of the current coding unit.

**[0142]** As another example, when the block shape information of the current coding unit indicates a non-square shape, the image decoding apparatus **100** may determine one of samples adjacent to a boundary for splitting a long side of the current coding unit in half, as the sample from which the certain information may be obtained. According to an embodiment, when the current coding unit is split into a plurality of coding units, the image decoding apparatus **100** may use the split shape mode information to determine a coding unit at a certain location from among the plurality of coding units. According to an embodiment, the image decoding apparatus **100** may obtain the split shape mode information from a sample at a certain location in a coding unit, and may split the plurality of coding units, which are generated by splitting the current coding unit, by using the split shape mode information, which is obtained from the sample of the certain location in each of the plurality of coding units. That is, a coding unit may be recursively split based on the split shape mode information, which is obtained from the sample at the certain location in each coding unit.

**[0143]** A process of recursively splitting a coding unit has been described above with reference to FIG. **5**, and thus detailed descriptions thereof will be omitted.

**[0144]** According to an embodiment, the image decoding apparatus **100** may determine one or more coding units by splitting the current coding unit, and may determine an order of decoding the one or more coding units, based on a certain block (e.g., the current coding unit).

**[0145]** FIG. **7** illustrates an order of processing a plurality of coding units when the image decoding apparatus determines the plurality of coding units by splitting a current coding unit, according to an embodiment.

**[0146]** According to an embodiment, the image decoding apparatus **100** may determine second coding units **710***a* and **710***b* by splitting a first coding unit **700** in a vertical direction, may determine second coding units **730***a* and **730***b* by splitting the first coding unit **700** in a horizontal direction, or may determine second coding units **750***a*, **750***b*, **750***c*, and **750***d* by splitting the first coding unit **700** in vertical and horizontal directions, based on split shape mode information. Referring to FIG. **7**, the image decoding apparatus **100** may determine to process the second coding units **710***a* and **710***b*, which are determined by splitting the first coding unit **700** in a vertical direction, in a horizontal direction order **710***c*. The image decoding apparatus **100** may determine to process the second coding units **730***a* and **730***b*, which are determined by splitting the first coding unit **700** in a horizontal direction, in a vertical direction order **730***c*.

**[0147]** The image decoding apparatus **100** may determine the second coding units **750***a*, **750***b*, **750***c*, and **750***d*, which are determined by splitting the first coding unit **700** in vertical and horizontal directions, according to a certain order (e.g., a raster scan order or Z-scan order **750***e*) by which coding units in a row are processed and then coding units in a next row are processed. According to an embodiment, the image decoding apparatus **100** may recursively split coding units. Referring to FIG. **7**, the image decoding apparatus **100** may determine the plurality of coding units **710***a* and **710***b*, **730***a* and **730***b*, or **750***a*, **750***b*, **750***c*, and **750***d* by splitting the first coding unit **700**, and may recursively split each of the determined plurality of coding units **710***a*, **710***b*, **730***a*, **730***b*, **750***a*, **750***b*, **750***c*, and **750***d*. A splitting method of the plurality of coding units **710***a* and **710***b*, **730***a* and **730***b*, or **750***a*, **750***b*, **750***c*, and **750***d* may correspond to a splitting method of the first coding unit **700**. Accordingly, each of the plurality of coding units **710***a* and **710***b*, **730***a* and **730***b*, or **750***a*, **750***b*, **750***c*, and **750***d* may be independently split into a plurality of coding units.

**[0148]** Referring to FIG. **7**, the image decoding apparatus **100** may determine the second coding units **710***a* and **710***b* by splitting the first coding unit **700** in a vertical direction, and may determine to independently split or not to split each of the second coding units **710***a* and **710***b*.

**[0149]** According to an embodiment, the image decoding apparatus **100** may determine third coding units **720***a* and **720***b* by splitting the left second coding unit **710***a* in a horizontal direction, and may not split the right second coding unit **710***b*. According to an embodiment, a processing order of coding units may be determined based on a process of splitting a coding unit. In other words, a processing order of split coding units may be determined based on a processing order of coding units immediately before being split. The image decoding apparatus **100** may determine a processing order of the third coding units **720***a* and **720***b* determined by splitting the left second coding unit **710***a*, independently of the right second coding unit **710***b*. Because the third coding units **720***a* and **720***b* are determined by splitting the left second coding unit **710***a* in a horizontal direction, the third coding units **720***a* and **720***b* may be processed in a vertical direction order **720***c*. Because the left and right second coding units **710***a* and **710***b* are processed in the horizontal direction order **710***c*, the right second coding unit **710***b* may be processed after the third coding units **720***a* and **720***b* included in the left second coding unit **710***a* are processed in the vertical direction order **720***c*.

**[0150]** A process of determining a processing order of coding units based on a coding unit before being split is not limited to the above-described example, and various methods may be used to independently process coding units, which are split and determined to various shapes, in a certain order.

**[0151]** FIG. **8** illustrates a process in which the image decoding apparatus determines that a current coding unit is to be split into an odd number of coding units, when the coding units are not processable in a certain order, according to an embodiment. According to an embodiment, the image decoding apparatus **100** may determine that the current coding unit is to be split into an odd number of coding units, based on obtained split shape mode information. Referring to FIG. **8**, a square first coding unit **800** may be split into non-square second coding units **810***a* and **810***b*, and the second coding units **810***a* and **810***b* may be independently split into third coding units **820***a* and **820***b*, and **820***c*, **820***d*, and **820***e*.

**[0152]** According to an embodiment, the image decoding apparatus **100** may determine the plurality of third coding units **820***a* and **820***b* by splitting the left second coding unit **810***a* in a horizontal direction, and may split the right second coding unit **810***b* into the odd number of third coding units **820***c*, **820***d*, and **820***e*. According to an embodiment, the image decoding apparatus **100** may determine whether any coding unit is split into an odd number of coding units, by determining whether the third coding units **820***a* and **820***b*, and **820***c*, **820***d*, and **820***e* are processable in a certain order. Referring to FIG. **8**, the image decoding apparatus **100** may determine the third coding units **820***a* and **820***b*, and **820***c*, **820***d*, and **820***e* by recursively splitting the first coding unit **800**. The image decoding apparatus **100** may determine whether any of the first coding unit **800**, the second coding units **810***a* and **810***b*, or the third coding units **820***a* and **820***b*, and **820***c*, **820***d*, and **820***e* are split into an odd number of coding units, based on at least one of the block shape information and the split shape mode information. For example, a right coding unit from among the second coding units **810***a* and **810***b* may be split into an odd number of third coding units **820***c*, **820***d*, and **820***e*.

**[0153]** A processing order of a plurality of coding units included in the first coding unit **800** may be a certain order (e.g., a Z-scan order **830**), and the image decoding apparatus **100** may determine whether the third coding units **820***c*, **820***d*, and **820***e*, which are determined by splitting the right second coding unit **810***b* into an odd number of coding units, satisfy a condition for processing in the certain order. According to an embodiment, the image decoding apparatus **100** may determine whether the third coding units **820***a* and **820***b*, and **820***c*, **820***d*, and **820***e* included in the first coding unit **800** satisfy the condition for processing in the certain order, and the condition relates to whether at least one of a width and height of the second coding units **810***a* and **810***b* is to be split in half along a boundary of the third coding units **820***a* and **820***b*, and **820***c*, **820***d*, and **820***e*. For example, the third coding units **820***a* and **820***b* determined when the height of the left second coding unit **810***a* of the non-square shape is split in half may satisfy the condition. It may be determined that the third coding units **820***c*, **820***d*, and **820***e* do not satisfy the condition because the boundaries of the third coding units **820***c*, **820***d*, and **820***e* determined when the right second coding unit **810***b* is split into three coding units are unable to split the width or height of the right second coding unit **810***b* in half. When the condition is not satisfied as described above, the image decoding apparatus **100** may decide disconnection of a scan order, and may determine that the right second coding unit **810***b* is to be split into an odd number of coding units, based on a result of the decision.

**[0154]** According to an embodiment, when a coding unit is split into an odd number of coding units, the image decoding apparatus **100** may put a certain restriction on a coding unit at a certain location from among the split coding units.

**[0155]** The restriction or the certain location has been described above through various embodiments, and thus detailed descriptions thereof will be omitted. FIG. **9** illustrates a process in which the image decoding apparatus determines at least one coding unit by splitting a first coding unit, according to an embodiment. According to an embodiment, the image decoding apparatus **100** may split the first coding unit **900**, based on split shape mode information, which is obtained by the receiver **110**. The square first coding unit **900** may be split into four square coding units, or may be split into a plurality of non-square coding units.

**[0156]** For example, referring to FIG. **9**, when the first coding unit **900** has a square shape and the split shape mode information indicates to split the first coding unit **900** into non-square coding units, the image decoding apparatus **100** may split the first coding unit **900** into a plurality of non-square coding units. In detail, when the split shape mode information indicates to determine an odd number of coding units by splitting the first coding unit **900** in a horizontal direction or a vertical direction, the image decoding apparatus **100** may split the square first coding unit **900** into an odd number of coding units, e.g., second coding units **910***a*, **910***b*, and **910***c* determined by splitting the square first coding unit **900** in a vertical direction or second coding units **920***a*, **920***b*, and **920***c* determined by splitting the square first coding unit **900** in a horizontal direction. According to an embodiment, the image decoding apparatus **100** may determine whether the second coding units **910***a*, **910***b*, **910***c*, **920***a*, **920***b*, and **920***c* included in the first coding unit **900** satisfy a condition for processing in a certain order, and the condition relates to whether at least one of a width and height of the first coding unit **900** is to be split in half along a boundary of the second coding units **910***a*, **910***b*, **910***c*, **920***a*, **920***b*, and **920***c*. Referring to FIG. **9**, because boundaries of the second coding units **910***a*, **910***b*, and **910***c* determined by splitting the square first coding unit **900** in a vertical direction do not split the width of the first coding unit **900** in half, it may be determined that the first coding unit **900** does not satisfy the condition for processing in the certain order. Also, because boundaries of the second coding units **920***a*, **920***b*, and **920***c* determined by splitting the square first coding unit **900** in a horizontal direction do not split the height of the first coding unit **900** in half, it may be determined that the first coding unit **900** does not satisfy the condition for processing in the certain order.

**[0157]** When the condition is not satisfied as described above, the image decoding apparatus **100** may decide disconnection of a scan order, and may determine that the first coding unit **900** is to be split into an odd number of coding units, based on a result of the decision.

**[0158]** According to an embodiment, when a coding unit is split into an odd number of coding units, the image decoding apparatus **100** may put a certain restriction on a coding unit at a certain location from among the split coding units.

**[0159]** The restriction or the certain location has been described above through various embodiments, and thus detailed descriptions thereof will be omitted.

**[0160]** According to an embodiment, the image decoding apparatus **100** may determine various-shaped coding units by splitting a first coding unit. Referring to FIG. **9**, the image decoding apparatus **100** may split the square first coding unit **900** or a non-square first coding unit **930** or **950** into various-shaped coding units. FIG. **10** illustrates that a shape into which a second coding unit is splittable is restricted when the second coding unit having a non-square shape, which is determined when the image decoding apparatus splits a first coding unit, satisfies a certain condition, according to an embodiment. According to an embodiment, the image decoding apparatus **100** may determine to split the square first coding unit **1000** into non-square second coding units **1010***a*, and **1010***b* or **1020***a* and **1020***b*, based on split shape mode information, which is obtained by the receiver **110**. The second coding units **1010***a* and **1010***b*, or **1020***a* and **1020***b* may be independently split. Accordingly, the image decoding apparatus **100** may determine to split or not to split each of the second coding units **1010***a* and **1010***b*, or **1020***a* and **1020***b* into a plurality of coding units, based on the split shape mode information of each of the second coding units **1010***a* and **1010***b*, or **1020***a* and **1020***b*. According to an embodiment, the image decoding apparatus **100** may determine third coding units **1012***a* and **1012***b* by splitting the non-square left second coding unit **1010***a*, which is determined by splitting the first coding unit **1000** in a vertical direction, in a horizontal direction.

**[0161]** When the left second coding unit **1010***a* is split in a horizontal direction, the image decoding apparatus **100** may restrict the right second coding unit **1010***b* not to be split in a horizontal direction in which the left second coding unit **1010***a* is split. When third coding units **1014***a* and **1014***b* are determined by splitting the right second coding unit **1010***b* in the same direction, because the left and right second coding units **1010***a* and **1010***b* are independently split in a horizontal direction, the third coding units **1012***a*, **1012***b*, **1014***a*, and **1014***b* may be determined.

**[0162]** This case serves equally as a case in which the image decoding apparatus **100** splits the first coding unit **1000** into four square second coding units **1030***a*, **1030***b*, **1030***c*, and **1030***d*, based on the split shape mode information, and may be inefficient in terms of image decoding.

**[0163]** According to an embodiment, the image decoding apparatus **100** may determine third coding units **1022***a* and **1022***b*, or **1024***a* and **1024***b* by splitting the non-square second coding unit **1020***a* or **1020***b*, which is determined by splitting the first coding unit **1000** in a horizontal direction, in a vertical direction. When a second coding unit (e.g., the upper second coding unit **1020***a*) is split in a vertical direction, for the above-described reason, the image decoding apparatus **100** may restrict the other second coding unit (e.g., the lower second coding unit **1020***b*) not to be split in a vertical direction in which the upper second coding unit **1020***a* is split. FIG. **11** illustrates a process in which the image decoding apparatus splits a square coding unit when split shape mode information indicates that the square coding unit is not to be split into four square coding units, according to an embodiment. According to an embodiment, the image decoding apparatus **100** may determine second coding units **1110***a* and **1110***b*, or **1120***a* and **1120***b*, for example. by splitting a first coding unit **1100**, based on split shape mode information.

**[0164]** The split shape mode information may include information about various methods of splitting a coding unit but, the information about various splitting methods may not include information for splitting a coding unit into four square coding units. According to such split shape mode information, the image decoding apparatus **100** may not split the square first coding unit **1100** into four square second coding units **1130***a*, **1130***b*, **1130***c*, and **1130***d*.

**[0165]** The image decoding apparatus **100** may determine the non-square second coding units **1110***a* and **1110***b*, or **1120***a* and **1120***b*, for example, based on the split shape mode information. According to an embodiment, the image decoding apparatus **100** may independently split the non-square second coding units **1110***a* and **1110***b*, or **1120***a* and **1120***b*, for example. Each of the second coding units **1110***a* and **1110***b*, or **1120***a* and **1120***b*, for example, may be recursively split in a certain order, and this splitting method may correspond to a method of splitting the first coding unit **1100**, based on the split shape mode information. For example, the image decoding apparatus **100** may determine square third coding units **1112***a* and **1112***b* by splitting the left second coding unit **1110***a* in a horizontal direction, and may determine square third coding units **1114***a* and **1114***b* by splitting the right second coding unit **1110***b* in a horizontal direction.

**[0166]** Furthermore, the image decoding apparatus **100** may determine square third coding units **1116***a*, **1116***b*, **1116***c*, and **1116***d* by splitting both of the left and right second coding units **1110***a* and **1110***b* in a horizontal direction. In this case, coding units having the same shape as the four square second coding units **1130***a*, **1130***b*, **1130***c*, and **1130***d* split from the first coding unit **1100** may be determined. As another example, the image decoding apparatus **100** may determine square third coding units **1122***a* and **1122***b* by splitting the upper second coding unit **1120***a* in a vertical direction, and may determine square third coding units **1124***a* and **1124***b* by splitting the lower second coding unit **1120***b* in a vertical direction.

**[0167]** Furthermore, the image decoding apparatus **100** may determine square third coding units **1126***a*, **1126***b*, **1126***c*, and **1126***d* by splitting both the upper and lower second coding units **1120***a* and **1120***b* in a vertical direction.

**[0168]** In this case, coding units having the same shape as the four square second coding units **1130***a*, **1130***b*, **1130***c*, and **1130***d* split from the first coding unit **1100** may be determined. FIG. **12** illustrates that a processing order between a plurality of coding units may vary according to a process of splitting a coding unit, according to an embodiment. According to an embodiment, the image decoding apparatus **100** may split a first coding unit **1200**, based on split shape mode information. When a block shape indicates a square shape and the split shape mode information indicates to split the first coding unit **1200** in at least one of horizontal and vertical directions, the image decoding apparatus **100** may determine second coding units (e.g., **1210***a* and **1210***b*, or **1220***a* and **1220***b*, for example.) by splitting the first coding unit **1200**.

**[0169]** Referring to FIG. **12**, the non-square second coding units **1210***a* and **1210***b*, or **1220***a* and **1220***b* determined by splitting the first coding unit **1200** in a horizontal direction or vertical direction may be independently split based on the split shape mode information of each coding unit. For example, the image decoding apparatus **100** may determine third coding units **1216***a*, **1216***b*, **1216***c*, and **1216***d* by splitting the second coding units **1210***a* and **1210***b*, which are generated by splitting the first coding unit **1200** in a vertical direction, in a horizontal direction, and may determine third coding units **1226***a*, **1226***b*, **1226***c*, and **1226***d* by splitting the second coding units **1220***a* and **1220***b*, which are generated by splitting the first coding unit **1200** in a horizontal direction, in a vertical direction. A process of splitting the second coding units **1210***a* and **1210***b*, or **1220***a* and **1220***b* has been described above with reference to FIG. **11**, and thus detailed descriptions thereof will be omitted. According to an embodiment, the image decoding apparatus **100** may process coding units in a certain order.

**[0170]** An operation of processing coding units in a certain order has been described above with reference to FIG. **7**, and thus detailed descriptions thereof will be omitted.

**[0171]** Referring to FIG. **12**, the image decoding apparatus **100** may determine four square third coding units **1216***a*, **1216***b*, **1216***c*, and **1216***d*, and **1226***a*, **1226***b*, **1226***c*, and **1226***d* by splitting the square first coding unit **1200**.

**[0172]** According to an embodiment, the image decoding apparatus **100** may determine processing orders of the third coding units **1216***a*, **1216***b*, **1216***c*, and **1216***d*, and **1226***a*, **1226***b*, **1226***c*, and **1226***d* based on a split shape into which the first coding unit **1200** is split. According to an embodiment, the image decoding apparatus **100** may determine the third coding units **1216***a*, **1216***b*, **1216***c*, and **1216***d* by splitting the second coding units **1210***a* and **1210***b* generated by splitting the first coding unit **1200** in a vertical direction, in a horizontal direction, and may process the third coding units **1216***a*, **1216***b*, **1216***c*, and **1216***d* in a processing order **1217** for initially processing the third coding units **1216***a* and **1216***c*, which are included in the left second coding unit **1210***a*, in a vertical direction and then processing the third coding unit **1216***b* and **1216***d*, which are included in the right second coding unit **1210***b*, in a vertical direction. According to an embodiment, the image decoding apparatus **100** may determine the third coding units **1226***a*, **1226***b*, **1226***c*, and **1226***d* by splitting the second coding units **1220***a* and **1220***b* generated by splitting the first coding unit **1200** in a horizontal direction, in a vertical direction, and may process the third coding units **1226***a*, **1226***b*, **1226***c*, and **1226***d* in a processing order **1227** for initially processing the third coding units **1226***a* and **1226***b*, which are included in the upper second coding unit **1220***a*, in a horizontal direction and then processing the third coding unit **1226***c* and **1226***d*, which are included in the lower second coding unit **1220***b*, in a horizontal direction.

**[0173]** Referring to FIG. **12**, the square third coding units **1216***a*, **1216***b*, **1216***c*, and **1216***d*, and **1226***a*, **1226***b*, **1226***c*, and **1226***d* may be determined by splitting the second coding units **1210***a* and **1210***b*, and **1220***a* and **1220***b*, respectively.

**[0174]** Although the second coding units **1210***a* and **1210***b* are determined by splitting the first coding unit **1200** in a vertical direction differently from the second coding units **1220***a* and **1220***b* which are determined by splitting the first coding unit **1200** in a horizontal direction, the third coding units **1216***a*, **1216***b*, **1216***c*, and **1216***d*, and **1226***a*, **1226***b*, **1226***c*, and **1226***d* split therefrom eventually show same-shaped coding units split from the first coding unit **1200**. As such, by recursively splitting a coding unit in different manners based on the split shape mode information, the image decoding apparatus **100** may process a plurality of coding units in different orders even when the coding units are eventually determined to have the same shape. FIG. **13** illustrates a process of determining a depth of a coding unit as a shape and a size of the coding unit change, when the coding unit is recursively split to determine a plurality of coding units, according to an embodiment. According to an embodiment, the image decoding apparatus **100** may determine a depth of a coding unit, based on a certain criterion.

**[0175]** For example, the certain criterion may be the length of a long side of the coding unit. When the length of a long side of a coding unit before being split is 2n times (n>0) the length of a long side of a split current coding unit, the image decoding apparatus **100** may determine that a depth of the current coding unit is increased from a depth of the coding unit before being split, by n. In the following descriptions, a coding unit having an increased depth is represented as a coding unit of a lower depth. Referring to FIG. **13**, according to an embodiment, the image decoding apparatus **100** may determine a second coding unit **1302** and a third coding unit **1304** of lower depths by splitting a square first coding unit **1300** based on block shape information indicating a square shape (e.g., the block shape information may be represented as '0: SQUARE'). Assuming that the size of the square first coding unit **1300** is 2N×2N, the second coding unit **1302** determined by splitting a width and height of the first coding unit **1300** in ½ may have a size of N×N.

**[0176]** Furthermore, the third coding unit **1304** determined by splitting a width and height of the second coding unit **1302** in ½ may have a size of N/2×N/2.

**[0177]** In this case, a width and height of the third coding unit **1304** are ¼ times those of the first coding unit **1300**. When a depth of the first coding unit **1300** is D, a depth of the second coding unit **1302**, the width and height of which are ½ times those of the first coding unit **1300**, may be D+1, and a depth of the third coding unit **1304**, the width and height of which are ¼ times those of the first coding unit **1300**, may be D+2.

**[0178]** According to an embodiment, the image decoding apparatus **100** may determine a second coding unit **1312** or **1322** and a third coding unit **1314** or **1324** of lower depths by splitting a non-square first coding unit **1310** or **1320** based on block shape information indicating a non-square shape (e.g., the block shape information may be represented as '1: NS_VER' indicating a non-square shape, a height of which is longer than a width, or as '2: NS_HOR' indicating a non-square shape, a width of which is longer than a height). The image decoding apparatus **100** may determine the second coding unit **1302**, **1312**, or **1322** by splitting at least one of a width and height of the first coding unit **1310** having a size of N×2N.

**[0179]** That is, the image decoding apparatus **100** may determine the second coding unit **1302** having a size of N×N or the second coding unit **1322** having a size of N×N/2 by splitting the first coding unit **1310** in a horizontal direction, or may determine the second coding unit **1312** having a size of N/2×N by splitting the first coding unit **1310** in horizontal and vertical directions. According to an embodiment, the image decoding apparatus **100** may determine a second coding unit (e.g., **1302**, **1312**, or **1322**) by splitting at least one of a width and height of the first coding unit **1320** having a size of 2N×N.

**[0180]** That is, the image decoding apparatus **100** may determine the second coding unit **1302** having a size of N×N or the second coding unit **1312** having a size of N/2×N by splitting the first coding unit **1320** in a vertical direction, or may determine the second coding unit **1322** having a size of N×N/2 by splitting the first coding unit **1320** in horizontal and vertical directions. According to an embodiment, the image decoding apparatus **100** may determine a third coding unit (e.g., **1304**, **1314**, or **1324**) by splitting at least one of a width and height of the second coding unit **1302** having a size of N×N.

**[0181]** That is, the image decoding apparatus **100** may determine the third coding unit **1304** having a size of N/2×N/2, the third coding unit **1314** having a size of N/4×N/2, or the third coding unit **1324** having a size of N/2×N/4 by splitting the second coding unit **1302** in vertical and horizontal directions. According to an embodiment, the image decoding apparatus **100** may determine a third coding unit (e.g., **1304**, **1314**, or **1324**) by splitting at least one of a width and height of the second coding unit **1312** having a size of N/2×N.

**[0182]** That is, the image decoding apparatus **100** may determine the third coding unit **1304** having a size of N/2×N/2 or the third coding unit **1324** having a size of N/2×N/4 by splitting the second coding unit **1312** in a horizontal direction, or may determine the third coding unit **1314** having a size of N/4×N/2 by splitting the second coding unit **1312** in vertical and horizontal directions. According to an embodiment, the image decoding apparatus **100** may determine a third coding unit (e.g., **1304**, **1314**, or **1324**) by splitting at least one of a width and height of the second coding unit **1322** having a size of N×N/2. That is, the image decoding apparatus **100** may determine the third coding unit **1304** having a size of N/2×N/2 or the third coding unit **1314** having a size of N/4×N/2 by splitting the second coding unit **1322** in a vertical direction, or may determine the third coding unit **1324** having a size of N/2×N/4 by splitting the second coding unit **1322** in vertical and horizontal directions.

**[0183]** According to an embodiment, the image decoding apparatus **100** may split a square coding unit (e.g., **1300**, **1302**, or **1304**) in a horizontal or vertical direction. For example, the image decoding apparatus **100** may determine the first coding unit **1310** having a size of N×2N by splitting the first coding unit **1300** having a size of 2N×2N in a vertical direction, or may determine the first coding unit **1320** having a size of 2N×N by splitting the first coding unit **1300** in a horizontal direction.

**[0184]** According to an embodiment, when a depth is determined based on the length of a longest side of a coding unit, a depth of a coding unit determined by splitting the first coding unit **1300** having a size of 2N×2N in a horizontal or vertical direction may be the same as the depth of the first coding unit **1300**.

**[0185]** According to an embodiment, a width and height of the third coding unit **1314** or **1324** may be ¼ times those of the first coding unit **1310** or **1320**. When a depth of the first coding unit **1310** or **1320** is D, a depth of the second coding unit **1312** or **1322**, the width and height of which are ½ times those of the first coding unit **1310** or **1320**, may be D+1, and a depth of the third coding unit **1314** or **1324**, the width and height of which are ¼ times those of the first coding unit **1310** or **1320**, may be D+2. FIG. **14** illustrates depths that are determinable based on shapes and sizes of coding units, and part indexes (PIDs) that are for distinguishing the coding units, according to an embodiment.

**[0186]** According to an embodiment, the image decoding apparatus **100** may determine various-shape second coding units by splitting a square first coding unit **1400**. Referring to FIG. **14**, the image decoding apparatus **100** may determine second coding units **1402***a* and **1402***b*, **1404***a* and **1404***b*, and **1406***a*, **1406***b*, **1406***c*, and **1406***d* by splitting the first coding unit **1400** in at least one of vertical and horizontal directions based on split shape mode information. That is, the image decoding apparatus **100** may determine the second coding units **1402***a* and **1402***b*, **1404***a* and **1404***b*, and **1406***a*, **1406***b*, **1406***c*, and **1406***d*, based on the split shape mode information of the first coding unit **1400**.

**[0187]** According to an embodiment, depths of the second coding units **1402***a* and **1402***b*, **1404***a* and **1404***b*, and **1406***a*, **1406***b*, **1406***c*, and **1406***d* that are determined based on the split shape mode information of the square first coding unit **1400** may be determined based on the length of a long side thereof. For example, because the length of a side of the square first coding unit **1400** equals the length of a long side of the non-square second coding units **1402***a* and **1402***b*, and **1404***a* and **1404***b*, the first coding unit **1400** and the non-square second coding units **1402***a* and **1402***b*, and **1404***a* and **1404***b* may have the same depth, e.g., D.

**[0188]** When the image decoding apparatus **100** splits the first coding unit **1400** into the four square second coding units **1406***a*, **1406***b*, **1406***c*, and **1406***d* based on the split shape mode information, because the length of a side of the square second coding units **1406***a*, **1406***b*, **1406***c*, and **1406***d* is ½ times the length of a side of the first coding unit **1400**, a depth of the second coding units **1406***a*, **1406***b*, **1406***c*, and **1406***d* may be D+1 which is deeper than the depth D of the first coding unit **1400** by 1. According to an embodiment, the image decoding apparatus **100** may determine a plurality of second coding units **1412***a* and **1412***b*, and **1414***a*, **1414***b*, and **1414***c* by splitting a first coding unit **1410**, a height of which is longer than a width, in a horizontal direction based on the split shape mode information. According to an embodiment, the image decoding apparatus **100** may determine a plurality of second coding units **1422***a* and **1422***b*, and **1424***a*, **1424***b*, and **1424***c* by splitting a first coding unit **1420**, a width of which is longer than a height, in a vertical direction based on the split shape mode information.

**[0189]** According to an embodiment, a depth of the second coding units **1412***a* and **1412***b*, and **1414***a*, **1414***b*, and **1414***c*, or **1422***a* and **1422***b*, and **1424***a*, **1424***b*, and **1424***c*, which are determined based on the split shape mode information of the non-square first coding unit **1410** or **1420**, may be determined based on the length of a long side thereof. For example, because the length of a side of the square second coding units **1412***a* and **1412***b* is ½ times the length of a long side of the first coding unit **1410** having a non-square shape, a height of which is longer than a width, a depth of the square second coding units **1412***a* and **1412***b* is D+1 which is deeper than the depth D of the non-square first coding unit **1410** by 1. Furthermore, the image decoding apparatus **100** may split the non-square first coding unit **1410** into an odd number of second coding units **1414***a*, **1414***b*, and **1414***c* based on the split shape mode information. The odd number of second coding units **1414***a*, **1414***b*, and **1414***c* may include the non-square second coding units **1414***a* and **1414***c* and the square second coding unit **1414***b*.

**[0190]** In this case, because the length of a long side of the non-square second coding units **1414***a* and **1414***c* and the length of a side of the square second coding unit **1414***b* are ½ times the length of a long side of the first coding unit **1410**, a depth of the second coding units **1414***a*, **1414***b*, and **1414***c* may be D+1 which is deeper than the depth D of the non-square first coding unit **1410** by 1. The image decoding apparatus **100** may determine depths of coding units split from the first coding unit **1420** having a non-square shape, a width of which is longer than a height, by using the above-described method of determining depths of coding units split from the first coding unit **1410**. According to an embodiment, the image decoding apparatus **100** may determine PIDs for identifying split coding units, based on a size ratio between the coding units when an odd number of split coding units do not have the same size. Referring to FIG. **14**, the coding unit **1414***b* of a center location among an odd number of split coding units **1414***a*, **1414***b*, and **1414***c* may have a width equal to that of the other coding units **1414***a* and **1414***c* and a height which is two times that of the other coding units **1414***a* and **1414***c*. That is, in this case, the coding unit **1414***b* at the center location may include two of the other coding unit **1414***a* or **1414***c*. Accordingly, when a PID of the coding unit **1414***b* at the center location is 1 based on a scan order, a PID of the coding unit **1414***c* located next to the coding unit **1414***b* may be increased by 2 and thus may be 3.

**[0191]** That is, discontinuity in PID values may be present. According to an embodiment, the image decoding apparatus **100** may determine whether an odd number of split coding units do not have the same size, based on whether discontinuity is present in PIDs for identifying the split coding units. According to an embodiment, the image decoding apparatus **100** may determine whether to use a specific splitting method, based on PID values for identifying a plurality of coding units determined by splitting a current coding unit. Referring to FIG. **14**, the image decoding apparatus **100** may determine an even number of coding units **1412***a* and **1412***b* or an odd number of coding units **1414***a*, **1414***b*, and **1414***c* by splitting the first coding unit **1410** having a rectangular shape, a height of which is longer than a width.

**[0192]** The image decoding apparatus **100** may use PIDs indicating respective coding units so as to identify the respective coding units. According to an embodiment, the PID may be obtained from a sample at a certain location of each coding unit (e.g., an upper left sample). According to an embodiment, the image decoding apparatus **100** may determine a coding unit at a certain location from among the split coding units, by using the PIDs for distinguishing the coding units. According to an embodiment, when the split shape mode information of the first coding unit **1410** having a rectangular shape, a height of which is longer than a width, indicates to split a coding unit into three coding units, the image decoding apparatus **100** may split the first coding unit **1410** into three coding units **1414***a*, **1414***b*, and **1414***c*. The image decoding apparatus **100** may assign a PID to each of the three coding units **1414***a*, **1414***b*, and **1414***c*. The image decoding apparatus **100** may compare PIDs of an odd number of split coding units to determine a coding unit at a center location from among the coding units. The image decoding apparatus **100** may determine the coding unit **1414***b* having a PID corresponding to a middle value among the PIDs of the coding units, as the coding unit at the center location from among the coding units determined by splitting the first coding unit **1410**. According to an embodiment, the image decoding apparatus **100** may determine PIDs for distinguishing split coding units, based on a size ratio between the coding units when the split coding units do not have the same size. Referring to FIG. **14**, the coding unit **1414***b* generated by splitting the first coding unit **1410** may have a width equal to that of the other coding units **1414***a* and **1414***c* and a height which is two times that of the other coding units **1414***a* and **1414***c*. In this case, when the PID of the coding unit **1414***b* at the center location is 1, the PID of the coding unit **1414***c* located next to the coding unit **1414***b* may be increased by 2 and thus may be 3. When the PID is not uniformly increased as described above, the image decoding apparatus **100** may determine that a coding unit is split into a plurality of coding units including a coding unit having a size different from that of the other coding units.

**[0193]** According to an embodiment, when the split shape mode information indicates to split a coding unit into an odd number of coding units, the image decoding apparatus **100** may split a current coding unit in such a manner that a coding unit of a certain location among an odd number of coding units (e.g., a coding unit of a center location) has a size different from that of the other coding units.

**[0194]** In this case, the image decoding apparatus **100** may determine the coding unit of the center location, which has a different size, by using PIDs of the coding units.

**[0195]** The PIDs and the size or location of the coding unit of the certain location to be determined are not limited to the above-described examples, and various PIDs and various locations and sizes of coding units may be used. According to an embodiment, the image decoding apparatus **100** may use a certain data unit where a coding unit starts to be recursively split. FIG. **15** illustrates that a plurality of coding units are determined based on a plurality of certain data units included in a picture, according to an embodiment.

**[0196]** According to an embodiment, a certain data unit may be defined as a data unit where a coding unit starts to be recursively split by using split shape mode information. That is, the certain data unit may correspond to a coding unit of an uppermost depth, which is used to determine a plurality of coding units split from a current picture. In the following descriptions, for convenience of explanation, the certain data unit is referred to as a reference data unit. According to an embodiment, the reference data unit may have a certain size and a certain shape.

**[0197]** According to an embodiment, a reference coding unit may include M×N samples. In some embodiments, M and N may be equal to each other, and may be integers represented as powers of 2. That is, the reference data unit may have a square or non-square shape, and may be split into an integer number of coding units.

**[0198]** According to an embodiment, the image decoding apparatus **100** may split the current picture into a plurality of reference data units. According to an embodiment, the image decoding apparatus **100** may split the plurality of reference data units, which are split from the current picture, by using the split shape mode information of each reference data unit.

**[0199]** The process of splitting the reference data unit may correspond to a splitting process using a quadtree structure. According to an embodiment, the image decoding apparatus **100** may pre-determine a minimum size allowed for the reference data units included in the current picture.

**[0200]** Accordingly, the image decoding apparatus **100** may determine reference data units having various sizes equal to or greater than the minimum size, and may determine one or more coding units by using the split shape mode information with reference to the determined reference data unit. Referring to FIG. **15**, the image decoding apparatus **100** may use a square reference coding unit **1500** or a non-square reference coding unit **1502**.

**[0201]** According to an embodiment, the shape and size of reference coding units may be determined based on various data units capable of including one or more reference coding units (e.g., sequences, pictures, slices, slice segments, tiles, tile groups, largest coding units, or the like). According to an embodiment, the receiver **110** of the image decoding apparatus **100** may obtain, from a bitstream, at least one of reference coding unit shape information and reference coding unit size information for each of the various data units. A process of splitting the square reference coding unit **1500** into one or more coding units has been described above in relation to the process of splitting the current coding unit **300** of FIG. **3**, and a process of splitting the non-square reference coding unit **1502** into one or more coding units has been described above in relation to the process of splitting the current coding unit **400** or **450** of FIG. **4**, and thus detailed descriptions thereof will be omitted. According to an embodiment, the image decoding apparatus **100** may use a PID for identifying the size and shape of reference coding units, to determine the size and shape of reference coding units according to some data units pre-determined based on a certain condition. That is, the receiver **110** may obtain, from the bitstream, the PID for identifying the size and shape of reference coding units with respect to each slice, slice segment, tile, tile group, or largest coding unit which is a data unit satisfying a certain condition (e.g., a data unit having a size equal to or smaller than a slice) among the various data units (e.g., sequences, pictures, slices, slice segments, tiles, tile groups, largest coding units, or the like). The image decoding apparatus **100** may determine the size and shape of reference data units for each data unit, which satisfies the certain condition, by using the PID.

**[0202]** When the reference coding unit shape information and the reference coding unit size information are obtained and used from the bitstream according to each data unit having a relatively small size, the efficiency of using the bitstream may not be high, and therefore, the PID may be obtained and used instead of directly obtaining the reference coding unit shape information and the reference coding unit size information. In this case, at least one of the size and shape of reference coding units corresponding to the PID for identifying the size and shape of reference coding units may be pre-determined. That is, the image decoding apparatus **100** may determine at least one of the size and shape of reference coding units included in a data unit serving as a unit for obtaining the PID, by selecting the pre-determined at least one of the size and shape of reference coding units based on the PID. According to an embodiment, the image decoding apparatus **100** may use one or more reference coding units included in a largest coding unit. That is, a largest coding unit split from a picture may include one or more reference coding units, and coding units may be determined by recursively splitting each reference coding unit.

**[0203]** According to an embodiment, at least one of a width and height of the largest coding unit may be integer times at least one of the width and height of the reference coding units. According to an embodiment, the size of reference coding units may be obtained by splitting the largest coding unit n times based on a quadtree structure. That is, the image decoding apparatus **100** may determine the reference coding units by splitting the largest coding unit n times based on a quadtree structure, and may split the reference coding unit based on at least one of the block shape information and the split shape mode information according to various embodiments. According to an embodiment, the image decoding apparatus **100** may obtain block shape information indicating the shape of a current coding unit or split shape mode information indicating a splitting method of the current coding unit, from the bitstream, and may use the obtained information.

**[0204]** The split shape mode information may be included in the bitstream related to various data units.

**[0205]** For example, the image decoding apparatus **100** may use the split shape mode information included in a sequence parameter set, a picture parameter set, a video parameter set, a slice header, a slice segment header, a tile header, or a tile group header. Furthermore, the image decoding apparatus **100** may obtain, from the bitstream, a syntax information (e.g., syntax element) corresponding to the block shape information or the split shape mode information according to each largest coding unit or each reference coding unit or a processing block, and may use the obtained syntax element. a method of determining a split rule according to an embodiment of the disclosure will be described in detail. The image decoding apparatus **100** may determine a split rule of an image. The split rule may be pre-determined between the image decoding apparatus **100** and the image encoding apparatus **200**.

**[0206]** The image decoding apparatus **100** may determine the split rule of the image, based on information obtained from a bitstream. The image decoding apparatus **100** may determine the split rule based on the information obtained from at least one of a sequence parameter set, a picture parameter set, a video parameter set, a slice header, a slice segment header, a tile header, and a tile group header. The image decoding apparatus **100** may determine the split rule differently according to frames, slices, tiles, temporal layers, largest coding units, or coding units. The image decoding apparatus **100** may determine the split rule based on a block shape of a coding unit. The block shape may include a size, a shape, a ratio of width and height, and a direction of the coding unit.

**[0207]** The image decoding apparatus **100** may pre-determine to determine the split rule based on the block shape of the coding unit. An embodiment is not limited thereto. The image decoding apparatus **100** may determine the split rule based on the information obtained from the received bitstream. The shape of the coding unit may include a square and a non-square.

**[0208]** When the lengths of the width and height of the coding unit are the same, the image decoding apparatus **100** may determine the shape of the coding unit to be a square. Also, when the lengths of the width and height of the coding unit are not the same, the image decoding apparatus **100** may determine the shape of the coding unit to be a non-square. The size of the coding unit may include various sizes, such as 4×4, 8×4, 4×8, 8×8, 16×4, 16×8, . . . and 256×256. The size of the coding unit may be classified based on the length of a long side of the coding unit, the length of a short side, or the area. The image decoding apparatus **100** may apply the same split rule to coding units classified as the same group. For example, the image decoding apparatus **100** may classify coding units having the same lengths of long sides as having the same size.

**[0209]** Also, the image decoding apparatus **100** may apply the same split rule to coding units having the same lengths of long sides. The ratio of the width and height of the coding unit may include 1:2, 2:1, 1:4, 4:1, 1:8, 8:1, 1:16, 16:1, 32:1, 1:32, or the like. Also, the direction of the coding unit may include a horizontal direction and a vertical direction. The horizontal direction may indicate a case in which the length of the width of the coding unit is longer than the length of the height thereof.

**[0210]** The vertical direction may indicate a case in which the length of the width of the coding unit is shorter than the length of the height thereof. The image decoding apparatus **100** may adaptively determine the split rule based on the size of the coding unit. The image decoding apparatus **100** may differently determine an allowable split shape mode based on the size of the coding unit. For example, the image decoding apparatus **100** may determine whether splitting is allowed based on the size of the coding unit. The image decoding apparatus **100** may determine a split direction according to the size of the coding unit.

**[0211]** The image decoding apparatus **100** may determine an allowable split type according to the size of the coding unit. The split rule determined based on the size of the coding unit may be a split rule pre-determined between the image encoding apparatus **200** and the image decoding apparatus **100**.

**[0212]** Also, the image decoding apparatus **100** may determine the split rule based on the information obtained from the bitstream. The image decoding apparatus **100** may adaptively determine the split rule based on a location of the coding unit.

**[0213]** The image decoding apparatus **100** may adaptively determine the split rule based on the location of the coding unit in the image. Also, the image decoding apparatus **100** may determine the split rule so that coding units generated via different splitting paths do not have the same block shape. An embodiment is not limited thereto, and the coding units generated via different splitting paths have the same block shape. The coding units generated via the different splitting paths may have different decoding processing orders.

**[0214]** FIG. **16** is a block diagram of an image encoding and decoding system.

**[0215]** An encoding end **1610** of an image encoding and decoding system **1600** may transmit an encoded bitstream of an image, and a decoding end **1650** of the image encoding and decoding system **1600** may receive a bitstream and decode the bitstream to output a reconstructed image. In some embodiments, the decoding end **1650** may be a configuration that is similar to the image decoding apparatus **100**.

**[0216]** In the encoding end **1610**, a prediction encoder **1615** outputs a reference image through inter-prediction and intra-prediction, and a transformer and quantizer **1616** quantizes residual data between the reference image and a current input image into a quantized transform coefficient and output the quantized transform coefficient. An entropy encoder **1625** encodes the quantized transform coefficient and outputs the encoded quantized transform coefficient as the bitstream. The quantized transform coefficient is reconstructed into data of a spatial domain via an inverse quantizer and inverse transformer **1630**, and the reconstructed data of the spatial domain may be output as a reconstructed image via a deblocking filter **1635** and a loop filter **1640**. The reconstructed image may be used as a reference image of a next input image in the prediction encoder **1615**.

**[0217]** Encoded image data among the bitstream received by the decoding end **1650** is reconstructed as residual data of a spatial domain via an entropy decoder **1655** and an inverse quantizer and inverse transformer **1660**. Image data of a spatial domain may be configured when a reference image output by a prediction decoder **1675** and the residual data are combined, and a deblocking filter **1665** and a loop filter **1670** may output a reconstructed image for a current original image by performing filtering on the image data of the spatial domain. The reconstructed image may be used as a reference image for a next original image by the prediction decoder **1675**.

**[0218]** The loop filter **1640** of the encoding end **1610** performs loop filtering by using filter information input according to a user input or system settings. Filter information used by the loop filter **1640** is output to the entropy encoder **1610** and transmitted to the decoding end **1650** along with the encoded image data. The loop filter **1670** of the decoding end **1650** may perform loop filtering based on the filter information input from the decoding end **1650**.

**[0219]** In the present disclosure, a 'tree structure' may refer to a hierarchical structure of one or more coding units formed according to whether a splitting mode of a coding unit is a quad split, a binary split, a ternary split, or a non-split. For example, the hierarchical structure of blocks generated from a current coding unit according to the splitting process of FIG. **5** is referred to as a tree structure.

**[0220]** In the present disclosure, 'availability of a block' refers to whether the block has already been encoded or decoded and thus information of the block may be obtained. In detail, in an encoding process, when a current block has already been encoded, a neighboring block may be encoded using encoding information of the current block, so the current block may be displayed as available. When the current block is not encoded, the current block may be displayed as unavailable. Likewise, in a decoding process, when a current block has already been decoded, a neighboring block may be decoded using encoding information of the current block, so the current block may be displayed as available. When the current block is not decoded, the current block may be displayed as unavailable.

**[0221]** In the present disclosure, 'availability of motion information of a block' refers to whether motion prediction for the block (prediction other than prediction according to an intra mode or intra block copy mode) may be performed and thus motion information (a motion vector, a prediction direction (L0-pred, L1-pred or Bi-pred), and a reference picture index) of the block may be obtained. In detail, in an encoding process, when motion prediction has already been performed on the current block and motion information of the current block exists, motion prediction of a neighboring block may be performed using the motion information of the current block, so the motion information of the current block may be displayed as available. In an encoding process, when motion prediction is not performed on the current block, the motion information of the current block may be displayed as unavailable. Likewise, in a decoding process, when motion prediction has already been performed on the current block and motion information of the current block exists, motion prediction of a neighboring block may be performed using the motion information of the current block, so the motion information of the current block may be displayed as available. In a decoding process, when motion prediction is not performed on the current block, the motion information of the current block may be displayed as unavailable.

**[0222]** In the present disclosure, a 'merge candidate' may correspond to a motion vector corresponding to a neighboring block of a current block. Because a prediction motion vector of the current block is determined from a motion vector of the neighboring block, each prediction motion vector may correspond to the neighboring block. Therefore, in the present disclosure, for convenience of explanation, a 'merge candidate' is described as corresponding to a motion vector of a neighboring block or corresponding to a neighboring block, and there is no difference in meaning between the two expressions.

**[0223]** In the present disclosure, an 'affine merge candidate' may correspond to control point vectors corresponding to a neighboring block or block group of a current block. Because control point vectors are determined from the motion vectors of the neighboring block or the control point vectors are determined based on motion vectors of neighboring blocks belonging to a block group, each of the control point vectors may correspond to a corresponding neighboring block or a corresponding block group. Therefore, in the present disclosure, for convenience of explanation, an 'affine merge candidate' is described as corresponding to control point vectors determined from a neighboring block or block group or corresponding to a neighboring block or a block group, and there is no difference in meaning between the two expressions.

**[0224]** In the present disclosure, a 'motion vector prediction (MVP) candidate' may correspond to motion vectors corresponding to a neighboring block of a current block. Because a prediction motion vector of the current block is determined from a motion vector of the neighboring block, each prediction motion vector may correspond to the neighboring block. Therefore, in the present disclosure, for convenience of explanation, a 'MVP candidate' is described as corresponding to a motion vector of a neighboring block or corresponding to a neighboring block, and there is no difference in meaning between the two expressions.

**[0225]** The 'merge candidate' is a neighboring block (or a motion vector of a neighboring block) used in a merge mode among inter prediction methods, whereas the 'MVP candidate' corresponds to a neighboring block (or a motion vector of a neighboring block) used in an AMVP mode among inter prediction methods. In the merge mode, the motion vector of the current block may be not only determined using a motion vector of the merge candidate, but also a prediction direction of and a reference picture index of the current block may be determined using a prediction direction (L0-pred, L1-pred, or bi-pred) of and a reference picture index of the merge candidate. On the other hand, in the AMVP mode, the motion vector of the current block may be determined using a motion vector of the MVP candidate, but a prediction direction of and a reference picture index of the current block may be determined separately from a prediction direction of and a reference picture index of the MVP candidate.

**[0226]** In the advanced motion vector prediction (AMVP) mode or merge mode, a motion vector of a block corresponding to a current block (or a co-located block or col-located block) in a reference picture (or a co-located reference picture) is used as a temporal motion vector predictor (temporal MVP) or a temporal motion vector candidate. When respective temporal locations (e.g., picture order count (POC) values) of a reference picture of a current block and a reference picture of a co-located block are different, a motion vector of the co-located block is used by being scaled (linearly) in accordance with the reference picture of the current block. When the co-located block is quantized to a high quantization parameter value and the co-located block is coded in a skip mode or in a merge mode, the accuracy of the motion vector of the co-located block may be reduced. When a block corresponding to a current block in a co-located reference picture (or a co-located block) is coded in an intra mode, the motion vector of the co-located block does not exist. As such, when the motion vector of the co-located block does not exist or has low accuracy and the motion vector of the co-located block is used as a temporal motion vector predictor or candidate of the current block, motion vector prediction performance of the current block may be deteriorated or the motion vector of the current block may be determined as an inaccurate value, resulting in overall deterioration of coding performance or efficiency.

**[0227]** Technology of increasing coding efficiency by generating a virtual reference picture corresponding to a current picture to be coded (e.g., decoded or encoded) via a frame rate up conversion (FRUC) technique and using the virtual reference picture as one reference picture during coding (e.g., decoding or encoding) of the current picture is being used. In this FRUC technique, a picture buffer of the same size as the current picture needs to be allocated to code the current picture, and additional complex processing in a pixel domain needs to be performed to generate a virtual reference picture. There are impractical issues in terms of complexity, such as external memory bandwidth requirements or processing power, when processing images of a high resolution, such as 4K/8K.

**[0228]** One or more embodiments of the present disclosure provide a method of and a device for generating motion information corresponding to a current picture by using information (e.g., a reconstructed sample, motion information, etc.) of coding (e.g., decoding or encoding)-completed pictures before the current picture, and using the motion information during coding (e.g., decoding or encoding) of the current picture, thereby increasing coding efficiency.

**[0229]** When the method according to the present disclosure is applied, the motion information corresponding to the current picture may be generated before the current picture is coded (e.g., decoded or encoded), and the generated motion information may be used as a more accurate motion vector predictor or candidate in an existing AMVP mode or merge mode or may be used in a new motion vector coding mode, thereby improving coding efficiency. When the method according to the present disclosure is applied, because motion information is generated in units of specific size blocks (e.g., 4×4, 8×8, 16×16, etc.) rather than in units of actual pixels, the same effect as FRUC may be obtained, and also it may be considered more efficient in terms of buffer efficiency and computational complexity.

**[0230]** In the present disclosure, a picture for which coding has been completed before the current picture is coded (e.g., decoded or encoded) is referred to as a "reconstruction picture" or a "reconstructed picture". For example, the reconstructed picture may include a picture included in a decoded picture buffer (DPB), or may include a reference picture included in a reference list (or a reference picture list) (e.g., reference list **0** or reference list **1**). Therefore, in the present disclosure, the reconstructed picture may be replaced with a reference picture. In the present disclosure, the number of reconstructed pictures used to generate motion information of the current picture may be one or more. That is, the number of reconstructed pictures used to generate motion information according to the present disclosure may be one or may be plural.

**[0231]** In the present disclosure, "motion information" may include at least one of motion vector information, prediction list utilization information, and prediction mode information. For example, the motion vector information may include at least one of an L**0** motion vector (or an x component and/or a y component of the L**0** motion vector) or an L**1** motion vector (or an x component and/or a y component of the L**1** motion vector). In the present disclosure, the L**0** motion vector may be referred to as first motion vector information or mvL**0**, and the L**1** motion vector may be referred to as second motion vector information or mvL**1**.

**[0232]** For example, the prediction list utilization information may include at least one of list **0** utilization information (e.g., predFlagL**0**) or list **1** utilization information (e.g., predFlagL**1**), and may be referred to as reference list utilization information. In the present disclosure, the list **0** utilization information may be referred to as first prediction list utilization information (or first reference list utilization information or first prediction list utilization flag information), and the list **1** utilization information may be referred to as second prediction list utilization information (or second reference list utilization information or second prediction list utilization flag information). In the present disclosure, when both list **0** and list **1** are used, this may be referred to as bi-directional prediction, and, when one of list **0** or list **1** is used, this may be referred to as uni-directional prediction.

**[0233]** For example, the prediction mode information may include at least one of information indicating whether a corresponding block is coded in an intra mode or coded in an inter mode (e.g., pred_mode_flag or CuPredMode) or information indicating whether the corresponding block is coded in a skip mode (e.g., cu_skip_flag or CuSkipFlag).

**[0234]** Motion information generated according to the proposed method of the present disclosure may be referred to as "generated motion information". In the present disclosure, the generated motion information may also be referred to as reference motion information, generated reference motion information, for example. Moreover, in the present disclosure, the generated motion information may be briefly referred to as generated information or generated data.

**[0235]** The motion information may be generated or stored in the form of a one-dimensional arrangement or two-dimensional arrangement in units of blocks of a specific size or in units of M×N pixels (e.g., 4×4, 8×8, or 16×16), and the motion information generated or stored in this form may be referred to as a motion information map. Generated motion information for (or of) a current picture may refer to motion information generated to correspond to blocks included in the current picture in units of blocks of a specific size (e.g., 4×4, 8×8, or 16×16), and may be referred to as a motion information map for (or of) the current picture. In the present disclosure, the generated motion information may be interchangeably used with a generated motion information map. For example, the generated motion information map may also be referred to as a generated reference motion map.

**[0236]** FIG. **17** is a block diagram of an image decoding apparatus **100** according to the present disclosure.

**[0237]** The apparatus **100** illustrated in FIG. **17** may be included in the image decoding apparatus **100** described above with reference to FIG. **1**, and may include a motion information generator **130** and a decoder **120**. The motion information generator **130** and the decoder **120** of the apparatus **100** may be implemented by at least one processor included in the apparatus **100**.

**[0238]** The motion information generator **130** may generate (reference) motion information for blocks included in a current picture, based on at least one of pixel data and/or motion information of at least one reconstructed picture according to the present disclosure. For example, the motion information generator **130** may generate motion information corresponding to the current picture before coding (e.g., decoding or encoding) the current picture. As a more specific example, the motion information generator **130** may generate motion information that is to be used to code the current picture before coding (e.g., decoding or encoding) an initial slice of the current picture. For example, the motion information generator **130** may generate motion information corresponding to the current picture, by using (or using as an input) motion information stored in at least one reconstructed picture (e.g., a reference picture). For example, the motion information generator **130** may generate the motion information corresponding to the current picture, by using (or using as an input) pixel data of the at least one reconstructed picture (e.g., a reference picture). For example, the motion information generator **130** may generate motion information corresponding to the current picture, by using (or using as an input) both the motion information and the pixel data stored in at least one reconstructed picture (e.g., a reference picture).

**[0239]** For example, the pixel data of the reconstructed picture may be used as an input of the motion information generator **130** without separate processing, or downsampled pixel data obtained by downsampling the pixel data of the reconstructed picture may be used as an input of the motion information generator **130** (e.g., see FIG. **28** and a related description thereof).

**[0240]** The input of the motion information generator **130** according to the present disclosure is not limited to the pixel data and the motion information of the reconstructed picture, and other pieces of information of the reconstructed picture may also be used as an input of the motion information generator **130**.

**[0241]** For example, the input of the motion information generator **130** may include at least one of quantization information, a predicted sample, and a residual sample of at least one reconstructed picture, and distance information from the current picture. For example, the quantization information may include quantization parameter information. For example, the predicted sample may include a sample predicted by intra prediction or inter prediction. For example, the residual sample may include a sample obtained by inverse transforming and inverse quantizing transform coefficient information signaled through a bitstream, or a difference value between a reference sample and a sample of a current block. For example, distance information from the current picture may include POC information of the reconstructed picture or information indicating a POC distance (or difference) between the reconstructed picture and the current picture.

**[0242]** The motion information generator **130** may generate motion information (or a motion information map) in units of blocks of a specific size (e.g., 4×4, 8×8, or 16×16). The generated motion information output by the motion information generator **130** may include at least one of motion information, intra-coding related information, residual coding related information, illumination change information, and statistical information in units of blocks of a specific size (e.g., 4×4, 8×8, or 16×16).

**[0243]** For example, as described above, the motion information may include at least one of motion vector information, prediction list utilization information, and prediction mode information for a block of a specific size (e.g., see a description of "motion information" of a "method and apparatus according to the present disclosure").

**[0244]** For example, the intra coding related information may include at least one of direct current (DC) transform coefficient information and intra prediction mode information (e.g., an intra DC mode, an intra planar mode, or intra angular mode) for a block of a specific size. For example, the residual coding related information may include information indicating whether residual coding (e.g., inverse transformation and/or inverse quantization) is performed on a block of a specific size. Alternatively, the residual coding related information may include information indicating whether non-zero transform coefficient information exists for a corresponding block. For example, the illumination change information may include information about the amount of change in illumination for illumination compensation with respect to a block of a specific size. For example, the statistical information may include probability information for entropy coding with respect to a block of a specific size, as information that is available in coding (e.g., encoding and/or decoding procedures).

**[0245]** The motion information generator **130** may process an input and generate an output, through various methods. For example, the motion information generator **130** may generate generated information or data, based on at least one of pixel data or motion information (and/or quantization information, a prediction sample, a residual sample, and distance information from the current picture) of at least one reconstructed picture by using an arithmetic operation (e.g., an average, a median, an interpolation, or filtering).

**[0246]** Alternatively, for example, the motion information generator **130** may generate the generated information or data, based on at least one of pixel data or motion information (and/or quantization information, a prediction sample, a residual sample, and distance information from the current picture) of at least one reconstructed picture by using an artificial neural network. For example, the artificial neural network usable in the motion information generator **130** according to the present disclosure may include a convolutional neural network (CNN), a fully connected neural network (FCNN), or a combination thereof. Respective structures of the convolutional neural network and the fully connected neural network are widely known, so detailed descriptions thereof will be omitted in this specification. Processing of the motion information generator **130** is not limited to the processing using the arithmetic operation and the processing using the artificial neural network, and various methods may be applied.

**[0247]** The decoder **120** of the apparatus **100** may decode the current picture, based on motion information generated by the motion information generator **130** and at least one piece of syntax information obtained from a bitstream (e.g., the bitstream obtained by the receiver **110**). For example, the decoder **120** may be configured to obtain motion information for the current block based on the motion information generated by the motion information generator **130** and the at least one piece of syntax information obtained from the bitstream, and to reconstruct the current block based on the obtained motion information. Details of utilization of the generated information or data (or the motion information map) in decoding according to the present disclosure will now be described in detail.

**[0248]** FIG. **18** is a flowchart of an image decoding method **1800** according to the present disclosure. The method **1800** illustrated in FIG. **18** may be performed by the apparatus **100**.

**[0249]** Referring to FIG. **18**, the apparatus **100** may generate (reference) motion information for blocks included in a current picture, based on at least one of pixel data or motion information of at least one reconstructed picture (operation **1810**). For example, operation **1810** may be performed in the motion information generator **130** of the apparatus **100**, and the description related to the motion information generator **130** given with reference to FIG. **17** is included in some embodiments.

**[0250]** The apparatus **100** may decode the current picture, based on the (reference) motion information generated in operation **1810** and at least one piece of syntax information obtained from a bitstream (operation **1820**). In more detail, in operation **1820**, the apparatus **100** may obtain motion information of for current block based on the motion information (or a motion information map) generated in operation **1810** and the at least one piece of syntax information obtained from the bitstream, and may reconstruct the current block based on the obtained motion information for the current block.

**[0251]** For example, reconstructing the current block by the apparatus **100** in operation **1820** may include determining whether the current block is coded in an inter mode or an intra mode, based on the motion information for the current block (e.g., prediction mode information (e.g., pred_mode_flag or CuPredMode)). For example, when the current block is coded in an inter mode, reconstructing the current block by the apparatus **100** in operation **1820** may include performing inter prediction on the current block based on the motion information (e.g., motion vector information and prediction list utilization information) for the current block to obtain a prediction sample (predicted block) for the current block. For example, when the current block is coded in an intra mode, reconstructing the current block by the apparatus **100** in operation **1820** may include performing intra prediction on the current block based on intra prediction mode information (e.g., an intra DC mode, an intra planar mode, and an intra angular mode) for the current block to obtain a prediction sample (or a predicted block) for the current block. For example, in operation **1820**, reconstructing the current block by the apparatus **100** in operation **1820** may include reconstructing the current block based on an obtained prediction sample (or a prediction block).

**[0252]** FIG. **19** illustrates an image encoding apparatus **200** according to the present disclosure. The image encoding apparatus **200** according to the present disclosure is not limited to the example of FIG. **19**, and may include other components not shown in FIG. **19**. For example, the image encoding apparatus **200** may include at least one processor, and a motion information generator **130** and an encoder **220** illustrated in FIG. **19** may be implemented by using at least one processor.

**[0253]** Referring to FIG. **19**, the apparatus **200** may include the motion information generator **130** and the encoder **220**. The motion information generator **130** may operate the same as the motion information generator **130** of FIG. **17**. Therefore, a detailed description of the motion information generator **130** includes, as a reference, FIG. **17** and a related description thereof.

**[0254]** The encoder **220** may determine motion information for a current block in a current picture corresponding to an original image, based on rate-distortion (RD) optimization, and may encode the determined motion information for the current block into a bitstream, based on the motion information (or a motion information map) generated by the motion information generator **130**.

**[0255]** For example, the encoder **220** may determine prediction mode information (e.g., pred_mode_flag or CuPredMode) indicating whether the current block is coded in an intra mode or in an inter mode by using pixel data of the current block and a reconstructed neighboring block of a reference picture and/or the current picture, based on RD optimization. For example, when the current block is coded in an intra mode, the encoder **220** may identify motion vector information (e.g., a first motion vector and/or a second motion vector) and prediction list utilization information (e.g., list **0** utilization information and/or list **1** utilization information) obtained through motion estimation (ME) using the pixel data of the current block and the reference picture. For example, based on the motion information (or the motion information map) generated by the motion information generator **130**, the encoder **220** may determine whether to encode the motion information for the current block determined based on RD optimization in an AMVP mode, in a merge mode, or in a skip mode, and may encode the motion information for the current block into a bitstream based on the determined mode.

**[0256]** The bitstream generated by the apparatus **200** may be stored in a storage medium or a recording medium, or transmitted through a wired or wireless network in units of network adaptation layer (NAL) units.

**[0257]** FIG. **20** is a flowchart of an image encoding method **2000** according to the present disclosure. The method **2000** illustrated in FIG. **20** may be performed by the apparatus **200**. Likewise, a bitstream generated by the method **2000** may be stored in a storage medium or a recording medium, or transmitted through a wired or wireless network in units of NAL units.

**[0258]** Referring to FIG. **20**, the apparatus **200** may generate (reference) motion information for blocks included in a current picture, based on at least one of pixel data or motion information of at least one reconstructed picture (operation **2010**). For example, operation **2010** may be performed in the motion information generator **130** of the apparatus **200**, and FIG. **19** and a related description thereof (i.e., FIG. **17** and a related description thereof) is included in some embodiments.

**[0259]** The apparatus **200** may encode the current picture into a bitstream, based on the motion information generated in operation **2010** (operation **2020**). For example, the apparatus **200** may determine motion information for a current block included in the current picture, by using pixel data of the original image (i.e., the current picture), based on RD optimization. As described above with reference to FIG. **19**, the motion information for the current block may include prediction mode information (e.g., pred_mode_flag or CuPredMode), motion vector information (e.g., a first motion vector and/or a second motion vector) and prediction list utilization information (e.g., list **0** utilization information and/or list **1** utilization information). The apparatus **200** may encode the determined motion information for the current block into a bitstream, based on the motion information generated in operation **2010** (operation **2020**). In operation **2020**, the apparatus **200** may determine the motion information for the current block by performing an operation of the encoder **220** of FIG. **19**, and may encode the motion information for the current block into the bitstream.

**[0260]** FIGS. **21**A-**21**C illustrate a method of generating motion information according to the present disclosure.

**[0261]** In the example of FIGS. **21**A-**21**C, a method of generating motion information (or a motion information map) **2140** corresponding to a current picture by using an artificial neural network **2110** is illustrated, but, in the method according to the present disclosure, motion information may be generated using another method (e.g., an arithmetic operation) rather than the artificial neural network. For convenience of explanation, the example of FIGS. **21**A-**21**C illustrates pixel data and/or a motion information map of two reconstructed pictures (e.g., a current reconstructed picture and/or at least one reconstructed picture) are illustrated as being input to the artificial neural network (ANN) (briefly, referred to as a neural network (NN)) **2110**. The number of reconstructed pictures input to the NN **2110** is not limited to two, and various numbers of reconstructed pictures and combinations thereof may be used as an input of the NN **2110**.

**[0262]** The example of FIGS. **21**A-**21**C may be performed by the image decoding apparatus **100** and the image encoding apparatus **200**, and, as a more specific example, may be performed by the motion information generator **130**. Moreover, the example of FIGS. **21**A-**21**C may be performed in operation **1810** of the image decoding method **1800** and operation **2010** of the image encoding method **2000**.

**[0263]** Referring to FIG. **21**A, the input of the NN **2110** may include pixel data of a reconstructed picture (e.g., POC=T−1 or a reference picture) and pixel data of a current reconstructed picture (e.g., POC=T) **2130**. The NN **2110** may be configured to generate a generated motion information map **2150** by using the pixel data of the reconstructed picture **2120** and the current reconstructed picture **2130** as the input. In the example of FIG. **21**A, the generated motion information map **2150** may be used as a motion information map for a next picture to be coded or may be used to refine a motion information map of a current picture.

**[0264]** Referring to FIG. **21**B, when the current picture (e.g., POC=T) is a bi-directional predicted image, the input of the NN **2110** may include a motion information map **2120** stored in an adjacent forward reference picture (e.g., POC=T−1) and the motion information map **2140** stored in a backward reference picture (e.g., POC=T+1). Alternatively, when the current picture (e.g., POC=T) is a bi-directional predicted picture, the input of the NN **2110** may include motion information (or a motion information map) stored in each reference picture by using, as the forward reference picture and the backward reference pictures, a reference picture in a specific order (e.g., a first reference picture in each reference list or a reference picture with a reference index of 0 in each reference list) in each of reference list **0** (or reference picture list **0**) and reference list **1** (or reference picture list **1**).

**[0265]** Alternatively, in the example of FIG. **21**B, the backward reference picture may be replaced with a reference picture of reference list **0**, and the forward reference picture may be replaced with a reference picture of reference list **1**. In other words, when the current picture (e.g., POC=T) is a bi-directional predicted picture, the input of the NN **2110** may include motion information (or a motion information map) stored in each reference picture by using, as the forward reference picture and the backward reference pictures, a reference picture in a specific order (e.g., a first reference picture in each reference list or a reference picture with a reference index of 0 in each reference list) in each of reference list **1** (or reference picture list **1**) and reference list **0** (or reference picture list **0**).

**[0266]** Referring to FIG. **21**C, when the current picture is a uni-directional predicted image, the input of the NN **2110** may include motion information (or A motion information map) of two reconstructed pictures that are close to the current picture in terms of a POC distance among reconstructed pictures (e.g., reference pictures) available in the current picture. In the example of FIG. **21**C, the backward reference picture may be replaced with a reconstructed picture (e.g., a reference picture) closest to the current picture, and the forward reference picture may be replaced with a reconstructed picture (e.g., a reference picture) second closest to the current picture.

**[0267]** FIG. **22** illustrates a method of normalizing the input of the motion information generator **130** (e.g., the NN **2110**) according to the present disclosure. The example of FIG. **22** is an example, and the proposed method of the present disclosure is not limited to the example of FIG. **22**.

**[0268]** In the present disclosure, when motion information (or a motion information map) is input to the motion information generator **130** (e.g., the NN **2110**), the motion information may be used by being normalized to a distance of K in units of POCs. For example, K may be set to be a POC distance (or a difference with a nearest reconstructed picture (e.g., a reference picture) of the current picture.

**[0269]** In more detail, the proposed method of the present disclosure may include scaling motion information (or a motion information map) of an input reconstructed picture, based on a POC distance (or difference) between the current picture and a reconstructed picture (e.g., a reference picture) nearest to the current picture and a POC distance (or a difference) between an input reconstructed picture and a reference picture of the input reconstructed picture.

**[0270]** Normalization of the motion information (or the motion information map) may be performed by the apparatuses **100** and **200**, and, in more detail, may be performed by the motion information generators **130** of the apparatuses **100** and **200**. Moreover, the normalization of the motion information (or the motion information map) may be performed in operation **1810** of the image decoding method **1800** and operation **2010** of the image encoding method **2000**.

**[0271]** For example, referring to FIG. **22**, when a POC of a current picture **2210** is 3, a POC of a nearest forward reference picture **2220** is 2, and a POC of a nearest backward reference picture **2230** is 4, a distance between the current picture and the nearest reconstructed picture is 1, and therefore K may be 1. When bi-directional prediction is performed for the forward reference picture **2220** (e.g., POC=2) and, as an example of the value of the motion information of a specific location of the reference picture **2220**, a forward MV indicates the reference picture **2240** with a POC of 0 and an MV value is (4, 8), and a backward MV indicates the reference picture **2230** with a POC of 4 and an MV value is (−10, −4), the motion information at the specific location has an MV of the POC distance of 2, and thus may be scaled (e.g., ½) to the POC distance of 1 so that the forward MV may be normalized to (2, 4) and the backward MV may be normalized to (−5, −2). Then, normalized motion information may be used as the input of the motion information generator **130** (e.g., the NN **2110**).

**[0272]** As another example, when motion information is input to the motion information generator **130**, POC information (or a POC map) of each motion information of a reconstructed picture (e.g., a reference picture) may also be input (e.g., see FIG. **23** and a related description thereof). A unit in which the input motion information exists may be a block of a specific size or may be M×N pixels. In detail, motion information may be stored in the reconstructed picture in units of 16×16, in units of 8×8, or in units of 4×4. For example, when motion information is stored in units of 8×8 and a 1920×1080 image is coded, motion information with a resolution of 240×135 needs to be processed. For example, when motion information is stored in units of 16×16, motion information with a resolution of 120×68 needs to be processed.

**[0273]** When motion information does not exist in a specific unit (e.g., when the corresponding location is coded in an intra mode), the motion information may be filled with a specific value, and, in detail, a unit in which motion information does not exist may be filled with a zero vector (e.g., see FIG. **23** and a related description thereof). Moreover, a map about whether motion information exists may be created and used as the input of the motion information generator **130** (e.g., the NN **2110**) (e.g., see FIG. **23** and a related description thereof).

**[0274]** FIG. **23** illustrates an input motion information map according to the present disclosure.

**[0275]** In the example of FIG. **23**, in order to clarify the explanation of the present disclosure, an explanation focuses on motion vector information included in motion information. A motion information map according to the present disclosure may include other information in addition to the motion vector information. As described above with reference to the 'proposed method and device of the present disclosure" and FIG. **17**, the input of the motion information generator **130** may include motion information including at least one of motion vector information, prediction list utilization information, and prediction mode information of a reconstructed picture, and at least one of quantization information, a prediction sample, a residual sample of the reconstructed picture, and distance information from the current picture.

**[0276]** As described above, the motion information map of the reconstructed picture may include motion information in the form of a one-dimensional arrangement or a two-dimensional arrangement in units of specific-sized blocks or in units of M×N pixels (e.g., 4×4, 8×8, or 16×16).

**[0277]** In the example of FIG. **23**, motion information maps of the reconstructed pictures **2120** and **2140** of FIG. **21**B are assumed and described. The present disclosure may be applied identically/similarly even when other reconstructed pictures are used. The motion information map illustrated in FIG. **23** may be a motion information map used as the input of the motion information generator **130** (e.g., the NN **2110**).

**[0278]** Referring to FIG. **23**, each of the motion information maps of the reconstructed pictures **2120** and **2140** of FIG. **21**B may include motion vector maps **2310** and **2330**, POC maps **2320** and **2340**, and a map **2350** about presence or absence of a motion vector. For example, the reconstructed picture **2120** may be a forward reference picture, and the reconstructed picture **2140** may be a backward reference picture.

**[0279]** In the example of FIG. **23**, the motion vector maps **2310** and **2330** may separately include a map **2310** for an L**0** motion vector (or a first motion vector) and a map **2330** for an L**1** motion vector (or a second motion vector). Alternatively, the motion vector maps **2310** and **2330** may include the L**0** motion vector (or the first motion vector) and the L**1** motion vector (or the second motion vector) as a single map. The motion vector maps **2310** and **2330** may include separate maps for an x component and a y component of a motion vector, or may include one map in the form of a vector of (x, y).

**[0280]** In the example of FIG. **23**, the POC maps **2320** and **2340** may include POC values of reference pictures indicated by the motion vectors of the motion vector maps **2310** and **2330** (or reference pictures referenced by a motion vector of a specific location of the reconstructed picture). When the motion information is not normalized (or scaled) based on a POC distance (or a difference) between a current picture and a nearest reconstructed picture as described above with reference to FIG. **22**, the motion information map may include the POC maps **2320** and **2340**. When the motion information is normalized (or scaled) based on the POC distance (or the difference) between the current picture and the nearest reconstructed picture, the motion information map may not include the POC maps **2320** and **2340** (or the POC maps **2320** and **2340** may be omitted from the motion information map).

**[0281]** In the example of FIG. **23**, when there is no motion vector at specific locations of the motion vector maps **2310** and **2330** of the reconstructed pictures **2120** and **2140** (e.g., when the location is coded in an intra mode), a motion vector at the location may be filled with a specific value (e.g., a zero vector) as described above. To this end, the motion information maps of the reconstructed pictures **2120** and **2140** may include the map **2350** about presence or absence of a motion vector. For example, the map **2350** may include a value indicating, in a directional mode, presence or absence of a motion vector. As a non-limiting example, the map **2350** may include a first value (e.g., 0) when the L**0** motion vector (or the first motion vector) exists, the map **2350** may include a second value (e.g., 1) when the L**1** motion vector (or the second motion vector) exists, the map **2350** may include a third value (e.g., 2) when both the L**0** motion vector and the L1 motion vector exist, and the map **2350** may include a fourth value (e.g., 3) when no motion vectors exist. Alternatively, as another example, a map about presence or absence of a motion vector may be included separately with respect to each of the motion vector maps **2310** and **2330**. In this case, the map may include flag information having a value (e.g., 0 or 1) indicating whether a corresponding motion vector (L**0** motion vector or L1 motion vector) exists.

**[0282]** FIG. **24** illustrates a generated motion information map according to the present disclosure.

**[0283]** In the example of FIG. **24**, in order to clarify the explanation of the present disclosure, an explanation focuses on motion vector information included in generated motion information. The motion information map according to the present disclosure may include other information in addition to the motion vector information. As described above with reference to the 'proposed method and device of the present disclosure" and FIG. **17**, generated information or data according to the present disclosure may include motion information including at least one of motion vector information, prediction list utilization information, and prediction mode information, and at least one of intra coding related information, residual coding related information, illumination change information, and statistical information.

**[0284]** As described above, the generated motion information map according to the present disclosure may include motion information in the form of a one-dimensional arrangement or a two-dimensional arrangement in units of specific-sized blocks or in units of M×N pixels (e.g., 4×4, 8×8, or 16×16).

**[0285]** In the example of FIG. **24**, a generated motion information map **2150** of FIGS. **21A-21C** is assumed and described. The present disclosure may be applied identically/similarly even when other motion information maps are generated. The motion information map illustrated in FIG. **24** may be a motion information map generated as an input of the motion information generator **130** (e.g., the NN **2110**).

**[0286]** Referring to FIG. **24**, the generated motion information map **2150** may include a motion vector map **2410**, a POC map **2420**, and a map **2430** about presence or absence of a motion vector. In the example of FIG. **24**, it is illustrated that the motion vector map **2410**, the POC map **2420**, and the map **2430** about presence or absence of a motion vector are included separately for the L**0** motion vector and the L**1** motion vector. One motion vector map **2410**, one POC map **2420**, and one map **2430** about presence or absence of a motion vector are included for the L**0** motion vector and the L1 motion vector.

**[0287]** In the example of FIG. **24**, the motion vector ma **2410** may include separate maps for an x component and a y component of a motion vector, or may include one map in the form of a vector of (x, y).

**[0288]** In the example of FIG. **24**, the POC map **2420** may include a reference picture's POC value indicated by the motion vector of the corresponding motion vector map **2410**. When input motion information is normalized (or scaled) based on a POC distance (or a difference) between a current picture and a nearest reconstructed picture as described above with reference to FIG. **22**, the POC map **2420** may be omitted from the generated motion information map. Alternatively, when the motion information generator **130** (e.g., the NN **2110**) is configured to output a normalized motion vector value, the POC map **2420** may be omitted from the generated motion information map.

**[0289]** In the example of FIG. **24**, the map **2430** about presence or absence of a motion vector may be output for each reference list (or each reference picture list), or presence or absence of a motion vector may be output as one map for reference lists L**0** and L**1**, or a directional mode of the motion vector may be expressed as one map and output. For example, the map **2430** may include a first value (e.g., 0) when the L**0** motion vector (or the first motion vector) exists, the map **2430** may include a second value (e.g., 1) when the L**1** motion vector (or the second motion vector) exists, the map **2430** may include a third value (e.g., 2) when both the L**0** motion vector and the L**1** motion vector exist, and the map **2430** may include a fourth value (e.g., 3) when no motion vectors exist. Alternatively, as another example, a map about presence or absence of a motion vector may be included separately with respect to each of the L**0** motion vector and the L**1** motion vector. In this case, the map may include flag information having a value (e.g., 0 or 1) indicating whether a corresponding motion vector (L**0** motion vector or L**1** motion vector) exists.

**[0290]** FIG. **25** illustrates a motion information generating model and a learning method according to the present disclosure.

**[0291]** In the present disclosure, the artificial neural network **2110** configured to output generated motion information may be referred to as a motion information generation model. The motion information generation model **2110** may include a convolutional neural network (CNN), a fully connected neural network (FCNN), or a combination of the CNN and the FCNN. In the example of FIG. **25**, the motion information generation model **2110** is illustrated as a CNN. The present disclosure is applicable identically/similarly even when other types of neural networks are used.

**[0292]** The motion information generating model and the learning method described above with reference to FIG. **25** may be performed by the apparatuses **100** and **200**, and, in more detail, may be performed by the motion information generators **130** of the apparatuses **100** and **200**. Moreover, the motion information generating model and the learning method described above with reference to FIG. **25** may be performed in operation **1810** of the image decoding method **1800** and operation **2010** of the image encoding method **2000**.

**[0293]** Also, in the example of FIG. **25**, the reconstructed picture (e.g., POC=T−1 or a reference picture) **2120** and the current reconstructed picture (e.g., POC=T) **2130** are used as input reconstructed pictures as in FIGS. **21A-21C**. The present disclosure is applicable identically/similarly even when other reconstructed pictures are used as inputs.

**[0294]** Referring to (a) of FIG. **25**, the motion information generation model **2110** according to the present disclosure may be trained using an objective function based on residual signal minimization, similar to motion vector map (or MV

Map) calculation of an existing video codec (e.g., motion estimation to minimize a residual image in a searching range).

**[0295]** An existing neural network (NN) is trained using a dataset (e.g., a synthetic video) having a label (or a ground truth label), whereas the motion information generation model **2110** according to the present disclosure is trained in a direction for minimizing a residual signal **2530** between a warped image **2520** obtained by warping the reconstructed picture **2120** by using obtained (sample level) motion information **2510** and an existing image (e.g., the reconstructed picture **2130**). Therefore, the motion information generation model **2110** according to the present disclosure is able to be trained using a real video, because the motion information generation model **2110** does not need a label (or a ground truth label). The motion information generation model **2110** according to the present disclosure has an advantage because it is also possible to implement a model trained based on supervised learning, such as an existing neural network (NN), in the form of fine-tuning.

**[0296]** For example, the objective function for training the motion information generation model **2110** may use a mean square error between the reconstructed picture **2130** and the warped image **2520**, or a sum or average of the squares of the samples of the residual image **2530**. In the present disclosure, motion information of a sample unit or sample level may be referred to as an optical flow or a flow, and a motion information map of a sample unit or sample level may be referred to as a motion field or an (optical) flow field.

**[0297]** The motion information generation model **2110** according to the present disclosure may also estimate an occlusion mask through warping using motion information generated in sample units or sample levels. A portion where the occlusion mask is activated may mean a portion where it is difficult to accurately determine presence or absence of motion information. Therefore, when the motion information generation model **2110** according to the present disclosure is used, a block where occlusion masks overlap each other may contribute to performance improvement by setting a mode in which motion information (or a motion information map or MV map) is not used. For example, a block determined as an occlusion area (or the block where occlusion masks overlap each other) may be determined as a block coded in an intra mode.

**[0298]** Referring to (b) of FIG. **25**, in an existing method, a searching range is set, and a block with a smallest error from a current block is found in the searching range, and thus a motion vector is calculated. Because a process of obtaining a block of the same size as the current block through interpolation or the like performed in a unit (e.g., in units of 1 pixels, in units of ½ pixels, or in units of ¼ pixels) of a motion vector in the searching range and calculating an error with the current block needs to be repeated, the complexity is high and the amount of calculation is large. Because the searching range is set, an accurate motion vector may not be found.

**[0299]** On the other hand, when the motion information generation model **2110** according to the present disclosure is applied, pixel data and/or motion information of the entire picture may be used as an input, so that more accurate motion information may be generated with lower complexity and used to code the current picture, thereby providing improved coding performance or efficiency.

**[0300]** FIG. **26** illustrates a method of generating motion information according to the present disclosure.

**[0301]** A method of generating motion information (or a motion information map) by using pixel data of a reconstructed picture as an input in the motion information generation model **2110** (e.g., see FIG. **25** and a related description thereof) according to the present disclosure will be described with reference to FIG. **26**. The motion information generating method to be described with reference to FIG. **26** may be performed by the apparatuses **100** and **200**, and, in more detail, may be performed by the motion information generators **130** of the apparatuses **100** and **200**. Moreover, the motion information generating method described above with reference to FIG. **26** may be performed in operation **1810** of the decoding method **1800** and operation **2010** of the image encoding method **2000**.

**[0302]** In the example of FIG. **26**, the reconstructed picture (e.g., POC=T−1 or a reference picture) **2120** and the reconstructed picture (e.g., POC=T) **2130** are used as input reconstructed pictures as in FIGS. **21**A-**21**C. The present disclosure is applicable identically/similarly even when other reconstructed pictures are used as inputs.

**[0303]** In the example of FIG. **26**, generation of a motion information map **2150** generated in units of 16×16 blocks is illustrated. This is only an example, and the motion information map **2150** may also be generated in units of other block sizes (e.g., 4×4 or 8×8).

**[0304]** Referring to FIG. **26**, the apparatuses **100** and **200** may input the reconstructed pictures **2120** and **2130** to the motion information generation model **2110** to generate a generated motion information map **2510** (of a sample unit or sample level). For example, when the reconstructed pictures **2120** and **2130** have a size (or resolution) of H×W, the generated motion information map **2510** (of a sample unit or sample level) may have the same size (or resolution) (e.g., H×W) as the input reconstructed picture.

**[0305]** The apparatuses **100** and **200** may convert the generated motion information map **2510** (of a sample unit or sample level) in accordance with a motion information map format of a video codec (e.g., the image decoding method **1800** and the image encoding method **2000**). For example, the apparatuses **100** and **200** may generate the generated motion information map **2150** for a video codec by performing aggregation, scaling, and/or format transformation (e.g., quantization) on the generated motion information map **2510** (of a sample unit or sample level) according to an output format of the motion information generation model **2110**.

**[0306]** For example, in the present disclosure, aggregation may include resizing the block size of the generated motion information map **2510** (of a sample unit or a sample level) in accordance with the block size of the motion information map **2150** used in a video codec (e.g., the image decoding method **1800** and the image encoding method **2000**).

**[0307]** FIG. **27** illustrates a process of aggregating motion information according to the present disclosure.

**[0308]** The example in FIG. **27** is for illustrative purposes only, and the present disclosure may be applied equally/similarly to various block sizes.

**[0309]** As described above, in the present disclosure, motion information of a sample unit or sample level may be referred to as an optical flow or a flow, and a motion information map of a sample unit or sample level may be referred to as a motion field or an (optical) flow field.

Therefore, in the present disclosure, the motion information of a sample unit or sample level generated by the motion information generation model **2110** or an output of the motion information generation model **2110** may be referred to as neural network (NN) motion information (MI) or a neural network (NN) motion information (MI) field or field flow.

**[0310]** Referring to FIG. **27**, when the generated motion information map **2150** includes motion information in a 16×16 block size and the motion information generation model **2110** outputs or generates the motion information map (or an NN MI field or flow) **2510** in units of 8×8 blocks, the apparatuses **100** and **200** may aggregate 2×2 motion information (or flow) into one piece of motion information in the motion information map **2510**. For example, aggregation may include processes, such as fixed position sampling, median filtering, for example. For example, fixed position sampling may include determining, as the aggregated motion information, motion information at a specific position (e.g., a top left end, a top right end, a bottom left end, or a bottom right end) among pieces of motion information that are to be aggregated. For example, median filtering may include determining, as a value of the aggregated motion information, a median among the pieces of motion information that are to be aggregated.

**[0311]** FIG. **27** illustrates an example in which the apparatuses **100** and **200** determine motion information at a specific position (e.g., an upper left portion) as aggregated motion information according to fixed position sampling.

**[0312]** Referring back to FIG. **26**, in contrast, when the block size or unit of the motion information map (or the NN MI field or flow) **2510** is larger than the block size of the generated motion information map **2150**, the apparatuses **100** and **200** may perform upsampling on the motion information map (or the NN MI field or flow) **2510**. For example, when the generated motion information map **2150** includes motion information in a 8×8 block size or unit and the motion information generation model **2110** outputs or generates the motion information map (or an NN MI field or flow) **2510** in a 8×8 block size or unit, the apparatuses **100** and **200** may upsample one piece of motion information (or a flow) to 2×2 motion information. For example, upsampling may be done by directly copying or repeating the value of motion information (or a flow), but various other types (e.g., bi-cubic and DCT-IF interpolation) of upsampling may also be used.

**[0313]** Referring to FIG. **26**, the apparatuses **100** and **200** may perform scaling on the motion information (or NN MI) of the generated motion information map **2510** (in a sample unit or sample level). For example, scaling according to the present disclosure may convert a resolution (or a scale) of the motion information (or NN MI) of the generated motion information map **2510** (in a sample unit or sample level) in accordance with a motion vector resolution (or a motion scale) of a video codec (e.g., the image decoding method **1800** and the image encoding method **2000**).

**[0314]** For example, while an existing video codec (e.g., an HEVC) is designed to store a motion vector in units of ¼ pixels, the motion information (or NN MI) of the generated motion information map **2510** may be output or generated in units of one pixel. Thus, in this case, the apparatuses **100** and **200** may perform scaling (e.g., quadrupling of a resolution or scale) on the motion information (or NN MI) of the generated motion information map **2510**. In this example, the apparatuses **100** and **200** may perform 4-fold scaling through a left shift by 2 bits with respect to the motion information (or NN MI) of the generated motion information map **2510**.

**[0315]** Referring to FIG. **26**, the apparatuses **100** and **200** may perform format transformation on the motion information (or NN MI) of the generated motion information map **2510** (in a sample unit or sample level). For example, because the weight of the motion information generation model **2110** may have a value in the form of a floating point, the output of the motion information generation model **2110** may have a value in the form of a floating point. On the other hand, in a video codec (e.g., the image decoding method **1800** and the image encoding method **2000**), motion information may be stored in the form of an integer (e.g., a 16-bit short integer). Accordingly, the apparatuses **100** and **200** may convert the format of the output motion information (or NN MI) of the motion information generation model **2110** in accordance with a motion information storage form (or format) of the video codec.

**[0316]** For example, while an existing codec (e.g., an HEVC) is designed to store a motion vector in the form of a 16-bit short integer, the motion information (or NN MI) of the generated motion information map **2510** may be output or generated in the form of a floating point. Thus, in this case, the apparatuses **100** and **200** may perform quantization on the motion information (or NN MI) of the generated motion information map **2510**. For example, quantization may include mapping a value represented in a floating point form to a nearest integer value.

**[0317]** FIG. **28** illustrates a method of downscaling an input reconstructed picture according to the present disclosure.

**[0318]** In the case of high-definition images, if an image size or resolution of a high-definition image may be reduced before the high-definition image is input to the motion information generation model **2110**, there may be benefits in terms of complexity and processing time. For example, in the case of a typical convolution operation performed in a convolutional neural network (CNN), computational complexity may increase in proportion to the input image size. That is, when the input image size is H×W, the computational complexity may increase in proportion to a product of a width W and a height H of an input image (i.e., in proportion to O (HW)). Therefore, when the width and height of the input image are each reduced to 1/N, a reduction in computational complexity of 1/(N×N) may be expected.

**[0319]** Accordingly, in the present disclosure, the apparatuses **100** and **200** may perform downsampling (DN) on (each of respective widths and respective heights) of input reconstructed pictures **2120** and **2130** by a down-sampling factor (e.g., 1/N) and input down-sampled reconstructed pictures **2810** and **2820** to the motion information generation model **2110**, to thereby generate motion information **2510**. For example, downsampling may use bi-linear interpolation and Lanczos interpolation, but there are various other downsampling methods that may be used.

**[0320]** When the input reconstructed pictures **2120** and **2130** are downsampled and used as inputs of the motion information generation model **2110** as in the present disclosure, an output of the motion information generation model **2110** may be used as it is, without resizing (or flow field resizing), such as aggregation, with respect to the motion information map **2510** generated by the motion information generation model **2110** (e.g., see related descriptions of FIGS. **25** and **26**), so that computational complexity and a processing time period may be further reduced.

**[0321]** The apparatuses **100** and **200** may perform scaling (e.g., see FIG. **25** and its related description) on the motion information **2510** (or NN MI) generated using a downsampled input image. For example, in addition to scaling the resolution (or scale) of the motion information (or NN MI) in accordance with the video codec, the apparatuses **100** and **200** may scale the motion information, based on a downsampling factor. For example, in the case of motion information unit scaling (MI unit scaling), the apparatuses **100** and **200** may perform scaling on the motion information **2510** (or NN MI) generated by reflecting the downsampling factor in an existing scaling factor (e.g., multiplying an inverse number of the downsampling factor). In more detail, in the examples of FIGS. **26** and **28**, because ⅛ downsampling is applied to an input reconstructed picture and the generated motion information **2510** is scaled 4 times in accordance with the video codec, the apparatuses **100** and **200** may perform 32 (=4*8) times scaling on the generated motion information **2510** (or NN MI), and the apparatuses **100** and **200** may perform 32 times scaling through a 5-bit left shift operation.

**[0322]** The apparatuses **100** and **200** may also perform format transformation (e.g., see FIG. **25** and its related description), such as quantization, on the motion information **2510** (or NN MI) generated using the downsampled input image.

**[0323]** In the example of FIG. **28**, an example is illustrated in which the input reconstrued picture has a size of 4K and or a resolution (3840×2160) and a width and a height of the input reconstructed picture are each downsampled by ⅛. The present disclosure is not limited to this example, and the present disclosure may be applied identically/similarly even when an input reconstructed picture of another size (e.g., 2K or 8K) is used or another downsampling factor (e.g., ½, ¼, or 1/16) is applied.

**[0324]** In the present disclosure, when a time interval between reconstructed pictures used as inputs of a motion information generation model or a time interval between a current picture and the reconstructed picture is large (or they are far apart from each other in terms of time), the accuracy of generated motion information may be reduced. Therefore, in the present disclosure, when the time interval between the reconstructed pictures or the time interval between the current picture and the reconstructed picture is greater than or equal to a specific value or a specific threshold, the apparatuses **100** and **200** may determine that a time interval between two pictures (or frames) is “far”.

**[0325]** For example, the specific threshold may be 1/25 seconds. Because 25 frame per second (FPS) is the frame rate of a typical video, 1/25 seconds may be a frame interval that the motion information generation model **2110** typically uses for learning. Therefore, when the specific threshold may be 1/25 seconds or higher, this may be a case where the specific threshold is equal to or greater than the interval between pictures (or frames) of a typical video, so the interval between two pictures (or frames) may be determined to be “far”. In the present disclosure, a threshold for determining “far” is not limited to 1/25 seconds, and various other values may be used.

**[0326]** Alternatively, a picture order count (POC) distance (or difference) instead of a time unit may be used as the specific threshold. For example, when the POC distance (or difference) is 4 or greater, the apparatuses **100** and **200** may determine that the time interval between two pictures (or frames) is “far”. Likewise, in the present disclosure, a POC distance (or difference) for determining “far” is not limited to 4, and various other values may be used.

**[0327]** When the time interval between two pictures (or frames) is determined to be “far”, the following rule may be added when the motion information (or the motion information map) generated by the motion information generation model **2110** is applied to coding (e.g., decoding or encoding) of the current picture, and other rules may be applied for each mode of a current block of the current picture (or frame). For example, when the current block is coded in a skip mode and the time interval between the current picture and the reconstructed picture or the time interval between the reconstructed pictures is determined to be “far”, the apparatuses **100** and **200** may not use the generated motion information in coding the current block. For example, when the current block is coded in a merge mode and the time interval between the current picture and the reconstructed picture or the time interval between the reconstructed pictures is determined to be “far”, the apparatuses **100** and **200** may exclude the generated motion information from a merge candidate. For example, when the current block is coded in an AMVP mode and the time interval between the current picture and the reconstructed picture or the time interval between the reconstructed pictures is determined to be “far”, the apparatuses **100** and **200** may add or insert the generated motion information as a candidate to or into a motion vector prediction (MVP) candidate list.

**[0328]** FIG. **29** illustrates reference picture resampling (RPR). RPR refers to a technique of coding a current picture by referring to a picture having a different size or resolution than the current picture, and is employed in the ITU-T H.266/VVC standard.

**[0329]** Even when RPR is applied, generated motion information (or a motion information map) according to the present disclosure may be applied. When RPR is applied, the apparatuses **100** and **200** may compare respective sizes or resolutions of two pictures (or frames) with each other, and, when the sizes or resolutions of the two pictures (or frames) are different from each other, may perform a process of aligning the sizes or resolutions of the two pictures (or frames).

**[0330]** In the present disclosure, the apparatuses **100** and **200** may downsample two pictures (or frames) (e.g., the reconstructed pictures **2120** and **2130**) to smaller sizes or resolutions, and then may generate the motion information (or the motion information map) **2510** by using the downsampled pictures. For example, the same method as a VVC standard may be used as a downsampling method. The present disclosure is not limited thereto, and various downsampling methods (e.g., bilinear interpolation and Lanczos interpolation) may be used. The method described above with reference to FIGS. **25** through **28** may be used as a method of generating the motion information (or the motion information map) **2510**.

**[0331]** The apparatuses **100** and **200** may perform operations (e.g., aggregation, scaling, and format transformation) for substituting the generated motion information (or the motion information map) **2510** for the generated motion information map **2150** (e.g., see FIGS. **25** through **28** and related descriptions thereof).

**[0332]** When there is a need to convert the generated motion information map **2150** to a higher resolution in order to code a picture of higher resolution, the apparatuses **100** and **200** may perform upsampling on the generated motion information map **2150**. For example, the same method as a VVC standard may be used as an upsampling method. The present disclosure is not limited thereto, and various upsampling methods (e.g., bi-cubic or DCT-IF interpolation) may be used.

**[0333]** A method of generating the motion information (or the motion information map) **2150**, according to the present disclosure, may be applied not only to coding (e.g., decoding or encoding) of a current picture but also to a frame rate up conversion (FRUC) technique. For example, the method of generating the motion information (or the motion information map) **2150**, according to the present disclosure, may be applied to generating motion information of a virtual reference picture generated according to the FRUC technique.

**[0334]** FIGS. **30** and **31** illustrate a method of utilizing a motion information generating method according to the present disclosure in FRUC.

**[0335]** The method to be described with reference to FIGS. **30** and **31** may be performed by the apparatuses **100** and **200** in the image decoding method **1800** and the image encoding method **2000**, and, in more detail, may be performed by the motion information generators **130** of the apparatuses **100** and **200**.

**[0336]** Referring to FIG. **30**, the apparatuses **100** and **200** may obtain motion information from two reconstructed pictures (e.g., an L**0** reference picture **3010** or an L**1** reference picture **3020**), and may generate a motion information map (or a MV map) for a target picture **3030** that is added by FRUC. For example, the apparatuses **100** and **200** may generate a motion information map **3050** by using the L**0** reference picture **3010** and the L**1** reference picture **3020** as inputs of the motion information generator **130** (e.g., the motion information generation model **2110**), and may perform interpolation or extrapolation on the generated motion vector map **2150**, based on a temporal position of the target picture **3030** to generate a final motion information map **3060** of the target picture **3030**.

**[0337]** Additionally, the apparatuses **100** and **200** may perform refinement on the final motion information map **3060** to generate a refined final motion information map **3070**. For example, the apparatuses **100** and **200** may generate the refined final motion information map **3070** by using, as inputs, not only the final motion information map **3060** but also a motion information map **3080** of the L**0** reference picture **3010** and a motion information map **3090** of the L**1** reference picture **3020**. For example, refinement may include operations such as sampling and weighted average. Alternatively, for example, the refinement may be performed using an artificial neural network model, and, in this case, the apparatuses **100** and **200** may generate the refined final motion information map **3070** by using, as inputs, at least one of the final motion information map **3060**, the motion information map **3080** of the L**0** reference picture **3010**, and the motion information map **3090** of the L**1** reference picture **3020**. For example, the artificial neural network may include a convolutional neural network (CNN), a fully-connected neural network (FCNN), or a combination thereof.

**[0338]** FIG. **31**A illustrates that the final motion information map **3060** is generated by performing interpolation on the generated motion information map **3050** when the target picture **3030** is located between the L**0** reference picture **3010** and the L**1** reference picture **3020**, and FIG. **31**B illustrates that the final motion information map **3060** is generated by performing extrapolation on the generated motion information map **3050** when the target picture **303** is located outside the L**0** reference picture **3010** and the L**1** reference picture **3020**. Although FIG. **31**B illustrates that extrapolation is performed on the generated motion information map **3050** when both the L**0** reference picture **3010** and the L**1** reference picture **3020** are temporally located in front of the target picture **3030**, the final motion information map **3060** may be generated by performing extrapolation on the generated motion information map **3050** even when both the L**0** reference picture **3010** and the L**1** reference picture **3020** are temporally located behind the target picture **3030**.

**[0339]** The motion information map **3060** generated by the method described above with reference to FIGS. **30**, **31**A, **31**B may be used to code (e.g., decode or encode) the target picture **3030**. For example, when a block to be coded in the target picture **3030** is coded in a skip mode, the apparatuses **100** and **200** may determine, as motion information of the block, motion information of a location corresponding to the block in the generated motion information map **3060**. For example, when the block to be coded in the target picture **3030** is coded in a merge mode, the apparatuses **100** and **200** may determine the motion information of the block by adding or inserting the motion information of the position corresponding to the block in the motion information map **3060** as one candidate to or into a merge candidate list. For example, when the block to be coded in the target picture **3030** is coded in an AMVP mode, the apparatuses **100** and **200** may determine the motion information of the block by adding or inserting the motion information of the location corresponding to the block in the motion information map **3060** as one MVP candidate to or into an MVP candidate list.

**[0340]** FIGS. **32**A and **32**B illustrate a method of refining a generated motion information map, according to the present disclosure.

**[0341]** The apparatuses **100** and **200** may perform refinement on the generated motion information (or motion information map) **2150** and **3060** in order to improve the accuracy of the generated motion information (or motion information map) **2150** and **3060**. The apparatuses **100** and **200** may include a motion information refiner **3230** configured to perform refinement on the generated motion information (or the motion information maps) **2150** and **3060** according to the present disclosure, and the motion information refiner **3230** may be implemented by at least one processor.

**[0342]** In the present disclosure, the motion information generator **3230** may operate in the same way as the motion information generator **130**. For example, the motion information refiner **3230** may generate refined motion information **3220**, based on the generated motion information (or the motion information maps) **2150** and **3060** and, additionally, at least one of pixel data and/or motion information of at least one reconstructed picture. The motion information refiner **3230** may process an input and generate an output according to various methods. For example, the motion information refiner **3230** may generate the refined motion information **3220** from input information by using an arithmetic operation (e.g., an average, a median, interpolation or filtering) or generate the refined motion information **3220** from the input information by using an artificial neural network **3210**. For example, the artificial neural network **3210** may use the same artificial neural network model as the artificial neural network (or the motion information generation model **2110**) of the motion information generator **130**, or may use a separate artificial neural network model.

**[0343]** The description of the motion information generator **130** or the motion information generation model **2110** given above with reference to FIGS. **17** through **31** may be equally applied to the motion information refiner **3230** or an artificial neural network model used in the motion information refiner **3230**. When an artificial neural network is used separately from the motion information generator **130** or the motion information generation model **2110** in the motion information refiner **3230**, the motion information generation model **2110** of the motion information generator **130** may be referred to as a first motion information generation model or a first artificial neural network and the motion information generation model **3210** of the motion information refiner **3230** may be referred to as a second motion information generation model or a second artificial neural network, in order to distinguish the two artificial neural networks from each other in the present disclosure. The description of the motion information generator **130** or the motion information generation model **2110** given above with reference to FIGS. **17** through **31** is included in some embodiments.

**[0344]** Referring to FIG. **32**A, an input of the motion information refiner **3230** may include the motion information maps **2150** and **3060** generated by the motion information generator **130** or the motion information generation model **2110**. The input of the motion information refiner **3230** may further include additional information, and, for example, the additional information may include at least one of pixel data or a motion information map of at least one reconstructed picture (e.g., a reference picture).

**[0345]** As a non-limiting example, the input of the motion information refiner **3230** may be a combination as follows.

- **[0346]** Motion information map of current picture (or current MV map)
- **[0347]** Motion information maps of reference picture (or reference MV maps)
- **[0348]** Motion information map of current picture (or current MV map)+motion information map(s) of reference picture (or reference MV map(s))
- **[0349]** Motion information map of current picture (or current MV map)+motion information map(s) of reference picture (or reference MV map(s))+pixel data of current reconstructed picture (or current recon)+pixel data of reference picture(s) (or reference picture(s))
- **[0350]** Motion information map of current picture (or current MV map)+pixel data of current reconstructed picture (or current recon)+pixel data of reference picture(s) (or reference picture(s))
- **[0351]** Motion information maps (or reference MV maps) of reference picture+pixel data of reference pictures (or reference pictures)

**[0352]** FIG. **32**B illustrates, as an example not limiting the present disclosure, generation of the refined motion information map **3220** by using the motion information maps **2150** and **3060** of the current picture, the pixel data of the current reconstructed picture **2130**, and the pixel data of the reference picture **2120** as inputs of the motion information generation model **3210** (or the second motion information generation model).

**[0353]** For example, the input of the motion information refiner **3230** may include at least one of quantization information, a prediction sample, a residual sample, and distance information from the current picture in addition to motion information (e.g., see FIG. **17** and its related description). For example, an output of the motion information refiner **3230** may include at least one of intra-coding related information, residual-coding related information, illumination change information, and statistical information in addition to refined motion information (e.g., see FIG. **17** and its related description).

**[0354]** Motion information (or a motion information map) generated in the present disclosure may be applied to video coding (e.g., decoding or encoding) in various ways.

- **[0355]** Motion information (or a motion information map) of a picture (or a frame) on which video coding is to be performed may be generated in advance before the picture (or the frame) is coded, and the generated motion information (or motion information map) may be utilized in coding the picture (or frame).

**[0356]** For example, when a current block of the picture to be coded is coded in a merge mode or an AMVP mode, motion information of a location corresponding to the current block in the generated motion information map may be used as a merge candidate or MVP candidate. For example, when the current block is coded in a skip mode, the motion information at the location corresponding to the current block in the generated motion information map may be determined as motion information of the current block.

- **[0357]** The motion information map of a current picture (or the current MV map) may be replaced with the generated motion information map and the generated motion information map may be used. For example, the motion information of the generated motion information map may be utilized as temporal motion information during subsequent picture (or frame) coding (e.g., decoding or encoding).
- **[0358]** When FRUC is applied, the generated motion information map may be used to generate a virtual reference picture of the picture (or frame) to be coded.
- **[0359]** A generated motion information map may be used when a dummy picture (or frame) for a dropped picture (or frame).

**[0360]** A method of using generated motion information (or motion information map) in video coding (e.g., decoding or encoding) will now be described in more detail.

**[0361]** In the present disclosure, new syntax information is defined to indicate whether the generated motion information (or motion information map) is used to code a current picture. For convenience of explanation, the new syntax information may be referred to as first information. In the present disclosure, although the first information is represented as use_generated_mv_flag, a name of the first information may be changed.

**[0362]** For example, the first information may indicate whether the generated motion information (or motion information map) is used in coding (e.g., decoding or encoding) a current block, according to a value of the first information. For example, when the first information has a value of **0**, the first information may indicate that the generated motion information (or motion information map) is not used in coding (e.g., decoding or encoding) the current block, and, when the first information has a value of 1, the first information may indicate that the generated motion information (or motion information map) is used in coding (e.g., decoding or encoding) the current block. As another example, the value of the first information may be set in the opposite manner (i.e., when the first information has a value of 0, the first information may indicate that the generated motion information is used in coding the current block, and when the first information has a value of 1, the first information may indicate that the generated motion information is not used in coding the current block).

**[0363]** For example, the current block may be a coding unit (CU) (or a coding block), and the first information may be signaled via a bitstream in units of CUs. As another example, the current block may be a coding tree unit (CTU) (or a coding tree block), may be a transform unit (TU) (or a transform block), or may be a prediction unit (PU) (or a prediction block), and the first information may be signalled through the bitstream in units of CTUs, TUs, or PUs. As another example, the first information may be signaled in units of sequence parameter sets (SPSs), picture parameter sets (PPSs), picture headers (PHs), or slice headers (SHs).

**[0364]** For convenience of explanation, it will now be assumed that the first information is signaled in units of CUs.

**[0365]** Referring to FIG. **33**, when decoding a current picture, the apparatus **100** may obtain first information with respect to the current block from a bitstream (operation **3310**).

**[0366]** When the first information indicates that the generated motion information (or motion information map) is not used in coding the current block (e.g., when a value of the first information is 0), the apparatus **100** may reconstruct or decode the current block by using syntax information of an existing video coding standard (operation **3320**). In the present disclosure, an existing video coding standard may refer to an ITU-T H.266 or Versatile Video Coding (VVC) standard, or an ITU-T H.265 or High Efficiency Video Coding (HEVC) standard. The syntax information of the existing video coding standard is described in detail in an ITU-T H.266/VVC standard document or an ITU-T H.265/HEVC standard document, and this specification includes the entire contents of those standard documents by reference.

**[0367]** When the first information indicates that the generated motion information (or motion information map) is used in coding the current block (e.g., when a value of the first information is 1), the apparatus **100** may determine whether there is generated motion information at a location corresponding to the current block. When it is determined that there is no generated motion information at a location corresponding to the current block, the apparatus **100** may determine intra prediction mode information (e.g., an intra DC mode, an intra plane mode, and an intra angle mode) for the current block by using pieces of intra mode related syntax information of the existing video coding standard, and perform intra prediction on the current block to obtain a predicted sample (operation **3330**).

**[0368]** When it is determined that there is generated motion information at a location corresponding to the current block, the apparatus **100** may determine the motion information at the location corresponding to the current block within the generated motion information map as motion information for the current block, and may perform inter prediction on the current block to obtain a prediction sample (operation **3340**). In operation **3340**, the apparatus **100** may not obtain (or may omit obtaining) relevant syntax information from the bitstream.

**[0369]** After obtaining a predicted sample (or predicted block) through intra prediction or inter prediction, the apparatus **100** may obtain a residual sample by performing inverse transformation and inverse quantization on the current block by using residual-coding related syntax information of the existing video coding standard, and may reconstruct the current block based on the predicted sample and the residual sample.

**[0370]** In the example of FIG. **33**, when encoding the current picture, the apparatus **200** may perform an operation corresponding to an operation of the apparatus **100** described above. For example, the apparatus **200** may determine whether to use the generated motion information (or motion information map) about the current picture to encode the current block and whether to encode the current block in an intra mode or encode the current block in an inter mode.

**[0371]** Based on a result of the determination, the apparatus **200** may encode the first information into the bitstream (operation **3310**). For example, when it is determined that the generated motion information (or motion information map) is not used in encoding the current block, the apparatus **200** may encode the first information into the bitstream with a value of 0, and may encode the syntax information of the existing video coding standard into the bitstream according to a result of the encoding of the current block (operation **3320**).

**[0372]** For example, when it is determined that the generated motion information (or motion information map) is used in encoding the current block, the apparatus **200** may encode the first information into the bitstream with a value of 1, and may determine whether the generated motion information at a location corresponding to the current block exists in the generated motion information map.

**[0373]** When it is determined that there is no generated motion information at a position corresponding to the current block, the apparatus **100** may determine intra prediction mode information for the current block, perform intra prediction on the current block to obtain a predicted sample, and may encode pieces of syntax information (related to an intra mode of the existing video coding standard) indicating the determined intra prediction mode into the bitstream (operation **3330**).

**[0374]** When it is determined that there is generated motion information at a location corresponding to the current block, the apparatus **100** may determine the motion information at the location corresponding to the current block within the generated motion information map as motion information for the current block, and may perform inter prediction on the current block to obtain a prediction sample (operation **3340**). In operation **3340**, the apparatus **100** may not encode (or may omit encoding) relevant syntax information into the bitstream.

**[0375]** The apparatus **200** may obtain a predicted sample (or predicted block) through intra prediction or inter prediction, and then perform residual coding according to the existing video coding standard and encode the related syntax information into the bitstream (operation **3350**).

**[0376]** As described above, the value of the first information may also be encoded in the opposite way.

**[0377]** The operation of the apparatus **100** described above with reference to FIG. **33** may be performed in operation **1820** of the image decoding method **1800**, and the operation of the apparatus **200** may be performed in operation **2020** of the image encoding method **2000**.

**[0378]** As described above, the generated motion information (or the generated motion information map) may be generated by the motion information generator **130** or the motion information refiner **3230**, and, for example, the generated motion information (or the generated motion information map) may be generated through an artificial neural network (e.g., the motion information generation models **2110** and **3210**) or may be generated according to various methods (e.g., an arithmetic operation) not using an artificial neural network.

**[0379]** FIGS. **34**A and **34**B illustrate a syntax structure and a coding method according to the present disclosure.

**[0380]** The syntax structure of FIGS. **34**A and **34**B is obtained by changing the syntax information in operation **3340** from the syntax structure of FIG. **33**. Accordingly, the descriptions of operations **3310**, **3320**, **3330**, and **3350** in the syntax structure and the coding method of FIG. **33** are identical to those of FIGS. **34**A and **34**B, and are incorporated in some embodiments.

**[0381]** In the present disclosure, when the generated motion information (or motion information map) is used, new syntax information indicating whether a motion vector difference (MVD) additionally exists in a bitstream may be defined. In the present disclosure, the syntax information may be referred to as second information. In the present disclosure, although the second information is represented as mvd_flag, a name of the second information may be changed.

**[0382]** For example, the second information may indicate whether syntax information related to the MVD is additionally present in the bitstream when the generated motion information (or motion information map) is used, according to a value of the second information. For example, when the second information has a value of 0, the second information may indicate that syntax information related to the MVD does not exist in the bitstream, and, when the second information has a value of 1, the second information may indicate that the syntax information related to the MVD exists in the bitstream. As another example, the value of the second information may be set in the opposite manner.

**[0383]** In the present disclosure, new syntax information generating a candidate list including motion information of a location corresponding to a current block and motion information of a location corresponding to at least one (spatial) neighboring block of the current block in the generated motion information map and indicating one piece of motion information in the candidate list may be defined, and may be referred to as third information in the present disclosure. In the present disclosure, although the third information is represented as mv_index, a name of the third information may be changed. For example, the third information may have a value less than or equal to the maximum number of candidate lists, and the third information may indicate a motion information candidate at a location corresponding to the value of the third information among the candidate lists.

**[0384]** Referring to FIG. **34**A, when it is determined that generated motion information at a location corresponding to the current block exists, the apparatus **100** may obtain the second information (e.g., mvd_flag) from the bitstream (operation **3410**).

**[0385]** When the second information indicates that MVD-related syntax information for the current block exists in the bitstream, the apparatus **100** may obtain the MVD-related syntax information from the bitstream, and obtain an MVD for the current block based on the obtained information (operation **3420**). The apparatus **100** may obtain a motion vector predictor (MVP) from the generated motion information map, obtain a motion vector for the current block based on the MVD obtained from the bitstream and the MVP obtained from the generated motion information map, and perform inter prediction on the current block by using the obtained motion vector to thereby obtain a predicted sample.

**[0386]** Moreover, in operation **3420**, the third information (e.g., mv_index) may be signaled via the bitstream to indicate the MVP in the generated motion information map. In this case, the apparatus **100** may obtain the third information from the bitstream, and may determine, as the MVP, a motion information candidate indicated by the third information from among the motion information candidates of the generated motion information map.

**[0387]** When the second information indicates that the MVD-related syntax information for the current block does not exist in the bitstream, the apparatus **100** may determine the motion information at the location corresponding to the current block within the generated motion information map as the motion information for the current block, and may perform inter prediction to obtain a prediction sample for the current block (operation **3430**). In operation **3430**, the apparatus **100** may not obtain (or may omit obtaining) relevant syntax information from the bitstream. Thereafter, the apparatus **100** may reconstruct the current block by performing the operation described above with reference to **3350**.

**[0388]** When encoding the current picture, the apparatus **200** may perform an operation corresponding to an operation of the apparatus **100** described above. For example, when it is determined that the generated motion information of a location corresponding to the current block exists, the apparatus **200** may determine whether to encode the MVD-related syntax information into the bitstream, and may encode the second information with a corresponding value into the bitstream, based on a result of the determination (operation **3410**). Based on the second information, the apparatus **200** may encode the MVD-related syntax information for the current block into the bitstream and encode the third information indicating the MVP into the bitstream (operation **3420**), or may encode the current block (in an inter mode) by using motion information at a location corresponding to the current block in the generated motion information map (operation **3430**). In operation **3430**, the apparatus **200** may not encode (or may omit encoding) relevant syntax information into the bitstream. Thereafter, the apparatus **200** may generate the bitstream by performing the operation described above with reference to **3350**.

**[0389]** Referring to FIG. **34**B, when it is determined that generated motion information at a location corresponding to the current block exists, the apparatus **100** may obtain the third information from the bitstream (operation **3440**). For example, the apparatus **100** may determine, as the motion information for the current block, the motion information indicated by the third information among the motion information of the location corresponding to the current block in the generated motion information map and the motion information of a location corresponding to at least one neighboring block of the current block (operation **3450**).

**[0390]** The apparatus **100** may perform inter prediction by using the motion information determined in operation **3450** to thereby obtain a predicted sample. Thereafter, the apparatus **100** may reconstruct the current block by performing the operation described above with reference to **3350**.

**[0391]** In FIG. **34**B, when encoding the current picture, the apparatus **200** may perform an operation corresponding to an operation of the apparatus **100** described above. For example, when it is determined that there is generated motion information at a position corresponding to the current block, the apparatus **200** may determine motion information corresponding to the motion information for the current block among the motion information of the location corresponding to the current block in the generated motion information map and the motion information of a location corresponding to at least one neighboring block of the current block (operation **3450**), and may encode the third information indicating the determined motion information into the bitstream (operation **3440**). Thereafter, the apparatus **200** may generate the bitstream by performing the operation described above with reference to **3350**.

**[0392]** The operation of the apparatus **100** described above with reference to FIGS. **34**A and **34**B may be performed in operation **1820** of the image decoding method **1800**, and the operation of the apparatus **200** may be performed in operation **2020** of the image encoding method **2000**.

**[0393]** New flag information may be determined so that one of the syntax structures and coding methods of FIGS. **34**A and **34**B may be selected, allowing one of the two modes to be selected. In the disclosure, this flag information may be referred to as fourth information. In the present disclosure, although the fourth information is represented as no_mvd_flag, a name of the fourth information may be changed.

**[0394]** Accordingly, a more accurate motion vector (MV) corresponding to the current block may be found by transmitting a small amount of side information, thereby increasing coding efficiency.

**[0395]** FIG. **35** illustrates a syntax structure and a coding method according to the present disclosure.

**[0396]** The syntax structure of FIGS. **34**A and **34**B is obtained by changing the syntax information in operation **3340** from the syntax structure of FIG. **33**. Accordingly, the descriptions of operations **3310**, **3320**, **3330**, and **3350** in the syntax structure and the coding method of FIG. **33** are identical to those of FIG. **35**, and are incorporated in some embodiments.

**[0397]** In the example of FIG. **35**, the fourth information (e.g., no_mvd_flag) may be used to select one of the method of FIG. **34**A and the method of FIG. **34**B. For example, the fourth information may indicate whether a method in which MVD-related syntax information is signaled (or the method illustrated in FIG. **34**(*a*)) is used or whether a method of determining the motion information indicated by the third information (e.g., mv_index) among the generated motion information map as the motion information of the current block (or the method illustrated in FIG. **34**B) is used, according to a value of the fourth information. For example, when the fourth information has a value of 1, the fourth information may indicate that the method in which MVD-related syntax information is signaled (or the method illustrated in FIG. **34**A) is used, and, when the fourth information has a value of 0, the fourth information may indicate that the method of determining the motion information indicated by the third information (e.g., mv_index) among the generated motion information map as the motion information of the current block (or the method illustrated in FIG. **34**B) is used.

**[0398]** As another example, when the fourth information has a value of 0, the fourth information may indicate that the method in which MVD-related syntax information is signaled (or the method illustrated in FIG. **34**A) is used, and, when the fourth information has a value of 1, the fourth information may indicate that the method of determining the motion information indicated by the third information (e.g., mv_index) among the generated motion information map as the motion information of the current block (or the method illustrated in FIG. **34**B) is used.

**[0399]** Based on the fourth information, the apparatuses **100** and **200** may decode or encode the current block by performing the operation described above with reference to **3420** of FIG. **34**A or performing the operation described above with reference to **3440** and **3450** of FIG. **34**B. Descriptions of operations **3420**, **3440**, and **3450** are included in some embodiments.

**[0400]** FIG. **36** illustrates a syntax structure and a coding method according to the present disclosure.

**[0401]** The syntax structure of FIG. **36** is obtained by changing the syntax information in operations **3330** and **3340** from the syntax structure of FIG. **33**. Accordingly, the descriptions of operations **3310**, **3320**, and **3350** in the syntax structure and the coding method of FIG. **33** are identical to those of FIG. **36**, and are incorporated in some embodiments.

**[0402]** Referring to FIG. **36**, when performing image decoding, the apparatus **100** may determine whether motion information (or motion information map) generated for the current picture is used to code the current block, based on the first information (e.g., use_generated_mv_flag) (operation **3310**). When the first information indicates that the generated motion information (or motion information map) is used in coding the current block (e.g., when a value of the first information is 1), the apparatus **100** may determine that the current block is not coded in an intra mode, determine the motion information of a location corresponding to the current block in the generated motion information map as the motion information for the current block, and perform inter prediction on the current block to obtain a predicted sample (operation **3610**). In operation **3610**, the apparatus **100** may not obtain (or may omit obtaining) relevant syntax information from the bitstream. Thereafter, the apparatus **100** may reconstruct the current block by performing the operation described above with reference to **3350**.

**[0403]** When there is no motion information at a location corresponding to the current block in the generated motion information map, the apparatus **100** may search for other motion information and use found motion information to decode the current block.

**[0404]** Unlike the example of FIG. **33**, in the example of FIG. **36**, when the current block is coded in an intra mode, the apparatus **100** may perform decoding according to the syntax information of the existing video coding standard. That is, in the example of FIG. **36**, when the current block is coded in an intra mode, the first information for the current block may be set to indicate that the generated motion information (or motion information map) is not used to code the current block.

**[0405]** In the example of FIG. **36**, when encoding the current picture, the apparatus **100** may perform an operation corresponding to an operation of the apparatus **100** described above. For example, the apparatus **200** may determine whether to use the generated motion information (or motion information map) of the current picture to encode the current block, and may encode the first information to the bitstream, based on a result of the determination.

**[0406]** When it is determined that the generated motion information (or motion information map) is used to encode the current block, the apparatus **200** may determine the motion information at the location corresponding to the current block within the generated motion information map as the motion information about the current block, and may perform inter prediction on the current block to obtain a predicted sample (operation **3610**). In operation **3610**, the apparatus **200** may not encode (or may omit encoding) relevant syntax information into the bitstream. Thereafter, the apparatus **200** may generate the bitstream by performing the operation described above with reference to **3350**.

**[0407]** The operation of the apparatus **100** described above with reference to FIG. **36** may be performed in operation **1820** of the image decoding method **1800**, and the operation of the apparatus **200** may be performed in operation **2020** of the image encoding method **2000**.

**[0408]** FIGS. **37**A and **37**B illustrate a syntax structure and a coding method according to the present disclosure.

**[0409]** In the example of FIGS. **37**A and **37**B, a coding method using motion information (or motion information map) generated according to the present disclosure may be set to operate instead of a skip mode. Therefore, in the example of FIGS. **37**A and **37**B, the first information may be signaled via the bitstream in case of an inter picture (or a P slice or a B slice). When the first information indicates that the generated motion information (or motion information map) is not used to code the current block, mode-related syntax information of a video coding standard excluding an existing skip mode may be used, and existing residual-coding related syntax information may also be used. On the other hand, in the example of FIGS. **37**A and **37**B, when the first information indicates that the generated motion information (or motion information map) is used to code the current block, syntax information for the current block may not be signaled through the bitstream, and the existing residual-coding related syntax information may also not exist in the bitstream.

**[0410]** In the example of FIG. **37**A, when decoding the current picture, the apparatus **100** may determine whether the current picture (or slice) is an inter picture (or a picture coded in an inter mode) (or a P slice or a B slice) (operation **3710**). When the current picture (or slice) is an inter picture (or a picture coded in an inter mode) (or a P slice or a B slice), the apparatus **100** may obtain the first information of the current block from the bitstream (operation **3720**).

**[0411]** When the first information indicates that the generated motion information (or motion information map) is not used in coding the current block (e.g., when a value of the first information is 0), the apparatus **100** may reconstruct or decode the current block by using every relevant syntax information excluding a skip mode of the existing video coding standard (operation **3730**).

**[0412]** When the first information indicates that the generated motion information (or motion information map) is used in coding the current block (e.g., when a value of the first information is 1), the apparatus **100** may determine that the current block is not coded in an intra mode, determine the motion information of a location corresponding to the current block in the generated motion information map as the motion information for the current block, and obtain a sample of the current block (operation **3740**). In operation **3740**, the apparatus **100** may not obtain (or may omit obtaining) relevant syntax information from the bitstream. In operation **3740**, residual coding is not performed.

**[0413]** When there is no motion information at a location corresponding to the current block in the generated motion information map, the apparatus **100** may search for other motion information and use found motion information to decode the current block. Also, in operation **3740**, when the current block is coded in an intra mode, the first information for the current block may be set to indicate that the generated motion information (or motion information map) is not used to code the current block.

**[0414]** In the example of FIGS. **37**A and **37**B, when encoding the current picture, the apparatus **200** may perform an operation corresponding to an operation of the apparatus **100** described above. For example, the apparatus **200** may determine whether the current picture (or slice) is an inter picture (or a picture coded in an inter mode) (or a P slice or a B slice), and, when the current picture (or slice) is an inter picture (or a picture coded in an inter mode) (or a P slice or a B slice), may encode the first information into the bitstream. Of course, the apparatus **200** may determine whether to use the generated motion information (or motion information map) of the current picture to encode the current block and whether to encode the current block in an intra mode or encode the current block in an inter mode.

**[0415]** When a mode (including an intra mode) excluding a skip mode is applied to the current block, the apparatus **200** may encode the first information so that the first information indicates that the generated motion information (or motion information map) is not used to encode the current block. On the other hand, when a skip mode is applied to the current block, the apparatus **200** may encode the first information so that the first information indicates that the generated motion information (or motion information map) is used to encode the current block.

**[0416]** For example, when it is determined that the generated motion information (or motion information map) is not used in encoding the current block, the apparatus **200** may encode the first information into the bitstream with a value of 0 (operation **3720**), and may encode the syntax information of the existing video coding standard into the bitstream according to a result of the encoding of the current block (operation **3730**).

**[0417]** For example, when it is determined that the generated motion information (or motion information map) is used to encode the current block, the apparatus **200** may obtain a sample for the current block by using the motion information at the location corresponding to the current block within the generated motion information map (operation **3740**). In operation **3740**, the apparatus **200** may not encode (or may omit encoding) relevant syntax information into the bitstream. In operation **3740**, residual coding is not performed.

**[0418]** FIG. **37**B is obtained by changing the operation **3740** of FIG. **37**A. Accordingly, the descriptions of operations **3710**, **3720**, and **3730** in the description of FIG. **37**A are identical to those of FIG. **37**B, and are incorporated in some embodiments.

**[0419]** In the example of FIG. **37**B, when the first information indicates that the generated motion information (or motion information map) is used to code the current block, a method in which the third information indicates the motion information in the generated motion information map by signaling the third information through the bitstream may be used instead of a method of using, without changes, the motion information at the location corresponding to the current block in the generated motion information map.

**[0420]** In the example of FIG. **37**B, when decoding the current picture, the apparatus **100** may determine whether the generated motion information map is used to code the current block, based on the first information (operation **3720**). When the first information indicates that the generated motion information map is used in coding the current block (e.g., when a value of the first information is 1), the apparatus **100** may obtain the third information (e.g., mv_index) from the bitstream (operation **3750**). For example, the apparatus **100** may obtain a sample of the current block by using the motion information indicated by the third information from among the motion information of the location corresponding to the current block in the generated motion information map and the motion information of a location corresponding to at least one neighboring block of the current block (operation **3760**).

**[0421]** In FIG. **37**B, when encoding the current picture, the apparatus **200** may perform an operation corresponding to an operation of the apparatus **100** described above. For example, when it is determined that the generated motion information map is used to code the current block, the apparatus **200** may determine motion information corresponding to the motion information for the current block among the motion information of the location corresponding to the current block in the generated motion information map and the motion information of a location corresponding to at least one neighboring block of the current block (operation **3760**), and may encode the third information indicating the determined motion information into the bitstream (operation **3750**).

**[0422]** The operation of the apparatus **100** described above with reference to FIG. **37** may be performed in operation **1820** of the image decoding method **1800**, and the operation of the apparatus **200** may be performed in operation **2020** of the image encoding method **2000**.

**[0423]** FIG. **38** illustrates a syntax structure and a coding method according to the present disclosure.

**[0424]** In FIG. **38**, when the current block is coded in a skip mode or coded in a merge mode, the first information (e.g., use_generated_mv_flag) may be signaled through the bitstream, and the merge mode may be applied to the current block or the generated motion information map may be used according to the first information. In the example of FIG. **38**, when the current block is coded in an AMVP mode, the generated motion information map is not used, and motion information is signaled in the AMVP mode according to the existing video coding standard.

**[0425]** In the example of FIG. **38**, when decoding the current picture, the apparatus **100** may determine whether the current picture (or slice) is an inter picture (or a picture coded in an inter mode) (or a P slice or a B slice) (operation **3810**). When the current picture (or slice) is an inter picture (or a picture coded in an inter mode) (or a P slice or a B slice), the apparatus **100** may obtain information (e.g., skip_flag) indicating whether the current block is a skip mode. When information indicating whether the current block is a skip mode has a value of 0, the information may indicate that the current block is not the skip mode, and, when the information indicating whether the current block is a skip mode has a value of 1, the information may indicate that the current block is the skip mode (or vice versa).

**[0426]** When the current block is not a skip mode, the apparatus **100** may determine whether the current block is an intra block or an inter block (or whether the current block is coded in an intra mode or in an inter mode), and may obtain, from the bitstream, information (e.g., merge_flag) indicating whether the current block is a merge mode (operation **3820**). When information indicating whether the current block is a merge mode has a value of 0, the information may indicate that the current block is not the merge mode, and, when the information indicating whether the current block is a merge mode has a value of 1, the information may indicate that the current block is the merge mode (or vice versa).

**[0427]** When the current block is a merge mode, the apparatus **100** may obtain first information from a bitstream (operation **3830**). When the first information indicates that the generated motion information map is not used to code the current block, the apparatus **100** may obtain syntax information (e.g., merge_index) related to the merge mode from the bitstream, obtain motion information for the current block in the merge mode, and perform inter prediction to obtain a predicted sample (operation **3840**).

**[0428]** When the first information indicates that the generated motion information map is used to code the current block, the apparatus **100** may determine the motion information at the location corresponding to the current block within the generated motion information map as the motion information about the current block, and may perform inter prediction to obtain a predicted sample (operation **3850**). Alternatively, the apparatus **100** may obtain third information (e.g., mv_index) from the bitstream, and obtain a predicted sample by performing inter prediction by using the motion information indicated by the third information from among the motion information of the location corresponding to the current block in the generated motion information map and the motion information of a location corresponding to at least one neighboring block of the current block (operation **3850**).

**[0429]** When the current block is not a merge mode, the apparatus **100** may obtain relevant syntax information from the bitstream according to the AMVP mode of the existing video coding standard, obtain motion information in the AMVP mode, and perform inter prediction to obtain a predicted sample (operation **3860**).

**[0430]** Thereafter, the apparatus **100** may obtain a residual sample by performing inverse transformation and inverse quantization on the current block by using residual-coding related syntax information of the existing video coding standard, and may reconstruct the current block based on the predicted sample obtained through inter prediction and the residual sample.

**[0431]** When the current block is a skip mode, the apparatus **100** may perform the operations described above in relation to the operations **3830**, **3840**, and **3850**. The descriptions of operations **3830**, **3840**, and **3450** are included in some embodiments. Consequently, in the example of FIG. **38**, the first information and a related coding method may be applied when the current block is a merge mode or a skip mode.

**[0432]** In the example of FIG. **38**, when encoding the current picture, the apparatus **200** may perform an operation corresponding to an operation of the apparatus **100** described above.

**[0433]** FIGS. **39**A and **39**B illustrate use of generated motion information in a merge mode according to the present disclosure.

**[0434]** In the present disclosure, the merge mode refers to a method of generating a candidate list for motion information of a current block including at least one of a spatial neighboring block and a temporal neighboring block of a current block in a current picture, and determining, as the motion information of the current block, a candidate indicated by index information signaled through a bitstream from the generated candidate list (e.g., see FIG. **16** and its related description).

**[0435]** Referring to FIG. **39**A, generation of the candidate list in the merge mode of the existing video coding standard (e.g., the ITU-T H.265/HEVC standard) is illustrated. In the example of FIG. **39**A, A**0**, A**1**, A**2**, B**0**, and B**1** represent merge candidates of spatial neighboring blocks of the current block within the current picture including the current block, and Col represents a merge candidate of a temporal neighboring block (or a corresponding block or a co-located block or col-located block) corresponding to the current block within a picture different from the current block. A location of a neighboring block that may be used as a merge candidate is illustrated in FIG. **40**.

**[0436]** In the example of FIG. **39**A, availableFlagX (X=A**0**, A**1**, A**2**, B**0**, B**1**, Col) is information indicating availability of motion information of A neighboring block of the current block (e.g., see FIG. **16** and its related description). When the motion information of the neighboring block is available (e.g., when availableFlagX has a value of 1), the neighboring block may be added or inserted to or into a merge candidate list. On the other hand, when the motion information of the neighboring block is not available (e.g., when availableFlagX has a value of 0), the neighboring block may be added or inserted to or into the merge candidate list. As the number of merge candidates added to the merge candidate list increases, more accurate motion information may be determined for the current block, thereby improving coding efficiency.

**[0437]** In the present disclosure, the generated motion information may be added as one of the merge candidates to thereby improve the coding performance of the merge mode. FIG. **39**B illustrates addition of the generated motion information as one of the merge candidates, according to the present disclosure.

**[0438]** Referring to FIG. **39**B, a merge candidate of the generated motion information (or motion information map) according to the present disclosure is indicated by GeneratedMV, but the name may be changed. When the motion information at the location corresponding to the current block in the generated motion information map exists, the merge candidate (e.g., GeneratedMV) of the generated motion information (or motion information map) may be determined to be available. On the other hand, when the motion information at the location corresponding to the current block in the generated motion information map does not exist, the merge candidate (e.g., GeneratedMV) of the generated motion information (or motion information map) may be determined to be not available.

**[0439]** When the current block is a merge mode, the apparatus **100** may determine whether the merge candidate (e.g., GeneratedMV) of the generated motion information (or motion information map) is available, and, when the merge candidate (e.g., GeneratedMV) of the generated motion information (or motion information map) is available, may add or insert the merge candidate (e.g., GeneratedMV) of the generated motion information (or motion information map) to or into the merge candidate list (operation **3910**). For example, the merge candidate (e.g. GeneratedMV) may be added or inserted as a candidate at the beginning of the merge candidate list, but the order may be changed and the merge candidate (e.g. GeneratedMV) may be added or inserted at any location.

**[0440]** Then, the apparatus **100** may obtain merge index information (e.g., merge_idx or merge_index) for the current block from the bitstream, and may determine a merge candidate corresponding to the merge index information in the merge candidate list as motion information for the current block. The apparatus **100** may perform inter prediction by using the determined motion information and obtain a predicted sample.

**[0441]** When encoding the current block, the apparatus **200** may perform an operation corresponding to an operation of the apparatus **100** described above. For example, the apparatus **200** may determine the motion information for the current block, generate a merge candidate list including the merge candidate (e.g., GeneratedMV) of the generated motion information (or motion information map), obtain merge index information indicating a merge candidate corresponding to the determined motion information of the current block, and encode the merge index information into a bitstream.

**[0442]** The operation of the apparatus **100** described above with reference to FIG. **39**B may be performed in operation **1820** of the image decoding method **1800**, and the operation of the apparatus **200** may be performed in operation **2020** of the image encoding method **2000**. FIGS. **41**A and **41**B illustrate a method of determining a location corresponding

**[0443]** to a current block in a generated motion information map, according to the present disclosure. As described above, in the generated motion information map according to the present disclosure, motion information may be generated and stored in units of specific-sized blocks or in units of M×N pixels (e.g., 4×4, 8×8, or 16×16). When the size of the current block is the same as the size of a unit in which motion information is generated/stored in the generated motion information map, the generated motion information at the location may be used to code (e.g., decode or encode) the current block.

**[0444]** In the example of FIGS. **41**A and **41**B, it is assumed that the current block is a CU (or a coding block). The present disclosure is not limited thereto, and the current block may be a CTU (or a coding tree block), a TU (or a transform block), or a PU (or a prediction block). A location of an upper left sample of the current block is represented as (Cx, Cy), and a width and a height of the current block are represented as W and H, respectively. Moreover, for convenience of explanation, a block size or unit in which motion information is generated/stored in the generated motion information map is referred to as a generated motion information storing unit.

**[0445]** FIG. **41**A illustrates a method of determining the location corresponding to the current block in the generated motion information map when the generated motion information storing unit is less than the size of the current block (or when the size of the current block is greater than the generated motion information storing unit). For the case illustrated in FIG. **41**A, the apparatuses **100** and **200** may select or determine motion information corresponding to a sample location of (Cx+W/2, Cy+H/2) in the generated motion information map as the motion information corresponding to the current block.

**[0446]** Referring to FIG. **41**A, as a non-limiting example, when the current block is (Cx, Cy)=(0, 0), W=32, H=32 and the generated motion information storing unit is 8×8, Cx+W/2=16 and Cy+H/2=16, and thus the apparatuses **100** and **200** may select or determine motion information corresponding to a (16, 16) sample location (e.g., a "C" location in FIG. **41**A) as the motion information corresponding to the current block.

**[0447]** FIG. **41**B illustrates a method of determining the location corresponding to the current block in the generated motion information map when a size of the current block is less than the generated motion information storing unit (or when the size of the current block is greater than the generated motion information storing unit). For the case illustrated in FIG. **41**B, the apparatuses **100** and **200** may select or determine motion information corresponding to a sample location of (Cx+W/2, Cy+H/2) in the generated motion information map as the motion information corresponding to the current block.

**[0448]** Referring to FIG. **41**B, as a non-limiting example, when the current block is (Cx, Cy)=(16, 16), W=4, H=8 and the generated motion information storing unit is 8×8, Cx+W/2=18 and Cy+H/2=20, and thus the apparatuses **100** and **200** may select or determine motion information corresponding to a (20, 16) sample location (e.g., a "C" location in FIG. **41**B) as the motion information corresponding to the current block.

**[0449]** FIG. **41**C illustrates a method of determining the location corresponding to the current block in the generated motion information map when the current block spans the generated motion information storing unit. Even in this case, the apparatuses **100** and **200** may select or determine motion information corresponding to a sample location of (Cx+W/2, Cy+H/2) in the generated motion information map as the motion information corresponding to the current block.

**[0450]** Referring to FIG. **41**C, as a non-limiting example, when the current block is (Cx, Cy)=(12, 16), W=8, H=32 and the generated motion information storing unit is 8×8, Cx+W/2=16 and Cy+H/2=20, and thus the apparatuses **100** and **200** may select or determine motion information corresponding to a (16, 20) sample location (e.g., a "C" location in FIG. **41**C) as the motion information corresponding to the current block.

**[0451]** When it is determined that there is no motion information generated at a location (e.g., the location C) determined through the method described above with reference to FIG. **41**, the apparatuses **100** and **200** may scan pieces of generated motion information within a block corresponding to the size of the current block in the generated motion information map in a serial order and set or determine motion information found first as the motion information of the current block.

**[0452]** For example, in the example of FIG. **41**A, the apparatuses **100** and **200** may search for motion information in the order of C→a→b→c→d in the generated motion information map and set or determine the motion information found first as the motion information of the current block. As another example, the apparatuses **100** and **200** may perform a search in a raster scan order from a top-left position block (a) after the location C and set or determine motion information found first as the motion information of the current block.

**[0453]** When motion information is not found from a block corresponding to the size of the current block in the generated motion information map, motion information may be searched for from a neighboring block location of the block corresponding to the size of the current block, and motion information found first may be set or determined as the motion information of the current block. All or some of adjacent blocks around the block corresponding to the size of the current block in the generated motion information map may be searched for in a series of scan orders. For example, when a location corresponding to an upper left side of the current block in the generated motion information map is (Cx, Cy), the width is W, and the height is H, pieces of motion information of locations may be searched for from the generated motion information map in the following order, and motion information found first may be set or determined as the motion information of the current block (e.g., see FIG. **41**D).

$$(Cx-1,\ Cy-1)(\text{e.g., } 4102) \rightarrow \left(Cx+w/2,\ Cy-1\right)(\text{e.g., } 4104) \rightarrow$$
$$(Cx+w,\ Cy-1)(\text{¶, } 4106) \rightarrow \left(Cx-1,\ Cy+h/2\right)(\text{e.g., } 4108) \rightarrow$$
$$\left(Cx+w,\ Cy+h/2\right)(\text{e.g., } 4110) \rightarrow (Cx-1,\ Cy+h)(\text{e.g., } 4112) \rightarrow$$
$$\left(Cx+w/2,\ Cy+h\right)(\text{e.g., } 4114) \rightarrow (Cx+w,\ Cy+h)(\text{e.g., } 4116)$$

**[0454]** When no motion information is found from the generated motion information map, the apparatuses **100** and **200** may set the generated motion information for the current block to be a value of 0 (e.g., a zero vector (0, 0)) or may determine that it is in an intra mode. Moreover, during the search process, a pruning process may be additionally performed to prevent duplicate candidates.

**[0455]** FIGS. **42**A-**42**C illustrate a method of performing subblock prediction by using generated motion information, according to the present disclosure.

**[0456]** FIG. **42**A illustrates execution of motion compensation (MC) in units of subblocks of the current block by using the generated motion information, FIG. **42**B illustrates execution of affine motion compensation by using the generated motion information as a control point vector of a 4-parameter model, and FIG. **42**C illustrates execution of affine motion compensation by using the generated motion information as a control point vector of a 6-parameter model.

**[0457]** Referring to FIG. **42**A, the apparatuses **100** and **200** may divide the current block into sub-blocks each having a size equal to or greater than a unit block size (e.g., 4×4, 8×8, or 16×16) of the generated motion information, and may perform motion compensation by allocating the generated motion information in units of subblocks.

**[0458]** As a non-limiting example, in the example of FIG. **42**A, assuming that the size of the current block is 32×32 and the unit block size of the generated motion information is 8×8, the current block is divided into subblocks each having the same size (e.g., 8×8) as or a greater size (e.g., 16×16) than the unit block size of the generated motion information, and then obtain motion information from the generated motion information map in units of subblocks and perform motion compensation with respect to the subblocks by using the obtained (generated) motion information.

**[0459]** Referring to FIG. **42**B, the apparatuses **100** and **200** may allocate generated motion information at locations corresponding to a first control point cp**0** and a second control point cp**1** from the generated motion information map as first control point motion information and second control point motion information, and may obtain motion information for each subblock of the current block by using an affine 4-parameter model. In the example of FIG. **42**B, the apparatuses **100** and **200** may obtain a subblock motion vector for each subblock, based on the affine 4-parameter model, by using Equation 1. In Equation 1, (cp**0***x*, cp**0***y*) represents a first control point motion vector obtained from the generated motion information map, (cp**1***x*, cp**1***y*) represents a second control point motion vector obtained from the generated motion information map, (mvx, mvy) represents a motion vector at a subblock location (x, y) of the current block, and w represents a width of the current block.

$$mv_x = \frac{cp_{1x} - cp_{0x}}{w}x - \frac{cp_{1y} - cp_{0y}}{w}y + cp_{0x} \qquad \text{[Equation 1]}$$
$$mv_y = \frac{cp_{1y} - cp_{0y}}{w}x + \frac{cp_{1x} - cp_{0x}}{w}y + cp_{0y}$$

**[0460]** The apparatuses **100** and **200** may perform motion compensation for each subblock by using the generated motion information and the motion vector for each subblock obtained based on the affine 4-parameter model.

**[0461]** Referring to FIG. **42**C, the apparatuses **100** and **200** may allocate generated motion information at locations corresponding to a first control point cp**0**, a second control point cp**1**, and a third control point cp**2** from the generated motion information map as first control point motion information, second control point motion information, and third control point motion information, and may obtain motion information for each subblock of the current block by using an affine 6-parameter model. In the example of FIG. **42**C, the apparatuses **100** and **200** may obtain a subblock motion vector for each subblock, based on the affine 6-parameter model, by using Equation 2. In Equation 1, (cp**0***x*, cp**0***y*) represents a first control point motion vector obtained from the generated motion information map, (cp**1***x*, cp**1***y*) represents a second control point motion vector obtained from the generated motion information map, (cp**2***x*, cp**2***y*) represents a third control point motion vector obtained from the generated motion information map, (mvx, mvy) represents a motion vector at a subblock location (x, y) of the current block, and w and h represent a width and a height of the current block.

$$mv_x = \frac{cp_{1x} - cp_{0x}}{w}x + \frac{cp_{2x} - cp_{0x}}{h}y + cp_{0x} \qquad \text{[Equation 2]}$$
$$mv_y = \frac{cp_{1y} - cp_{0y}}{w}x + \frac{cp_{2y} - cp_{0y}}{h}y + cp_{0y}$$

**[0462]** The apparatuses **100** and **200** may perform motion compensation for each subblock by using the generated motion information and the motion vector for each subblock obtained based on the affine 6-parameter model.

**[0463]** The above-described embodiments of the disclosure can be written as computer-executable programs, and the written computer-executable programs can be stored in a medium.

**[0464]** The medium may continuously store the computer-executable programs, or temporarily store the computer-executable programs for execution or downloading. Also, the medium may be any one of various recording media or storage media in which a single piece or plurality of pieces of hardware are combined, and the medium is not limited to a medium directly connected to a computer system, but may be distributed on a network. Examples of the medium include a magnetic medium (e.g., a hard disk, a floppy disk, or a magnetic tape), an optical medium (e.g., a compact disk-read-only memory (CD-ROM) or a digital versatile disk (DVD), a magneto-optical medium (e.g., a floptical disk), and a ROM, a random-access memory (RAM), and a flash memory, which are configured to store program instructions. The machine-readable storage medium may be provided as a non-transitory storage medium. The 'non-transitory storage medium' is a tangible device and means that it does not contain a signal (e.g., electromagnetic waves). This term does not distinguish a case in which data is stored semi-permanently in a storage medium from a case in which data is temporarily stored. For example, the non-transitory recording medium may include a buffer in which data is temporarily stored.

**[0465]** Other examples of the medium include recording media and storage media managed by application stores distributing applications or by websites, servers, and the like supplying or distributing other various types of software.

**[0466]** According to an embodiment, methods according to various provided embodiments may be provided by being included in a computer program product. The computer program product, which is a commodity, may be traded between sellers and buyers. Computer program products are distributed in the form of device-readable storage media (e.g., compact disc read only memory (CD-ROM)), or may be distributed (e.g., downloaded or uploaded) through an application store (e.g., Play Store™) or between two user devices (e.g., smartphones) directly and online. In the case of online distribution, at least a portion of the computer program product (e.g., a downloadable app) may be stored at least temporarily in a device-readable storage medium, such as a memory of a manufacturer's server, a server of an application store, or a relay server, or may be temporarily generated.

**[0467]** While one or more embodiments of the disclosure have been described with reference to the figures, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope as defined by the following claims.

What is claimed is:

**1**. An image decoding method performed by an apparatus, the method comprising:

generating reference motion information for at least one block included in a current picture, based on at least one of: pixel data of at least one reconstructed picture, or motion information of the at least one reconstructed picture;

obtaining motion information for a current block among the at least one block included in the current picture, based on the reference motion information and at least one syntax information obtained from a bitstream; and

reconstructing the current block, based on the motion information for the current block.

**2**. The image decoding method of claim **1**, wherein

the generating of the reference motion information for the at least one block included in the current picture comprises:

generating the reference motion information for the at least one block included in the current picture based on inputting at least one of: the pixel data of the at least one reconstructed picture, or the motion information of the at least one reconstructed picture to a neural network.

**3**. The image decoding method of claim **1**, wherein

the motion information for the current block includes at least one of first motion vector information, second motion vector information, first prediction list utilization information, second prediction list utilization information, or prediction mode information.

**4**. The image decoding method of claim **1**, further comprising:

performing at least one of:

aggregating the reference motion information,

scaling the reference motion information, or

transforming a format of the reference motion information.

**5**. The image decoding method of claim **1**, wherein

the generating of the reference motion information for the at least one block included in the current picture comprises:

down-sampling the at least one reconstructed picture; and

generating the reference motion information for the at least one block included in the current picture, based on at least one of: pixel data of the downsampled reconstructed picture or motion information of the downsampled reconstructed picture.

**6**. The image decoding method of claim **1**, wherein

the obtaining of the motion information for the current block comprises:

based on a time interval between the reconstructed picture and the current picture being equal to or greater than a certain value, excluding the reference motion information from a motion information candidate for the current block.

**7**. The image decoding method of claim **1**, wherein

the generating of the reference motion information for the at least one block included in the current picture comprises:

based on reference picture resampling being applied,

downsampling the at least one reconstructed picture, based on a reference picture of low resolution.

**8**. The image decoding method of claim **1**, further comprising

generating motion information for an additional picture based on performing interpolation or extrapolation, based on the motion information of the at least one reconstructed picture.

**9**. The image decoding method of claim **1**, wherein

the at least one syntax information includes information indicating whether the reference motion information is used to obtain the motion information for the current block.

**10**. The image decoding method of claim **1**, wherein

the obtaining of the motion information for the current block comprises:

constructing a merge candidate list including the reference motion information; and

obtaining the motion information for the current block, based on the merge candidate list.

**11**. The image decoding method of claim **1**, wherein

the reference motion information includes motion information at a location corresponding to:

an x-coordinate based on adding half a width of the current block to an upper left sample location of the current block in a map of the reference motion information, and

a y-coordinate based on adding half a height of the current block to the upper left sample location of the current block in the map of the reference motion information.

**12**. The image encoding method of claim **1**,

wherein the obtaining of the motion information for the current block comprises:

dividing the current block into a plurality of subblocks; and

assigning the reference motion information to the plurality of subblocks, and

wherein the reconstructing of the current block comprises performing motion compensation on a subblock, based on reference motion information for the subblock.

**13**. The image decoding method of claim **1**, wherein

the obtaining of the motion information for the current block comprises obtaining a plurality of control point motion vectors for the current block, based on the reference motion information, and

the reconstructing of the current block comprises performing motion compensation on a subblock of the current block based on the plurality of the control point motion vectors for the current block.

**14**. An image encoding method performed by an apparatus, comprising:

generating reference motion information for at least one block included in a current picture, based on at least one of: pixel data of at least one reconstructed picture, or motion information of the at least one reconstructed picture;

determining motion information for a current block among the at least one block included in the current picture; and

encoding the motion information for the current block into a bitstream, based on the reference motion information.

**15**. A non-transitory computer-readable storage medium storing a bitstream encoded by an image encoding method, the image encoding method comprising:

generating reference motion information for at least one block included in a current picture, based on at least one of: pixel data of at least one reconstructed picture, or motion information of the at least one reconstructed picture;

determining motion information for a current block among the at least one block included in the current picture; and

encoding the motion information for the current block into a bitstream, based on the reference motion information.

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